
Geometry of Quantum Computing

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Introduction

The subject of this thesis is an introduction to the geometrical modeling of some problems of quantum computing. Quantum computing is an emergent field of computer science and engineering, based on R. Feynman's suggestion in a seminal talk in 1981 [23], for a computer functioning according to non-binary logic. The quantum computer uses the unique qualities of quantum mechanics to solve problems beyond the ability of the most powerful classical computers.

Quantum computation can be translated into geometric terms, where quantum states are represented either by pure states $|\psi\rangle$ in the projective Hilbert space $\mathbb{P}(\mathcal{H})$, with $\mathcal{H} = (\mathbb{C}^2)^{\otimes n}$, or more generally by density operators

$$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|,$$

and the quantum logic gates correspond to transformations in $U(2^n)$. The universality of the quantum gate CNOT

$$(x, y) \rightarrow (x, x \oplus y)$$

plays a central role in quantum computing, but many problems arise because of the no-cloning theorem. The threshold theorem facilitates the situation, providing a safe and low-damage way for the development of fault tolerant algorithms.

The symmetric logarithmic derivative L provides the ability to define the quantum Fisher matrix

$$F_Q = \text{Tr}(\rho L^2),$$

and examine the strict analogue between the classical and respective quantum formulations. The Fubini-Study metric

$$ds^2 = \frac{(1 + z_i \bar{z}_i) dz_j d\bar{z}_j - \bar{z}_j z_i dz_j d\bar{z}_i}{(1 + z_i \bar{z}_i)^2}$$

deals on $\mathbb{P}(\mathcal{H})$ as a natural way to measure the "distance" between two pure quantum states. Moreover, it is seen as the real part of the quantum geometric tensor

$$\mathcal{Q}_{\mu\nu} = \frac{\langle \partial_\mu \psi | \partial_\nu \psi \rangle}{\langle \psi | \psi \rangle} - \frac{\langle \partial_\mu \psi | \psi \rangle \langle \psi | \partial_\nu \psi \rangle}{\langle \psi | \psi \rangle^2},$$

a gauge-invariant complex tensor that allows the local realization of quantum states' geometry.

Finally, quantum circuits, constructed using quantum logic gates that act on quantum states, are often very complicated and thus it is hard to predict the outputs. First introduced in [13], in 2008, ZX-calculus is a new way to efficiently represent quantum circuits and simplify them easily, following specific axioms.

In this thesis we examine the relationship between quantum computing, quantum information geometry and quantum groups, in the form of Hopf algebras. Making use of the models that are produced by these connections can allow researchers to produce and hopefully answer quantum computing questions.

In particular, the thesis is divided into four chapters.

Chapter 1 of the thesis is an introductory chapter, with graduate-level, standard arguments needed for the thesis. We review briefly some notions of differential geometry, Lie groups, probability theory and statistics. Hopf algebras and F -Hopf algebras are defined, while the small quantum group $u_q(\mathfrak{sl}_2)$ is presented.

Chapter 2 establishes the notion of qubit, density operator and quantum logic gates, discussing the function of quantum algorithms comparing to the classical ones. We deal with the importance of the universal gate CNOT and the problems that are caused by the no-cloning theorem. The issue is facilitated using the threshold theorem.

In Chapter 3, we introduce some concepts of classical information geometry. We present their quantum counterparts, of the quantum information matrix, symmetric logarithmic derivative and examine the respective description between them. Moreover, we deal with the quantum geometric tensor, and Fubini-Study metric as the Kähler metric of the projective Hilbert space.

In the fourth and final chapter, we discuss ZX -calculus as a method to effectively represent quantum logic gates and quantum circuits, and we explore the connection between ZX -rules and axioms of Hopf algebraic structures, making quantum circuits be seen as objects of Hopf algebras.

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Chapter 1

Preliminaries

In this chapter, we introduce the fundamental definitions and results that will be used throughout the thesis. Proofs are omitted.

1.1 Linear Algebra

Definition 1.1. A metric space is a pair (X, d) , where X is a set and $d : X \times X \rightarrow \mathbb{R}$ is a distance function, with the following properties, $\forall x, y, z \in X$.

1. $d(x, y) \geq 0$,
2. $d(x, y) = 0 \iff x = y$,
3. $d(x, y) = d(y, x)$,
4. $d(x, y) \leq d(x, z) + d(z, y)$.

Definition 1.2. A sequence (x_n) in a metric space (X, d) is said to be Cauchy if for every $\varepsilon > 0$ there is an $N = N(\varepsilon)$ such that $d(x_m, x_n) < \varepsilon$, for every $m, n > N$.

The space X is said to be complete if every Cauchy sequence in X converges (that is, has a limit which is an element of X).

Definition 1.3. A Hilbert space $\mathcal{H}$ is a complete inner product space.

Theorem 1.4. Every finite-dimensional vector space with an inner product is Hilbert.

Example 1.5. $\mathbb{C}^2$ is a Hilbert space.

Definition 1.6. The special linear group over a field $\mathbb{k}$ is $\text{SL}(n, \mathbb{k}) = \{X \in \text{GL}(n, \mathbb{k}) \mid \det X = 1\}$.

Definition 1.7. A complex square matrix $U \in M_n(\mathbb{C})$ is called *unitary* if $U^\dagger U = I_n$, where $U^\dagger$ is the conjugate transpose of U .

Proposition 1.8. For $A \in M_{m \times n}(\mathbb{C})$, $B \in M_{n \times m}(\mathbb{C})$, it is $(AB)^\dagger = B^\dagger A^\dagger$.

Proposition 1.9. For $A, B \in M_{m \times n}(\mathbb{C})$, it is $(A \otimes B)^\dagger = A^\dagger \otimes B^\dagger$.

Definition 1.10. The unitary group is $U(n) = \{U \in GL(n, \mathbb{C}) \mid U^\dagger U = I_n\}$.

Definition 1.11. Suppose a linear operator T on vector space V with scalar product $\langle \cdot, \cdot \rangle$. The operator T is called

1. self-adjoint, if $T^\dagger = T$.
2. positive semi-definite, if $\langle Tv, v \rangle \geq 0$, $\forall v \in V$.
3. projection operator, if $T^2 = T$.

Definition 1.12. A Hermitian inner product on a complex vector space V is a complex-valued sesquilinear form on V with the following properties $\forall u, v, w \in V$ and $\forall \alpha \in \mathbb{C}$.

1. $\langle u + v, w \rangle = \langle u, w \rangle + \langle v, w \rangle$,
2. $\langle u, v + w \rangle = \langle u, v \rangle + \langle u, w \rangle$,
3. $\langle \alpha u, v \rangle = \alpha \langle u, v \rangle$,
4. $\langle u, \alpha v \rangle = \bar{\alpha} \langle u, v \rangle$,
5. $\langle u, v \rangle = \overline{\langle v, u \rangle}$,
6. $\langle u, u \rangle \in \mathbb{R}_{\geq 0}$, with equality only if $u = 0$.

Example 1.13. The standard Hermitian inner product is the form

$$h(z, w) = \sum_{i=1}^n z_i \bar{w}_i$$

on $\mathbb{C}^n$, where $z = (z_1, \dots, z_n)$ and $w = (w_1, \dots, w_n)$. Note that by writing $z_k = x_k + iy_k$, it is possible to consider $\mathbb{C}^n \sim \mathbb{R}^{2n}$, in which case $\text{Re}(h)$ is the Euclidean inner product and $\text{Im}(h)$ is a nondegenerate alternating bilinear form, i.e. a symplectic form (see Definition 3.18).

Definition 1.14. Let $\mathcal{H}_1$ and $\mathcal{H}_2$ be complex Hilbert spaces with inner products $\langle \cdot, \cdot \rangle_1$ and $\langle \cdot, \cdot \rangle_2$, respectively. On the tensor product $\mathcal{H}_1 \otimes \mathcal{H}_2$ we define the inner product

$$\begin{aligned} \langle \cdot, \cdot \rangle : (\mathcal{H}_1 \otimes \mathcal{H}_2) \times (\mathcal{H}_1 \otimes \mathcal{H}_2) &\rightarrow \mathbb{C}, \\ (v_1 \otimes v_2, w_1 \otimes w_2) &\mapsto \langle v_1 \otimes v_2, w_1 \otimes w_2 \rangle := \langle v_1, w_1 \rangle_1 \langle v_2, w_2 \rangle_2 \end{aligned}$$

Definition 1.15. We define the bra-ket $\langle \cdot | \cdot \rangle$ on a complex vector space V to be the inner product $\langle \cdot, \cdot \rangle$ on V . In particular, for $u, v \in V$ it is

$$\langle u | v \rangle := \langle u, v \rangle.$$

$\langle \cdot |$ is called the bra and $|\cdot \rangle$ is the ket. It is equivalent to write $\langle u | v \rangle = \langle u | |v \rangle$.

Proposition 1.16. Let a complex Hilbert space $\mathcal{H}$. For $u, v, v_1, v_2 \in \mathcal{H}$, $\alpha_1, \alpha_2 \in \mathbb{C}$, the following properties of bra-ket hold:

1. $\langle u | v \rangle = \langle v | u \rangle^*$
2. $|v|^2 = \langle v | v \rangle$
3. $\langle u | = |u \rangle^*$
4. $|v \rangle = \alpha_1 |v_1 \rangle + \alpha_2 |v_2 \rangle \iff \langle v | = \alpha_1^* \langle v_1 | + \alpha_2^* \langle v_2 |$.

Proposition 1.17. Let a complex Hilbert space $\mathcal{H}$. Let $u, v \in \mathcal{H}$, $\alpha \in \mathbb{C}$ and $T : \mathcal{H} \rightarrow \mathcal{H}$ a linear operator on $\mathcal{H}$. The following hold.

1. $T(|u \rangle + |v \rangle) = T|u \rangle + T|v \rangle$.
2. $|Tu \rangle = T|u \rangle$
3. $T(\alpha |u \rangle) = \alpha T|u \rangle$
4. $\langle T^\dagger u | v \rangle = \langle u | Tv \rangle$
5. $\langle v | T^\dagger |u \rangle = (\langle u | T |v \rangle)^*$
6. $(T|v \rangle)^\dagger = \langle v | T^\dagger$.

Proposition 1.18. Let a complex Hilbert space $\mathcal{H}$ and $u, v, w, z \in \mathcal{H}$. The following hold.

1. $(|u \rangle \otimes |v \rangle)^* = |u \rangle^* \otimes |v \rangle^*$
2. $(\langle u | \otimes \langle v |)(|w \rangle \otimes |z \rangle) = \langle u | w \rangle \langle v | z \rangle$.

Definition 1.19. A Hermitian map A on a finite-dimensional Hermitian product space V is a map such that $A^\dagger = A$.

Theorem 1.20 (Spectral theorem). If A is a Hermitian map on a finite-dimensional Hermitian space V , then there exists an orthonormal basis of V consisting of eigenvectors of A . Each eigenvalue of A is real.

Definition 1.21. A primitive n -th root of unity is a $q \in \mathbb{C}$ with $q^n = 1$ and $q^k \neq 1$, for integers $1 \leq k < n$.

1.2 Manifolds

The following concepts are used for Chapter 3. This section follows [28, 63, 67].

Definition 1.22. Let M be a set. A chart on M is a pair (U, ϕ) , where $U \subset M$ and ϕ is a bijection from U to an open subset $\phi(U) \subset \mathbb{R}^n$ of some Euclidean space.

Two charts (U_1, ϕ_1) and (U_2, ϕ_2) are said to be smoothly compatible iff $\phi_1(U_1 \cap U_2)$ and $\phi_2(U_1 \cap U_2)$ are both open in $\mathbb{R}^n$ and the transition map

$$\phi_2 \circ \phi_1^{-1} : \phi_1(U_1 \cap U_2) \rightarrow \phi_2(U_1 \cap U_2)$$

is a diffeomorphism.

A smooth atlas on M is a collection $\mathcal{A}$ of charts on M any two of which are smoothly compatible and such that the sets U cover M , as (U, ϕ) ranges over $\mathcal{A}$, i.e. for every $p \in M$ there is a chart $(U, \phi) \in \mathcal{A}$ with $p \in U$.

A maximal smooth atlas is an atlas which contains every chart that is smoothly compatible with each of its members.

A smooth manifold or n -manifold, where $n = \dim M$ is a pair consisting of a set M and a maximal atlas $\mathcal{A}$ on M .

Definition 1.23. Let M and N be smooth manifolds with or without boundary. A smooth embedding of M into N is a smooth immersion $F : M \rightarrow N$ that is also a topological embedding, i.e., a homeomorphism onto its image $F(M) \subseteq N$ in the subspace topology.

Definition 1.24. Let M be a smooth manifold with or without boundary. An embedded submanifold of M is a subset $S \subseteq M$ that is a manifold (without boundary) in the subspace topology, endowed with a smooth structure with respect to which the inclusion map $S \hookrightarrow M$ is a smooth embedding.

Definition 1.25. A Riemannian metric on a smooth manifold M is a symmetric positive definite smooth covariant 2-tensor field g . A smooth manifold M equipped with a Riemannian metric g is called a Riemannian manifold, and is denoted by (M, g) .

Definition 1.26. Let V be a finite-dimensional real vector space. If k is a positive integer, a covariant k -tensor on V is an element of the k -fold tensor product $V^* \otimes \cdots \otimes V^*$, which we typically think of as a real-valued multilinear function of k elements of V :

$$\alpha : V \times \cdots \times V \rightarrow \mathbb{R}.$$

The number k is called the rank of α .

Definition 1.27. An order- k tensor $A \in \mathsf{T}^k(\mathbb{C}^n)$ is symmetric if $\sigma(A) = A$ for all permutations $\sigma \in \mathfrak{S}_k$. The set of symmetric tensors in $\mathsf{T}^k(\mathbb{C}^n)$ will be denoted by $\mathsf{S}^k(\mathbb{C}^n)$.

Definition 1.28. The gradient of a map $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is $\nabla f : \mathbb{R}^n \rightarrow \mathbb{R}^n$, with

$$\nabla f(p) = \begin{pmatrix} \frac{\partial f}{\partial x_1}(p) \\ \vdots \\ \frac{\partial f}{\partial x_n}(p) \end{pmatrix}, \quad p = (x_1, \dots, x_n) \in \mathbb{R}^n.$$

Definition 1.29. A function $F = f + ig : U \subset \mathbb{C} \rightarrow \mathbb{C}$ is called holomorphic if it satisfies the Cauchy–Riemann equations:

$$\frac{\partial f}{\partial x} = \frac{\partial g}{\partial y} \quad \text{and} \quad \frac{\partial f}{\partial y} + \frac{\partial g}{\partial x} = 0.$$

Definition 1.30. Let $d : \mathcal{A}_{\mathbb{C}}^k(U) \rightarrow \mathcal{A}_{\mathbb{C}}^{k+1}(U)$ be the complex linear extension of the usual exterior differential. Then

$$\partial : \mathcal{A}^{p,q}(U) \rightarrow \mathcal{A}^{p+1,q}(U) \quad \bar{\partial} : \mathcal{A}^{p,q}(U) \rightarrow \mathcal{A}^{p,q+1}(U)$$

are defined as $\partial := \Pi^{p+1,q} \circ d$ and $\bar{\partial} := \Pi^{p,q+1} \circ d$. For any local function f the exterior differential is defined as

$$d(f) = \sum_i \frac{\partial f}{\partial x_i} dx_i + \sum_i \frac{\partial f}{\partial y_i} dy_i = \sum_i \frac{\partial f}{\partial z_i} dz_i + \sum_i \frac{\partial f}{\partial \bar{z}_i} d\bar{z}_i. \quad (1)$$

Thus, f is holomorphic if and only if $\bar{\partial}f = 0$. Moreover, using Equation (1) the operators ∂ and $\bar{\partial}$ can be expressed explicitly as follows:

$$\partial(f \, dz_{i_1} \wedge \dots \wedge dz_{i_p} \wedge d\bar{z}_{j_1} \wedge \dots \wedge d\bar{z}_{j_q}) = \sum_{k=1}^n \frac{\partial f}{\partial z_k} dz_k \wedge dz_{i_1} \wedge \dots \wedge dz_{i_p} \wedge d\bar{z}_{j_1} \wedge \dots \wedge d\bar{z}_{j_q}$$

and

$$\bar{\partial}(f \, dz_{i_1} \wedge \dots \wedge dz_{i_p} \wedge d\bar{z}_{j_1} \wedge \dots \wedge d\bar{z}_{j_q}) = \sum_{l=1}^n \frac{\partial f}{\partial \bar{z}_l} d\bar{z}_l \wedge dz_{i_1} \wedge \dots \wedge dz_{i_p} \wedge d\bar{z}_{j_1} \wedge \dots \wedge d\bar{z}_{j_q}.$$

Lemma 1.31. For the differential operators ∂ and $\bar{\partial}$ one has:

1. $d = \partial + \bar{\partial}$.
2. $\partial^2 = \bar{\partial}^2 = 0$ and $\partial\bar{\partial} = -\bar{\partial}\partial$.
3. They satisfy the Leibniz rule, i.e. for $\alpha \in \mathcal{A}^{p,q}(U)$ and $\beta \in \mathcal{A}^{r,s}(U)$,

$$\begin{aligned} \partial(\alpha \wedge \beta) &= \partial(\alpha) \wedge \beta + (-1)^{p+q} \alpha \wedge \partial(\beta), \\ \bar{\partial}(\alpha \wedge \beta) &= \bar{\partial}(\alpha) \wedge \beta + (-1)^{p+q} \alpha \wedge \bar{\partial}(\beta). \end{aligned}$$

Definition 1.32. Let M be a connected n -dimensional manifold, g be a Riemannian metric on M , and ∇ be the Levi-Civita connection of g . Let p, p' be points in M joined by a smooth path γ . Then parallel transport along γ using ∇ defines an isometry between the tangent spaces $T_p M, T_{p'} M$ at p and p' respectively.

The holonomy group $\text{Hol}_p(M, g)$ of g is the group of isometries of $T_p M$ generated by parallel transport around closed loops based at p in M . That is,

$$\text{Hol}_p(M, g) = \{P_\gamma \in \text{GL}(T_p M) : \gamma \text{ is a loop based at } p\}.$$

$\text{Hol}(M, g)$ can be viewed as a subgroup of the orthogonal group $\text{O}(n)$, where $n = \dim(M)$, defined up to conjugation by elements of $\text{O}(n)$. Then, $\text{Hol}(M, g)$ is independent of the base point p in M , up to conjugation.

1.3 Lie Groups and Algebras

The following notions are needed for Subsection 1.6.3 [38, 43, 64].

Definition 1.33. A Lie group G is a group and a manifold so that the following hold:

1. the multiplication map $\mu : G \times G \rightarrow G$ given by $\mu(g, g') = gg'$ is smooth,
2. the inverse map $\iota : G \rightarrow G$ given by $\iota(g) = g^{-1}$ is smooth.

Example 1.34. A very interesting example of a Lie group is $\text{GL}(n, \mathbb{k}) = \{X \in M_n(\mathbb{k}) \mid \det(X) \neq 0\}$. To verify that $\text{GL}(n, \mathbb{k})$ is a Lie group, we first observe that the multiplication is smooth, because it is a polynomial map in the coordinates.

In order to check that the inverse map is smooth, it is required to have the standard linear algebra formula

$$X^{-1} = \text{adj}(X) / \det X,$$

where the $\text{adj}(X)$ is the transpose of the matrix of cofactors.

In particular, the coordinates of $\text{adj}(X)$ are polynomial functions in the coordinates of X and $\det X$ is a nonvanishing polynomial on $\text{GL}(n, \mathbb{k})$ so the inverse is a smooth map.

Proposition 1.35. A closed subgroup of a Lie group is a Lie group in its own right with respect to the relative topology.

Example 1.36. $\text{SL}(n, \mathbb{k})$ is a Lie group since it is a closed subgroup of the Lie group $\text{GL}(n, \mathbb{k})$.

Definition 1.37. Let $\mathbb{k}$ be a field. An algebra $\mathfrak{g}$ (not necessarily associative) is a vector space over $\mathbb{k}$ with a product $[X, Y]$ that is linear in each variable.

The algebra is a Lie algebra if the product satisfies also $\forall X, Y, Z \in \mathfrak{g}$:

1. $[X, X] = 0$ for all $X \in \mathfrak{g}$, and hence $[X, Y] = -[Y, X]$,
2. the Jacobi identity $[[X, Y], Z] + [[Y, Z], X] + [[Z, X], Y] = 0$.

Example 1.38. Let $\mathfrak{gl}(n, \mathbb{k}) = \{X \in M_n(\mathbb{k})\}$ be the associative algebra of all $n \times n$ matrices with entries in the field $\mathbb{k}$, and define a bracket product by $[X, Y] = XY - YX$. Then, $\mathfrak{gl}(n, \mathbb{k})$ becomes a Lie algebra.

Example 1.39. The Lie algebra of $\mathrm{SL}(n, \mathbb{k})$ is $\mathfrak{sl}(n, \mathbb{k}) = \{X \in \mathfrak{gl}(n, \mathbb{k}) \mid \mathrm{Tr}(X) = 0\}$.

1.4 Probability and Statistics

This part contains useful definitions for Section 3.1. We follow [70].

Definition 1.40. Let X be a discrete random variable. The probability mass function of X is given by

$$p_X(x) = \mathbb{P}(X = x).$$

Definition 1.41. Let X be a discrete random variable. The cumulative distribution function of X is given by

$$F_X(x) = \mathbb{P}(X \leq x).$$

Definition 1.42. Let $X_1, \dots, X_n$ be independent and each has the same marginal distribution with cumulative distribution function F . Then, we say that $X_1, \dots, X_n$ are independent and identically distributed and we write $X_1, \dots, X_n \sim F$.

Definition 1.43. Let $X_1, \dots, X_n \sim F$ be an independent and identically distributed sample, with probability function F on the real line. The empirical distribution function $\hat{F}_n(X)$ is the cumulative distribution function that puts mass $\frac{1}{n}$ at each data point X_i :

$$\hat{F}_n(X) = \frac{\sum_{i=1}^n I(X_i \leq x)}{n},$$

where

$$I(X_i \leq x) = \begin{cases} 1, & \text{if } X_i \leq x, \\ 0, & \text{if } X_i > x. \end{cases}$$

Definition 1.44. Let X be a discrete random variable with finite or countably infinite range $R_X = \{x_1, x_2, x_3, \dots\}$. The expected value of X is defined as

$$\mathbb{E}(X) = \sum_{x_k \in R_X} x_k \mathbb{P}(X = x_k).$$

Remark 1.45. The mean value $\mathbb{E}(X)$ of a random variable $X = [X_1 \ X_2 \ \dots \ X_n]^t$ is often written as the column vector $\mu = \mathbb{E}(X) \in \mathbb{R}^n$, where

$$\mu = [\mu_1 \ \dots \ \mu_n]^t = [\mathbb{E}(X_1) \ \dots \ \mathbb{E}(X_n)]^t.$$

Definition 1.46. Let X be a random variable with mean value $\mu = \mathbb{E}(X)$. The variance of X is defined as

$$\text{Var}(X) = \mathbb{E}[(X - \mu)^2].$$

Definition 1.47. Let $X = [X_1 \ X_2 \ \dots \ X_n]^t \in \mathbb{R}^n$ be a random vector with mean vector $\mu = \mathbb{E}(X) \in \mathbb{R}^n$. The covariance matrix of X is defined via:

$$\Sigma = \text{Cov}(X) = \mathbb{E}[(X - \mu)(X - \mu)^t].$$

It is

$$\begin{aligned} \Sigma_{ij} &= \text{Cov}(X_i, X_j) = \mathbb{E}[(X_i - \mu_i)(X_j - \mu_j)], \\ \Sigma_{ii} &= \text{Cov}(X_i, X_i) = \mathbb{E}[(X_i - \mu_i)^2] = \text{Var}(X_i). \end{aligned}$$

So,

$$\text{Cov}(X) = \begin{pmatrix} \text{Var}(X_1) & \dots & \text{Cov}(X_1, X_n) \\ \dots & \ddots & \dots \\ \text{Cov}(X_n, X_1) & \dots & \text{Var}(X_n) \end{pmatrix}.$$

The covariance matrix is symmetric, positive semi-definite, with real and non-negative eigenvalues.

Definition 1.48. Suppose $f(x)$ and $g(x)$ are two functions defined on some subset of the real numbers. We write $f(x) = \mathcal{O}(g(x))$, for $x \rightarrow \infty$, if and only if there exist x_0 and M such that

$$|f(x)| \leq M|g(x)|, \quad \forall x > x_0.$$

In the notation, we often just use $f(x) = \mathcal{O}(g(x))$, omitting the reference to the limit.

1.5 Algebras and Coalgebras

We will give some useful algebraic definitions, that will be needed in Section 4.2 [10, 46].

Definition 1.49. Let R be a ring with multiplicative identity 1_R . A (left) R -module A is an abelian group $(A, +)$, together with an operation $\cdot : R \times A \rightarrow A$, such that, for all $r, s \in R$ and $a, b \in A$

$$1. \ (r + s) \cdot a = r \cdot a + s \cdot a,$$

$$2. \ r \cdot (a + b) = r \cdot a + r \cdot b,$$

$$3. \ (rs) \cdot a = r \cdot (s \cdot a),$$

$$4. \ 1_R \cdot a = a.$$

Definition 1.50. An R -algebra is an R -module A with R -linear maps

$$m : A \otimes_R A \rightarrow A \quad \text{and} \quad \eta : R \rightarrow A,$$

called the product and the unit, respectively, with the properties

$$m \circ (m \otimes \text{id}_A) = m \circ (\text{id}_A \otimes m),$$

$$m \circ (\eta \otimes \text{id}_A) = \text{id}_A = m \circ (\text{id}_A \otimes \eta).$$

Now we will give the definition of coalgebra. Intuitively, a coalgebra over a ring can be understood as a dualisation of an algebra over a ring. Coalgebras by themselves are equally fundamental objects as are algebras. From a geometric perspective, algebras can be viewed as functions on spaces and coalgebras as objects that encode additional structure of such spaces (e.g. group or monoid structures).

Definition 1.51. An R -coalgebra is an R -module C with R -linear maps

$$\Delta : C \rightarrow C \otimes_R C \quad \text{and} \quad \varepsilon : C \rightarrow R,$$

called (coassociative) coproduct and counit, respectively, with the properties

$$(\text{id}_C \otimes \Delta) \circ \Delta = (\Delta \otimes \text{id}_C) \circ \Delta,$$

$$(\text{id}_C \otimes \varepsilon) \circ \Delta = \text{id}_C = (\varepsilon \otimes \text{id}_C) \circ \Delta,$$

which can be expressed by commutativity of the diagrams

$$\begin{array}{ccc} C & \xrightarrow{\Delta} & C \otimes_R C \\ \Delta \downarrow & & \downarrow \text{id}_C \otimes \Delta \\ C \otimes_R C & \xrightarrow{\Delta \otimes \text{id}_C} & C \otimes_R C \otimes_R C \end{array} \qquad \begin{array}{ccc} C & \xrightarrow{\Delta} & C \otimes_R C \\ \Delta \downarrow & \searrow \text{id}_C & \downarrow \varepsilon \otimes \text{id}_C \\ C \otimes_R C & \xrightarrow{\text{id}_C \otimes \varepsilon} & C \end{array}$$

1.6 Unnormalized and F -Hopf Algebras

In this section, we recall some formal notions regarding unnormalized Hopf algebras and F -algebras [12, 37, 53, 55].

1.6.1 Unnormalized Hopf Algebras

Below we define the term of unnormalized morphism between two algebras. It is a relaxed definition of algebra morphism. We use it to define unnormalized bialgebras and unnormalized Hopf algebras.

Definition 1.52. Let $\mathbb{k}$ be a field and (A, m_A, η_A) and (B, m_B, η_B) be two $\mathbb{k}$ -algebras. Let a fixed nonzero scalar $\diamond \in \mathbb{k} \setminus \{0\}$. An unnormalized algebra morphism f up to $\diamond$ is a linear map $f : A \rightarrow B$ that satisfies

$$f \circ m_A = \diamond m_B \circ (f \otimes f), \quad \diamond f \circ \eta_A = \eta_B.$$

When $\diamond = 1$, we have the usual case of (normalized) algebra morphism.

Now, we define the unnormalized Hopf algebra. The axioms are presented in a graphical language, with string diagrams to be read from left to right. We use this graphical representation to denote the operations between algebras and coalgebras. This is the way we represent the circuits in ZX-calculus in general (for the detailed construction see Chapter 4).

Definition 1.53. An unnormalized Hopf algebra is a $\mathbb{k}$ -vector space A , equipped with

i) a multiplication and a unit

$$m_{\bullet} := \text{diagram} : A \otimes A \rightarrow A \quad \eta_{\bullet} := \text{diagram} : \mathbb{k} \rightarrow A$$

satisfying

$$\text{diagram} = \text{diagram} \quad \text{Associativity}$$

where the diagram shows that $m_{\bullet} \circ (m_{\bullet} \otimes \text{id}_A) = m_{\bullet} \circ (\text{id}_A \otimes m_{\bullet})$. Also, the operations need to satisfy

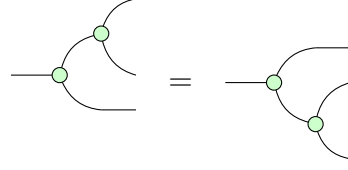
$$\text{diagram} = \text{diagram} = \text{diagram} \quad \text{Unitality}$$

that is, $m_{\bullet} \circ (\eta_{\bullet} \otimes \text{id}_A) = \text{id}_A = m_{\bullet} \circ (\text{id}_A \otimes \eta_{\bullet})$. Hence, $(A, m_{\bullet}, \eta_{\bullet})$ is an algebra (see Definition 1.50),

ii) a comultiplication and counit

$$\Delta_{\bullet} = \text{diagram} : A \rightarrow A \otimes A \quad \varepsilon_{\bullet} := \text{diagram} : A \rightarrow \mathbb{k}$$

satisfying



Coassociativity

that is, $(\Delta_{\bullet} \otimes \text{id}_A) \circ \Delta_{\bullet} = (\text{id}_A \otimes \Delta_{\bullet}) \circ \Delta_{\bullet}$ and



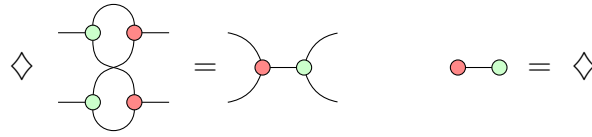
Cointeraction

that is, $(\varepsilon_{\bullet} \otimes \text{id}_A) \circ \Delta_{\bullet} = \text{id}_A = (\text{id}_A \otimes \varepsilon_{\bullet}) \circ \Delta_{\bullet}$. Thus, $(A, \Delta_{\bullet}, \varepsilon_{\bullet})$ is a coalgebra (see Definition 1.51),

iii) a normalization constant $\diamond \in \mathbb{k} \setminus \{0\}$ such that $\Delta_{\bullet}$ and $\varepsilon_{\bullet}$ are unnormalized algebra morphisms:



(2)



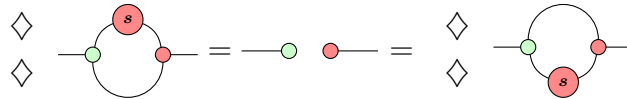
(3)

Equivalently, in algebraic form,

$$\begin{aligned} \diamond \Delta_{\bullet}(\eta_{\bullet}) &= \eta_{\bullet} \otimes \eta_{\bullet}, & \diamond \varepsilon_{\bullet}(m_{\bullet}) &= \varepsilon_{\bullet} \otimes \varepsilon_{\bullet} \\ \diamond m_{\bullet, A \otimes A}(\Delta_{\bullet} \otimes \Delta_{\bullet}) &= \Delta_{\bullet}(m_{\bullet}), & \diamond &= \varepsilon_{\bullet}(\eta_{\bullet}). \end{aligned}$$

hence, $(A, m_{\bullet}, \eta_{\bullet}, \Delta_{\bullet}, \varepsilon_{\bullet})$ is a bialgebra [53, 54],

iv) and a linear map $S := -\textcircled{s} : A \rightarrow A$ satisfying the antipode axioms



(4)

$$\text{i.e. } \diamond^2 m_{\bullet}(\text{id}_A \otimes S) \Delta_{\bullet} = \eta_{\bullet}(\varepsilon_{\bullet}) = \diamond^2 m_{\bullet}(S \otimes \text{id}_A) \Delta_{\bullet}.$$

The usual notion of (normalized) Hopf algebra is recovered by setting $\diamond = 1$.

Remark 1.54. In the Computer Science literature, a bialgebra is what an unnormalised (or scaled) bialgebra is in Quantum and Braided ZX-calculus, that we will see in Chapter 4. That is, in Computer Science we work up to the normalization constant, so we usually omit the word "unnormalized". [14, 54].

Proposition 1.55. Given an unnormalized Hopf algebra $(\mathcal{H}, m_{\bullet}, \eta_{\bullet}, \Delta_{\bullet}, \varepsilon_{\bullet}, S)$, we obtain a normalized Hopf algebra $(\mathcal{H}, m_{\bullet}, \eta_{\bullet}, \Delta'_{\bullet}, \varepsilon'_{\bullet}, S')$, by setting $\Delta'_{\bullet} := \diamond \Delta_{\bullet}$, $\varepsilon'_{\bullet} := \diamond^{-1} \varepsilon_{\bullet}$ and $S' := \diamond^2 S$.

Proof. $(\mathcal{H}, m_{\bullet}, \eta_{\bullet})$ is an algebra by assumption. Let $\Delta'_{\bullet}, \varepsilon'_{\bullet}$ as above. Then, the comultiplication is coassociative, since

$$\begin{aligned} (\Delta'_{\bullet} \otimes \text{id}_{\mathcal{H}}) \circ \Delta'_{\bullet} &= (\diamond \Delta_{\bullet} \otimes \text{id}_{\mathcal{H}}) \circ (\diamond \Delta_{\bullet}) = \diamond^2 (\Delta_{\bullet} \otimes \text{id}_{\mathcal{H}}) \circ \Delta_{\bullet} \\ &= \diamond^2 (\text{id}_{\mathcal{H}} \otimes \Delta_{\bullet}) \circ \Delta_{\bullet} = (\text{id}_{\mathcal{H}} \otimes \diamond \Delta_{\bullet}) \circ (\diamond \Delta_{\bullet}) = (\text{id}_{\mathcal{H}} \otimes \Delta'_{\bullet}) \circ \Delta'_{\bullet}. \end{aligned}$$

Also, it is counital, because

$$\begin{aligned} (\varepsilon'_{\bullet} \otimes \text{id}_{\mathcal{H}}) \circ \Delta'_{\bullet} &= (\diamond^{-1} \varepsilon_{\bullet} \otimes \text{id}_{\mathcal{H}}) \circ (\diamond \Delta_{\bullet}) = \diamond^{-1} \diamond (\varepsilon_{\bullet} \otimes \text{id}_{\mathcal{H}}) \circ \Delta_{\bullet} \\ &= \diamond^{-1} \diamond (\text{id}_{\mathcal{H}} \otimes \varepsilon_{\bullet}) \circ \Delta_{\bullet} = (\text{id}_{\mathcal{H}} \otimes \diamond^{-1} \varepsilon_{\bullet}) \circ (\diamond \Delta_{\bullet}) = (\text{id}_{\mathcal{H}} \otimes \varepsilon'_{\bullet}) \circ \Delta'_{\bullet}. \end{aligned}$$

Hence, $(\mathcal{H}, \Delta'_{\bullet}, \varepsilon'_{\bullet})$ is a coalgebra. Let $\diamond$ be the normalization constant of $(\mathcal{H}, m_{\bullet}, \eta_{\bullet}, \Delta'_{\bullet}, \varepsilon'_{\bullet}, S')$. Then,

$$\diamond = \varepsilon'_{\bullet}(\eta_{\bullet}) = \diamond^{-1} \varepsilon_{\bullet}(\eta_{\bullet}) = \diamond^{-1} \diamond = 1,$$

hence, $\diamond = 1$. So, we have

$$\begin{aligned} \diamond \Delta'_{\bullet}(\eta_{\bullet}) &= \Delta'_{\bullet}(\eta_{\bullet}) = \diamond \Delta_{\bullet}(\eta_{\bullet}) = \eta_{\bullet} \otimes \eta_{\bullet}, \\ \diamond \varepsilon'_{\bullet}(m_{\bullet}) &= \varepsilon'_{\bullet}(m_{\bullet}) = \diamond^{-1} \varepsilon_{\bullet}(m_{\bullet}) = \diamond^{-2} \diamond \varepsilon_{\bullet}(m_{\bullet}) = \diamond^{-2} \varepsilon_{\bullet} \otimes \varepsilon_{\bullet} = \varepsilon'_{\bullet} \otimes \varepsilon'_{\bullet}. \end{aligned}$$

So, the Equations (2) are satisfied. Also,

$$\begin{aligned} m_{\bullet, \mathcal{H} \otimes \mathcal{H}}(\Delta'_{\bullet} \otimes \Delta'_{\bullet}) &= m_{\bullet, \mathcal{H} \otimes \mathcal{H}}(\diamond \Delta_{\bullet} \otimes \diamond \Delta_{\bullet}) \\ &= \diamond^2 m_{\bullet, \mathcal{H} \otimes \mathcal{H}}(\Delta_{\bullet} \otimes \Delta_{\bullet}) = \diamond \Delta_{\bullet}(m_{\bullet}) = \Delta'_{\bullet}(m_{\bullet}) \end{aligned}$$

that is, the Equations (3) are satisfied too, and therefore, $(\mathcal{H}, m_{\bullet}, \eta_{\bullet}, \Delta'_{\bullet}, \varepsilon'_{\bullet})$ is a bialgebra. Now, for the antipode axioms we have

$$\begin{aligned} \diamond^2 m_{\bullet}(\text{id}_{\mathcal{H}} \otimes S') \Delta'_{\bullet} &= m_{\bullet}(\text{id}_{\mathcal{H}} \otimes S') \Delta'_{\bullet} = m_{\bullet}(\text{id}_{\mathcal{H}} \otimes \diamond^2 S) \diamond \Delta_{\bullet} \\ &= \diamond^3 m_{\bullet}(\text{id}_{\mathcal{H}} \otimes S) \Delta_{\bullet} = \diamond \eta_{\bullet}(\varepsilon_{\bullet}) = \eta_{\bullet}(\diamond \varepsilon_{\bullet}) = \eta_{\bullet}(\varepsilon'_{\bullet}) \end{aligned}$$

and similarly,

$$\diamond^2 m_{\bullet}(S' \otimes \text{id}_{\mathcal{H}}) \Delta'_{\bullet} = \diamond^3 m_{\bullet}(S \otimes \text{id}_{\mathcal{H}}) \Delta_{\bullet} = \diamond \eta_{\bullet}(\varepsilon_{\bullet}) = \eta_{\bullet}(\varepsilon'_{\bullet}).$$

Therefore, $(\mathcal{H}, m_{\bullet}, \eta_{\bullet}, \Delta'_{\bullet}, \varepsilon'_{\bullet}, S')$ is a normalized Hopf algebra. □

Example 1.56. Let G be a group with neutral element $e \in G$. We equip the group algebra $\mathbb{k}G$ with the maps

$$\begin{aligned} m_{\bullet}(g \otimes h) &:= gh, & \eta_{\bullet}(1_{\mathbb{k}}) &:= e, \\ \Delta_{\bullet}(g) &= g \otimes g, & \varepsilon_{\bullet}(g) &= 1_{\mathbb{k}} \end{aligned}$$

and $S(g) = g^{-1}$, defined for all $g, h \in G$ and extended $\mathbb{k}$ -linearly to $\mathbb{k}G$.

$m_{\bullet}$ is associative in $\mathbb{k}G$, since $\forall g, h, l \in G$ (and due to associativity of multiplication in G):

$$\begin{aligned} m_{\bullet} \circ (m_{\bullet}(g \otimes h) \otimes \text{id}_{\mathbb{k}G}(l)) &= m_{\bullet}(gh \otimes l) = (gh)l = g(hl) = m_{\bullet}(g \otimes hl) \\ &= m_{\bullet} \circ (\text{id}_{\mathbb{k}G}(g) \otimes m_{\bullet}(h \otimes l)) \end{aligned}$$

and also, $\eta_{\bullet}$ satisfies the unit property:

$$\begin{aligned} m_{\bullet} \circ (\eta_{\bullet}(1_{\mathbb{k}}) \otimes \text{id}_{\mathbb{k}G}(g)) &= m_{\bullet}(e \otimes g) = eg = \text{id}_{\mathbb{k}G}(g) = ge = m_{\bullet}(g \otimes e) \\ &= m_{\bullet} \circ (\text{id}_{\mathbb{k}G}(g) \otimes \eta_{\bullet}(1_{\mathbb{k}})). \end{aligned}$$

Thus, since the associativity and unitality are satisfied on basis elements, then by linearity they hold on all of $\mathbb{k}G$. This argument is used again on all the following steps. Therefore, $(\mathbb{k}G, m_{\bullet}, \eta_{\bullet})$ is an associative unital algebra. Similarly,

$$\begin{aligned} (\Delta_{\bullet} \otimes \text{id}_{\mathbb{k}G}) \circ \Delta_{\bullet}(g) &= (\Delta_{\bullet} \otimes \text{id}_{\mathbb{k}G})(g \otimes g) = \Delta_{\bullet}(g) \otimes g = (g \otimes g) \otimes g = g \otimes (g \otimes g) \\ &= g \otimes \Delta_{\bullet}(g) = (\text{id}_G \otimes \Delta_{\bullet})(g \otimes g) = (\text{id}_{\mathbb{k}G} \otimes \Delta_{\bullet}) \circ \Delta_{\bullet}(g) \end{aligned}$$

and

$$\begin{aligned} (\varepsilon_{\bullet} \otimes \text{id}_{\mathbb{k}G}) \circ \Delta_{\bullet}(g) &= (\varepsilon_{\bullet} \otimes \text{id}_{\mathbb{k}G})(g \otimes g) = \varepsilon_{\bullet}(g) \otimes g = 1_{\mathbb{k}} \otimes g \cong \text{id}_{\mathbb{k}G}(g) \\ &\cong g \otimes \varepsilon_{\bullet}(g) = (\text{id}_{\mathbb{k}G} \otimes \varepsilon_{\bullet})(g \otimes g) = (\text{id}_{\mathbb{k}G} \otimes \varepsilon_{\bullet}) \circ \Delta_{\bullet}(g). \end{aligned}$$

So, $(\mathbb{k}G, \Delta_{\bullet}, \varepsilon_{\bullet})$ is a coassociative counital coalgebra. Now,

$$\Delta_{\bullet} \circ \eta_{\bullet}(1_{\mathbb{k}}) = \Delta_{\bullet}(e) = e \otimes e = \eta_{\bullet}(1_{\mathbb{k}}) \otimes \eta_{\bullet}(1_{\mathbb{k}})$$

and also,

$$m_{\bullet}(\Delta_{\bullet}(g) \otimes \Delta_{\bullet}(h)) = \Delta_{\bullet}(g) \Delta_{\bullet}(h) = (g \otimes g)(h \otimes h) = gh \otimes gh = \Delta_{\bullet}(gh) = \Delta_{\bullet} \circ m_{\bullet}(g \otimes h).$$

That is, $\Delta_{\bullet}$ preserves multiplication and $e \otimes e$ is the unit in $\mathbb{k}G \otimes \mathbb{k}G$. Hence $\Delta_{\bullet}$ is an algebra homomorphism. In the same way,

$$\varepsilon_{\bullet} \circ m_{\bullet}(g \otimes h) = \varepsilon_{\bullet}(gh) = 1_{\mathbb{k}} = 1_{\mathbb{k}} \cdot 1_{\mathbb{k}} = \varepsilon_{\bullet}(g) \varepsilon_{\bullet}(h)$$

that is, $\varepsilon_{\bullet}$ preserves multiplication. Moreover,

$$\varepsilon_{\bullet} \circ \eta_{\bullet}(1_{\mathbb{k}}) = \varepsilon_{\bullet}(e) = 1_{\mathbb{k}}.$$

Therefore, $(\mathbb{k}G, m_{\bullet}, \eta_{\bullet}, \Delta_{\bullet}, \varepsilon_{\bullet})$ is a bialgebra. Finally,

$$\begin{aligned} m_{\bullet}(\text{id}_A \otimes S)\Delta_{\bullet}(g) &= m_{\bullet}(\text{id}_A \otimes S)(g \otimes g) = m_{\bullet}(g \otimes g^{-1}) = gg^{-1} = e = \eta_{\bullet}(1_{\mathbb{k}}) \\ &= \eta_{\bullet}(\varepsilon_{\bullet}(g)) = g^{-1}g = m_{\bullet}(g^{-1} \otimes g) = m_{\bullet}(S \otimes \text{id}_A)(g \otimes g) \\ &= m_{\bullet}(S \otimes \text{id}_A)\Delta_{\bullet}(g). \end{aligned}$$

Hence, S is an antipode on $\mathbb{k}G$. Therefore, $(\mathbb{k}G, m_{\bullet}, \eta_{\bullet}, \Delta_{\bullet}, \varepsilon_{\bullet}, S)$ is a normalized Hopf algebra.

1.6.2 F -Hopf algebras

Definition 1.57. A vector space A is called F -algebra if there is an associative unital algebra structure $(m = \rhd, \eta)$ on A and a coassociative counital coalgebra structure $(\Delta = \lhd, \varepsilon)$ on A such that the following holds:

$$\text{Diagram 1} = \text{Diagram 2} = \text{Diagram 3} \quad (5)$$

Definition 1.58. A pair $(A, m_{\bullet}, \eta_{\bullet}, \Delta_{\bullet}, \varepsilon_{\bullet})$, $(A, m_{\bullet}, \eta_{\bullet}, \Delta_{\bullet}, \varepsilon_{\bullet})$ of F -algebras on the same vector space A is called an F -bialgebra if

$$(A, m_{\bullet}, \eta_{\bullet}, \Delta_{\bullet}, \varepsilon_{\bullet}) \quad , \quad (A, m_{\bullet}, \eta_{\bullet}, \Delta_{\bullet}, \varepsilon_{\bullet})$$

are bialgebras. The pair is called an F -Hopf algebra if the latter are Hopf algebras.

Example 1.59. Let G be a finite group with neutral element $e \in G$. Then $A = \mathbb{k}G$ is an F -Hopf algebra with

$$m_{\bullet}(g \otimes h) = gh, \quad \eta_{\bullet}(1_{\mathbb{k}}) = e, \quad \Delta_{\bullet}(g) = \sum_{h, l \in G, hl=g} h \otimes l, \quad \varepsilon_{\bullet}(g) = \delta_{e, g}$$

and

$$m_{\bullet}(g \otimes h) = \delta_{g, h}g, \quad \eta_{\bullet}(1_{\mathbb{k}}) = \sum_{g \in G} g, \quad \Delta_{\bullet}(g) = g \otimes g, \quad \varepsilon_{\bullet}(g) = 1_{\mathbb{k}}.$$

The $(A, m_{\bullet}, \eta_{\bullet}, \Delta_{\bullet}, \varepsilon_{\bullet})$ is the Hopf algebra from Example 1.56 with antipode $S(g) = g^{-1}$. Now, for $(A, m_{\bullet}, \eta_{\bullet}, \Delta_{\bullet}, \varepsilon_{\bullet})$ and for $g, h, l \in G$ we have:

$$\delta_{\delta_{g, h} g, l} = \begin{cases} \delta_{g, l}, & \text{if } g = h, \\ \delta_{0, l}, & g \neq h \end{cases} = \begin{cases} 1, & \text{if } g = h = l, \\ 0, & g = h \neq l \\ 0, & g \neq h \end{cases} = \begin{cases} 1, & \text{if } g = h = l, \\ 0, & \text{else} \end{cases} = \delta_{g, h} \delta_{g, l}$$

where, for $\delta_{0,l} = 0$, we used the fact that $l \in G$ and therefore, it cannot be $l = 0$ as a group element. Now, making use of the notation $m_\bullet(g \otimes h) := g \bullet h$ and using the above computations for δ and its symmetric property, it follows for the associativity of $m_\bullet$ that:

$$\begin{aligned} (g \bullet h) \bullet \text{id}_{\mathbb{k}G}(l) &= (\delta_{g,h} g) \bullet l = \delta_{\delta_{g,h} g, l} \delta_{g,h} g = \delta_{g,h} \delta_{g,l} g = \delta_{h,l} \delta_{h,g} g \\ &= \delta_{\delta_{h,l} h, g} g = \delta_{g, \delta_{h,l} h} g = g \bullet (\delta_{h,l} h) = \text{id}_{\mathbb{k}G}(g) \bullet (h \bullet l). \end{aligned}$$

Also, for unitality:

$$\begin{aligned} \eta_\bullet(1_{\mathbb{k}}) \bullet \text{id}_{\mathbb{k}G}(h) &= \left(\sum_{g \in G} g \right) \bullet h = \sum_{g \in G} (g \bullet h) = \sum_{g \in G} \delta_{g,h} g = h = \text{id}_{\mathbb{k}G}(h) = \sum_{g \in G} \delta_{h,g} h \\ &= \sum_{g \in G} \delta_{h,g} g = \sum_{g \in G} (h \bullet g) = h \bullet \sum_{g \in G} g = \text{id}_{\mathbb{k}G}(h) \bullet \eta_\bullet(1_{\mathbb{k}}) \end{aligned}$$

then the above hold by linearity on all of $\mathbb{k}G$. Therefore, $(\mathbb{k}G, m_\bullet, \eta_\bullet)$ is an associative unital algebra. Now, and using the bilinearity of tensor product, and the associativity of G as a group, we have:

$$\begin{aligned} (\Delta_\bullet \otimes \text{id}_{\mathbb{k}G}) \circ \Delta_\bullet(g) &= (\Delta_\bullet \otimes \text{id}_{\mathbb{k}G}) \left(\sum_{\substack{h, l \in G, \\ hl = g}} h \otimes l \right) = \sum_{\substack{h, l \in G, \\ hl = g}} \Delta_\bullet(h) \otimes \text{id}_{\mathbb{k}G}(l) \\ &= \sum_{\substack{h, l \in G, \\ hl = g}} \left(\sum_{\substack{k, n \in G, \\ kn = h}} k \otimes n \right) \otimes l = \sum_{\substack{k, n, l \in G, \\ knl = g}} k \otimes n \otimes l = \sum_{\substack{h' \in G, \\ h'k'n' = g}} h' \otimes \sum_{k', n' \in G} k' \otimes n' \\ &= \sum_{\substack{h', l' \in G, \\ h'l' = g}} h' \otimes \sum_{\substack{k', n' \in G, \\ k'n' = l'}} k' \otimes n' = \sum_{\substack{h', l' \in G, \\ h'l' = g}} h' \otimes \Delta_\bullet(l') = (\text{id}_{\mathbb{k}G} \otimes \Delta_\bullet) \circ \Delta_\bullet(g), \end{aligned}$$

and also,

$$\begin{aligned} (\varepsilon_\bullet \otimes \text{id}_{\mathbb{k}G}) \circ \Delta_\bullet(g) &= (\varepsilon_\bullet \otimes \text{id}_{\mathbb{k}G}) \left(\sum_{\substack{h, l \in G, \\ hl = g}} h \otimes l \right) = \sum_{\substack{h, l \in G, \\ hl = g}} \varepsilon_\bullet(h) \otimes l = \sum_{\substack{h, l \in G, \\ hl = g}} \delta_{e,h} \otimes l \\ &\cong \sum_{\substack{l \in G, \\ el = g}} l = g = \text{id}_{\mathbb{k}G}(g) = \sum_{\substack{h, l \in G, \\ hl = g}} h \delta_{e,l} = \sum_{\substack{h, l \in G, \\ hl = g}} h \varepsilon_\bullet(l) = (\text{id}_{\mathbb{k}G} \otimes \varepsilon_\bullet) \left(\sum_{\substack{h, l \in G, \\ hl = g}} h \otimes l \right) \\ &= (\text{id}_{\mathbb{k}G} \otimes \varepsilon_\bullet) \circ \Delta_\bullet(g). \end{aligned}$$

So, $(\mathbb{k}G, \Delta_\bullet, \varepsilon_\bullet)$ is a coassociative counital coalgebra. Now,

$$\begin{aligned} \Delta_\bullet \circ \eta_\bullet(1_{\mathbb{k}}) &= \Delta_\bullet \left(\sum_{g \in G} g \right) = \sum_{g \in G} \Delta_\bullet(g) = \sum_{g \in G} \sum_{\substack{h, l \in G, \\ hl=g}} h \otimes l \\ &= \sum_{h, l \in G} h \otimes l = \left(\sum_{h \in G} h \right) \otimes \left(\sum_{l \in G} l \right) = \eta_\bullet(1_{\mathbb{k}}) \otimes \eta_\bullet(1_{\mathbb{k}}) \end{aligned}$$

and

$$\begin{aligned} \varepsilon_\bullet(m_\bullet(g \otimes h)) &= \varepsilon_\bullet(\delta_{g,h} g) = \delta_{e, \delta_{g,h} g} = \delta_{g,h} \delta_{e,g} = \delta_{e,g} \delta_{e,h} = \varepsilon_\bullet(g) \varepsilon_\bullet(h) \cong \varepsilon_\bullet(g) \otimes \varepsilon_\bullet(h) \\ &= (\varepsilon_\bullet \otimes \varepsilon_\bullet)(g \otimes h). \end{aligned}$$

Also,

$$\begin{aligned} m_{\bullet A \otimes A}((\Delta_\bullet \otimes \Delta_\bullet)(g \otimes h)) &= m_\bullet(\Delta_\bullet(g) \otimes \Delta_\bullet(h)) = \Delta_\bullet(g) \bullet \Delta_\bullet(h) \\ &= \sum_{\substack{k, l \in G \\ kl=g}} k \otimes l \bullet \sum_{\substack{k', l' \in G \\ k'l'=h}} k' \otimes l' = \sum_{\substack{k, l \in G \\ kl=g}} \sum_{\substack{k', l' \in G \\ k'l'=h}} (k \bullet k') \otimes (l \bullet l') \\ &= \sum_{\substack{k, l \in G \\ kl=g}} \sum_{\substack{k', l' \in G \\ k'l'=h}} \delta_{k,k'} \delta_{l,l'} (k \otimes l) = \delta_{g,h} \sum_{\substack{k, l \in G \\ kl=g}} k \otimes l \\ &= \Delta_\bullet(\delta_{g,h} g) = \Delta_\bullet(g \bullet h) \end{aligned}$$

and lastly,

$$\varepsilon_\bullet(\eta_\bullet(1_{\mathbb{k}})) = \varepsilon_\bullet \left(\sum_{g \in G} g \right) = \sum_{g \in G} \varepsilon_\bullet(g) = \sum_{g \in G} \delta_{e,g} = 1_{\mathbb{k}}.$$

That is, $(A, m_\bullet, \eta_\bullet, \Delta_\bullet, \varepsilon_\bullet)$ is a bialgebra. Moreover, S satisfies the antipode axioms:

$$\begin{aligned} m_\bullet(\text{id}_A \otimes S) \Delta_\bullet(g) &= m_\bullet(\text{id}_A \otimes S) \left(\sum_{\substack{h, l \in G \\ hl=g}} h \otimes l \right) = m_\bullet \left(\sum_{\substack{h, l \in G \\ hl=g}} h \otimes l^{-1} \right) \\ &= \sum_{\substack{h, l \in G \\ hl=g}} h \bullet l^{-1} = \sum_{\substack{h, l \in G \\ hl=g}} \delta_{h, l^{-1}} h = \delta_{e,g} \sum_{h \in G} h \\ &= \varepsilon_\bullet(g) \eta_\bullet(1_{\mathbb{k}}) = \eta_\bullet(\varepsilon_\bullet(g)) \end{aligned}$$

and similarly,

$$m_{\bullet}(S \otimes \text{id}_A)\Delta_{\bullet}(g) = \sum_{\substack{h,l \in G \\ hl=g}} h^{-1} \bullet l = \sum_{\substack{h,l \in G \\ hl=g}} \delta_{h^{-1},l} h^{-1} = \delta_{e,g} \sum_{h^{-1} \in G} h^{-1} = \eta_{\bullet}(\varepsilon_{\bullet}(g)).$$

Hence, $(A, m_{\bullet}, \eta_{\bullet}, \Delta_{\bullet}, \varepsilon_{\bullet})$ is a normalized Hopf algebra. Therefore, the pair $(A, m_{\bullet}, \eta_{\bullet}, \Delta_{\bullet}, \varepsilon_{\bullet})$, $(A, m_{\bullet}, \eta_{\bullet}, \Delta_{\bullet}, \varepsilon_{\bullet})$ is an F -Hopf algebra.

Remark 1.60. [54] Any finite-dimensional Hopf algebra admits a right-invariant integral $\int : A \rightarrow \mathbb{k}$ and a left integral element $\Lambda \in A$ characterized by

$$\left(\int \otimes \text{id} \right) \circ \Delta = \eta \int, \quad \int \Lambda = 1_{\mathbb{k}}, \quad h\Lambda = \Lambda\varepsilon(h),$$

for all $h \in A$.

The $\int$, Λ are each unique up to scale. Also, we are free to fix a relative normalization. The fact that their existence is always possible is part of the theory of Hopf algebra Fourier transform [7]. Here, Λ corresponds to a left-invariant integral on A^* .

The following subsection is the interesting case of the small quantum group $u_q(\mathfrak{sl}_2)$, that is a fundamental example of a finite-dimensional Hopf algebra [54, 58].

1.6.3 The Small Quantum Group $u_q(\mathfrak{sl}_2)$

Definition 1.61. Let q be a primitive n -th root of unity. The small quantum group $\mathcal{H} = u_q(\mathfrak{sl}_2)$ is defined as follows: As an algebra, it is generated by elements E , F , and an invertible element K , that is, every element of $\mathcal{H}$ is a finite sum

$$\sum_{i,j,k=0}^{n-1} c_{ijk} K^i F^j E^k, \quad c_{ijk} \in \mathbb{k}$$

with $1_{\mathcal{H}} = E^0 F^0 K^0$ and subject to the following relations respectively:

$$E^n = F^n = 0, \quad K^n = 1, \quad KE = q^{-1}EK, \quad KF = qFK, \quad [E, F] = \frac{K - K^{-1}}{q - q^{-1}}$$

where the multiplication $m_{\bullet} : \mathcal{H} \otimes \mathcal{H} \rightarrow \mathcal{H}$ is given by $m_{\bullet}(a \otimes b) = ab$, for $a, b \in \mathcal{H}$ and the unit is $\eta_{\bullet}(1_{\mathbb{k}}) = 1_{\mathcal{H}}$.

Definition 1.62. Let q be a primitive n -th root of unity and the algebra $\mathcal{H} = u_q(\mathfrak{sl}_2)$ as defined above. We define on $\mathcal{H}$ the comultiplication $\Delta_{\bullet}$, the counit $\varepsilon_{\bullet}$ and the antipode S by

$$\begin{aligned} \Delta_{\bullet}(K) &= K \otimes K, & \Delta_{\bullet}(E) &= E \otimes K + 1 \otimes E, & \Delta_{\bullet}(F) &= F \otimes 1 + K^{-1} \otimes F, \\ \varepsilon_{\bullet}(K) &= 1, & \varepsilon_{\bullet}(E) &= \varepsilon_{\bullet}(F) = 0, & S(K) &= K^{-1}, & S(E) &= -EK^{-1}, & S(F) &= -KF. \end{aligned}$$

Theorem 1.63. *Using the above notation, the structure $(\mathcal{H}, m_{\bullet}, \eta_{\bullet}, \Delta_{\bullet}, \varepsilon_{\bullet}, S)$ is a Hopf algebra.*

Proof. $(\mathcal{H}, m_{\bullet}, \eta_{\bullet})$ is an algebra. Now, we will prove that $(\mathcal{H}, \Delta_{\bullet}, \varepsilon_{\bullet})$ is a coalgebra. Since $\Delta_{\bullet}$ is defined on the generators E, F, K and extended, it is enough to verify the needed axioms on the generators. We have:

$$\begin{aligned}
 (\Delta_{\bullet} \otimes \text{id}) \circ \Delta_{\bullet}(K) &= (\Delta_{\bullet} \otimes \text{id})(K \otimes K) = \Delta_{\bullet}(K) \otimes K = K \otimes K \otimes K = K \otimes \Delta_{\bullet}(K) \\
 &= (\text{id} \otimes \Delta_{\bullet})(K \otimes K) = (\text{id} \otimes \Delta_{\bullet}) \circ \Delta_{\bullet}(K), \\
 (\Delta_{\bullet} \otimes \text{id}) \circ \Delta_{\bullet}(E) &= (\Delta_{\bullet} \otimes \text{id})(E \otimes K + 1 \otimes E) = \Delta_{\bullet}(E) \otimes K + \Delta_{\bullet}(1) \otimes E \\
 &= (E \otimes K + 1 \otimes E) \otimes K + (1 \otimes 1) \otimes E \\
 &= E \otimes K \otimes K + 1 \otimes (E \otimes K + 1 \otimes E) = E \otimes \Delta_{\bullet}(K) + 1 \otimes \Delta_{\bullet}(E) \\
 &= (\text{id} \otimes \Delta_{\bullet})(E \otimes K + 1 \otimes E) = (\text{id} \otimes \Delta_{\bullet}) \circ \Delta_{\bullet}(E), \\
 (\Delta_{\bullet} \otimes \text{id}) \circ \Delta_{\bullet}(F) &= (\Delta_{\bullet} \otimes \text{id})(F \otimes 1 + K^{-1} \otimes F) = \Delta_{\bullet}(F) \otimes 1 + \Delta_{\bullet}(K^{-1}) \otimes F \\
 &= (F \otimes 1 + K^{-1} \otimes F) \otimes 1 + (K^{-1} \otimes K^{-1}) \otimes F \\
 &= F \otimes 1 \otimes 1 + K^{-1} \otimes F \otimes 1 + K^{-1} \otimes K^{-1} \otimes F \\
 &= F \otimes 1 \otimes 1 + K^{-1} \otimes (F \otimes 1 + K^{-1} \otimes F) \\
 &= F \otimes \Delta_{\bullet}(1) + K^{-1} \otimes \Delta_{\bullet}(F) \\
 &= (\text{id} \otimes \Delta_{\bullet})(F \otimes 1 + K^{-1} \otimes F) \\
 &= (\text{id} \otimes \Delta_{\bullet}) \circ \Delta_{\bullet}(F).
 \end{aligned}$$

Hence Δ is coassociative. Also,

$$\begin{aligned}
 (\varepsilon_{\bullet} \otimes \text{id}) \circ \Delta_{\bullet}(K) &= (\varepsilon_{\bullet} \otimes \text{id})(K \otimes K) = \varepsilon_{\bullet}(K)K = K = K\varepsilon_{\bullet}(K) = (\text{id} \otimes \varepsilon_{\bullet}) \circ \Delta_{\bullet}(K), \\
 (\varepsilon_{\bullet} \otimes \text{id}) \circ \Delta_{\bullet}(E) &= (\varepsilon_{\bullet} \otimes \text{id})(E \otimes K + 1 \otimes E) = \varepsilon_{\bullet}(E)K + \varepsilon_{\bullet}(1)E \\
 &= E = E\varepsilon_{\bullet}(K) + 1\varepsilon_{\bullet}(E) = (\text{id} \otimes \varepsilon_{\bullet}) \circ \Delta_{\bullet}(E), \\
 (\varepsilon_{\bullet} \otimes \text{id}) \circ \Delta_{\bullet}(F) &= (\varepsilon_{\bullet} \otimes \text{id})(F \otimes 1 + K^{-1} \otimes F) = \varepsilon_{\bullet}(F)1 + \varepsilon_{\bullet}(K^{-1})F \\
 &= F = F\varepsilon_{\bullet}(1) + K^{-1}\varepsilon_{\bullet}(F) = (\text{id} \otimes \varepsilon_{\bullet}) \circ \Delta_{\bullet}(F).
 \end{aligned}$$

So, it is counital. Therefore $(\mathcal{H}, \Delta_{\bullet}, \varepsilon_{\bullet})$ is a coalgebra. Now, to show that $\mathcal{H}$ is a bialgebra, it remains to prove that $\Delta_{\bullet}$ and $\varepsilon_{\bullet}$ are algebra morphisms, i.e. all the relations are preserved and well defined under the action of $\Delta_{\bullet}, \varepsilon_{\bullet}$. Again, it suffices to work on generators:

$$\begin{aligned}
 \Delta_{\bullet}(\eta(1_{\mathbb{k}})) &= \Delta_{\bullet}(1_{\mathcal{H}}) = \Delta_{\bullet}(K^n) = \Delta_{\bullet}(K)^n = (K \otimes K)^n = K^n \otimes K^n = 1 \otimes 1 = \eta_{\bullet}(1_{\mathbb{k}}) \otimes \eta_{\bullet}(1_{\mathbb{k}}), \\
 \varepsilon_{\bullet}(\eta(1_{\mathbb{k}})) &= \varepsilon_{\bullet}(1_{\mathcal{H}}) = \varepsilon_{\bullet}(K^n) = \varepsilon_{\bullet}(K)^n = 1_{\mathbb{k}},
 \end{aligned}$$

$$\begin{aligned}
 \Delta_{\bullet}(KE) &= \Delta_{\bullet}(K)\Delta_{\bullet}(E) = (K \otimes K)(E \otimes K + 1 \otimes E) = KE \otimes K^2 + K \otimes KE \\
 &= q^{-1}EK \otimes K^2 + K \otimes q^{-1}EK = q^{-1}(E \otimes K + 1 \otimes E)(K \otimes K) \\
 &= q^{-1}\Delta_{\bullet}(E)\Delta_{\bullet}(K) = \Delta_{\bullet}(q^{-1}KE),
 \end{aligned}$$

$$\begin{aligned}
 \Delta_{\bullet}(KF) &= \Delta_{\bullet}(K)\Delta_{\bullet}(F) = (K \otimes K)(F \otimes 1 + K^{-1} \otimes F) = KF \otimes K + KK^{-1} \otimes KF \\
 &= qFK \otimes K + 1 \otimes qFK = q(F \otimes 1 + K^{-1} \otimes F)(K \otimes K) \\
 &= q\Delta_{\bullet}(F)\Delta_{\bullet}(K) = \Delta_{\bullet}(qFK).
 \end{aligned}$$

For the commutator, we compute

$$\begin{aligned}
 [\Delta_{\bullet}(E), \Delta_{\bullet}(F)] &= \Delta_{\bullet}(E)\Delta_{\bullet}(F) - \Delta_{\bullet}(F)\Delta_{\bullet}(E) \\
 &= (E \otimes K + 1 \otimes E)(F \otimes 1 + K^{-1} \otimes F) - (F \otimes 1 + K^{-1} \otimes F)(E \otimes K + 1 \otimes E) \\
 &= EF \otimes K + EK^{-1} \otimes KF + F \otimes E + K^{-1} \otimes EF \\
 &\quad - FE \otimes K - FK^{-1} \otimes FE - K^{-1}E \otimes FK - K^{-1} \otimes FE \\
 &= (EF - FE) \otimes K + (EK^{-1} \otimes KF - K^{-1}E \otimes FK) + K^{-1} \otimes (EF - FE) \\
 &= (EF - FE) \otimes K + K^{-1} \otimes (EF - FE) \\
 &= \frac{K - K^{-1}}{q - q^{-1}} \otimes K + K^{-1} \otimes \frac{K - K^{-1}}{q - q^{-1}} = \frac{K \otimes K - K^{-1} \otimes K^{-1}}{q - q^{-1}} \\
 &= \frac{\Delta_{\bullet}(K) - \Delta_{\bullet}(K^{-1})}{q - q^{-1}} = \Delta_{\bullet}([E, F]).
 \end{aligned}$$

It is immediate that $\varepsilon_{\bullet}$ also preserves the defining relations:

$$\begin{aligned}
 \varepsilon_{\bullet}(E^n) &= \varepsilon(F^n) = 0, & \varepsilon_{\bullet}(K^n) &= 1, \\
 \varepsilon_{\bullet}(KE) &= \varepsilon_{\bullet}(q^{-1}EK), & \varepsilon_{\bullet}(KF) &= \varepsilon_{\bullet}(qFK), & \varepsilon_{\bullet}([E, F]) &= \varepsilon_{\bullet}\left(\frac{K - K^{-1}}{q - q^{-1}}\right).
 \end{aligned}$$

Thus, $(\mathcal{H}, m_{\bullet}, \eta_{\bullet}, \Delta_{\bullet}, \varepsilon_{\bullet})$ is a bialgebra. Finally, we verify that S is an antipode. It suffices to verify that:

$$\begin{aligned}
 m_{\bullet}(S \otimes \text{id}) \circ \Delta_{\bullet}(K) &= m_{\bullet}(K^{-1} \otimes K) = 1 = \eta_{\bullet} \circ \varepsilon_{\bullet}(K) = 1 = m_{\bullet}(K \otimes K^{-1}) \\
 &= m_{\bullet}(\text{id} \otimes S)\Delta_{\bullet}(K), \\
 m_{\bullet}(S \otimes \text{id}) \circ \Delta_{\bullet}(E) &= m_{\bullet}(S(E) \otimes K + S(1) \otimes E) = (-EK^{-1})K + E = \eta_{\bullet} \circ \varepsilon_{\bullet}(E) \\
 &= EK^{-1} - EK^{-1} = m_{\bullet}(E \otimes K^{-1} + 1 \otimes (-EK^{-1})) = m_{\bullet}(\text{id} \otimes S)\Delta_{\bullet}(E), \\
 m_{\bullet}(S \otimes \text{id}) \circ \Delta_{\bullet}(F) &= m_{\bullet}(S(F) \otimes 1 + S(K^{-1}) \otimes F) = (-KF) \cdot 1 + K \cdot F = \eta_{\bullet} \circ \varepsilon_{\bullet}(F) \\
 &= F - F = m_{\bullet}(F \otimes 1 + K^{-1} \otimes (-KF)) = m_{\bullet}(\text{id} \otimes S)\Delta_{\bullet}(F).
 \end{aligned}$$

Therefore, S is an antipode. Hence, $(\mathcal{H}, m_{\bullet}, \eta_{\bullet}, \Delta_{\bullet}, \varepsilon_{\bullet}, S)$ is a Hopf algebra. $\square$

Definition 1.64. The right integral $\int$ on $\mathcal{H}$ and left integral Λ on $\mathcal{H}$ are defined by

$$\int K^i F^j E^k = \delta_{1,i} \delta_{n-1,j} \delta_{n-1,k}, \quad \Lambda = \Lambda_K F^{n-1} E^{n-1}, \quad \text{where} \quad \Lambda_K = \sum_{r=0}^{n-1} K^r,$$

so that $\int \Lambda = 1_{\mathbb{k}}$. They are often called integral and cointegral respectively [2].

Lemma 1.65. Using the above notation, in $u_q(\mathfrak{sl}_2)$, we have $S(\Lambda) = \Lambda$ [22, 55].

Definition 1.66. Let $\mathcal{H}$ be a Hopf algebra over $\mathbb{k}$. We use the Sweedler notation for $h \in \mathcal{H}$, to express h as an element of $\mathcal{H} \otimes \mathcal{H}$:

$$\Delta(h) = \sum_i h_{1,i} \otimes h_{2,i} =: h_{(1)} \otimes h_{(2)}.$$

Remark 1.67. The Sweedler notation is just a way to write h so that it looks like an element of $\mathcal{H} \otimes \mathcal{H}$, without having to actually choose a basis in $\mathcal{H}$.

We now construct a new Hopf algebra structure on $\mathcal{H}$, using the integral $\int$ and cointegral Λ defined above.

Theorem 1.68. Using the above notation, the associated $\tilde{\mathcal{H}} := (\mathcal{H}, m_{\bullet}, \eta_{\bullet}, \Delta_{\bullet}, \varepsilon_{\bullet}, S)$ is a Hopf algebra, with the associated product $\bullet := m_{\bullet}$ and coproduct $\Delta_{\bullet}$ given by

$$h_{\bullet} h' = h_{(1)} \int (S(h_{(2)}) h'), \quad \Delta_{\bullet}(h) = \Lambda_{(1)} \otimes (S(\Lambda_{(2)})) h \quad (6)$$

for all $h, h' \in u_q(\mathfrak{sl}_2)$, where we use Sweedler notation $\Delta_{\bullet}(h) = h_{(1)} \otimes h_{(2)}$ and $\Delta_{\bullet}(\Lambda) = \Lambda_{(1)} \otimes \Lambda_{(2)}$.

Below, we will see the construction of $\tilde{\mathcal{H}}$ for the $n = 2$ case.

Observation 1.69. For $n = 2$, the second root of unity is $q = -1$. There, it is $q - q^{-1} = 0$, and we do the convention to omit the $q - q^{-1}$ factors in the relations and in $\mathcal{R}$. This amounts to a different normalization of E, F (by setting $\tilde{E} = (q - q^{-1})E$, etc.) [4]. So, all the relevant products become:

$$\begin{aligned} K^2 &= 1, \quad E^2 = F^2 = 0, \quad [E, F] = K - K^{-1} = 0, \\ \{E, K\} &= EK + KE = EK + q^{-1}EK = 0, \quad \{F, K\} = FK + KF = FK + qFK = 0, \\ \Delta(K) &= K \otimes K, \quad \Delta(E) = E \otimes K + 1 \otimes E, \quad \Delta(F) = F \otimes 1 + K \otimes F, \\ \varepsilon(K) &= 1, \quad \varepsilon(E) = \varepsilon(F) = 0, \\ S(K) &= K^{-1} = K, \quad S(E) = -EK = (-1)^{-1}EK = KE, \quad S(F) = -KF, \\ \Lambda &= \Lambda_K EF, \quad \text{where} \quad \Lambda_K = 1 + K. \end{aligned} \quad (7)$$

This is the initial Hopf algebra; we now use it to compute the induced structure. From these we work out the values of $(\text{id} \otimes S)\Delta_\bullet$ on our monomial basis.

In particular, it is $(\text{id} \otimes S)\Delta_\bullet(\Lambda) = \Lambda_{(1)} \otimes S(\Lambda_{(2)}) = \sum_i \Lambda_{(1)i} \otimes S(\Lambda_{(2)i})$. So, it is

$$\begin{aligned}\Delta_\bullet(\Lambda) &= \Delta_\bullet(1 + K)\Delta_\bullet(E)\Delta_\bullet(F) = (1 \otimes 1 + K \otimes K)(E \otimes K + 1 \otimes E)(F \otimes 1 + K \otimes F) \\ &= (1 \otimes 1 + K \otimes K)(EF \otimes K + EK \otimes KF + F \otimes E + K \otimes EF) \\ &= EF \otimes K + EK \otimes KF + F \otimes E + K \otimes EF \\ &\quad + KEF \otimes 1 - E \otimes F + KF \otimes KE + 1 \otimes KEF.\end{aligned}$$

Therefore, using Equation (7), we compute

$$\begin{aligned}\Lambda_{(1)} \otimes S(\Lambda_{(2)}) &= EF \otimes S(K) + EK \otimes S(KF) + F \otimes S(E) + K \otimes S(EF) \\ &\quad + KEF \otimes S(1) - E \otimes S(F) + KF \otimes S(KE) + 1 \otimes S(KEF) \\ &= EF \otimes K + EK \otimes F + F \otimes KE + K \otimes EF \\ &\quad + KEF \otimes 1 + E \otimes KF - KF \otimes E + 1 \otimes KEF \\ &= F \otimes KE - KE \otimes F + EF \otimes K + K \otimes EF \\ &\quad + E \otimes KF - KF \otimes E + KEF \otimes 1 + 1 \otimes KEF.\end{aligned}$$

Now, we let $x = (1 + K)E$, $y = (1 + K)F$, $t = (K - 1)EF$, and find $\Delta_\bullet(t)$, $\Delta_\bullet(x)$, $\Delta_\bullet(y)$ for the associated Hopf algebra. Using Equation (7), we obtain

$$\begin{aligned}KEt &= Ft = K Ft = Et = E Ft = KE Ft = 0, \\ Kt &= K(K - 1)EF = (1 - K)EF = -(K - 1)EF = -t.\end{aligned}$$

Hence,

$$\begin{aligned}\Delta_\bullet(t) &= \Lambda_{(1)} \otimes (S(\Lambda_{(2)}))t \\ &= F \otimes KEt - KE \otimes Ft + EF \otimes Kt + K \otimes E Ft \\ &\quad + E \otimes K Ft - KF \otimes Et + KEF \otimes t + 1 \otimes KE Ft \\ &= -EF \otimes t + KEF \otimes t = (K - 1)EF \otimes t = t \otimes t.\end{aligned}$$

We omit the similar straightforward computations for the rest. We have:

$$\Delta_\bullet(t) = \Lambda_{(1)} \otimes (S(\Lambda_{(2)}))t = t \otimes t, \quad \Delta_\bullet(x) = x \otimes t + \Lambda \otimes x, \quad \Delta_\bullet(y) = y \otimes t + \Lambda \otimes y.$$

This computation shows that in the $q = -1$ case, the induced Hopf algebra structure simplifies drastically compared to the general case. Also,

$$\begin{aligned}x \bullet x &= y \bullet y = 0, \quad t \bullet t = \Lambda, \\ x \bullet y &= -y \bullet x = 1 + K, \quad x \bullet t = -t \bullet x = (K - 1)E, \quad y \bullet t = -t \bullet y = (1 - K)F,\end{aligned}$$

that is, x, y are nilpotent and x, y, t are anticommutative.

Definition 1.70. *The quasitriangular structure on $\mathcal{H}$ is defined as*

$$\mathcal{R} = \frac{1}{n} \sum_{r,a,b=0}^{n-1} \frac{(-1)^r (q - q^{-1})^r q^{-ab}}{[r]_{q^{-1}}!} F^r K^a \otimes E^r K^b, \quad \text{where } [r]_q = \frac{1 - q^r}{1 - q}.$$

Observation 1.71. *The associated Hopf algebra $\tilde{\mathcal{H}} := (\mathcal{H}, \bullet, \eta_{\bullet}, \Delta_{\bullet}, \varepsilon_{\bullet}, S)$ is isomorphic to $c_q[SL_2]^{op}$ for the appropriate reduced quantum group dual to $u_q(\mathfrak{sl}_2)$, where $c_q[SL_2]$ is the coordinate algebra of the quantum group $SL_q(n)$ and $\cdot_{op}$ is the reverse multiplication [9]. For the proof, one uses that*

$$\begin{aligned} \mathcal{R} &= \frac{1}{2} \sum_{r,a,b=0}^1 \frac{(-1)^r (-1)^{-ab}}{[r]_{(-1)^{-1}}!} (F^r \otimes E^r) (K^a \otimes K^b) \\ &= \frac{1}{2} \sum_{a,b=0}^1 \left[\frac{(-1)^0}{[0]_{-1}!} (F^0 \otimes E^0) + \frac{-1}{[1]_{-1}!} (F \otimes E) \right] (-1)^{-ab} (K^a \otimes K^b) \\ &= (1 \otimes 1 - F \otimes E) \frac{1}{2} \sum_{a,b=0}^1 (-1)^{-ab} (K^a \otimes K^b) \\ &= (1 \otimes 1 - F \otimes E) \mathcal{R}_K, \quad \text{where } \mathcal{R}_K = \frac{1}{2} (1 \otimes 1 + 1 \otimes K + K \otimes 1 - K \otimes K). \end{aligned}$$

We will not expand further, as it is beyond of the scope of this thesis. One can read more in [61].

Chapter 2

An Overview of Quantum Computing

In the following section we will give the basics of quantum computing theory, establishing a dictionary.

2.1 The Qubit

In a classical computer, the *bit* is the fundamental unit of information. A bit has a space of states that consists simply of two points, 0 and 1, representing *off* and *on*. The state space of a bit in probability theory is really a line segment, since the bit may be *on* with some probability between zero and one. But in mechanics, a classical bit can only be either *on* or *off*.



In quantum theory there is no distinction between probability theory and mechanics. A qubit (quantum bit) differs from a classical bit, since its state is represented by any superposition of two independent states $|0\rangle$ and $|1\rangle$ that correspond to the classical 0 and 1. That is, at any time the state of the quantum bit can be anywhere in the line segment between *on* and *off*.

We continue by constructing the qubit space.

Definition 2.1. The computational basis $\{|0\rangle, |1\rangle\}$ of the 1-qubit space is the standard basis of $\mathbb{C}^2$, where $|0\rangle = (1, 0)$ and $|1\rangle = (0, 1)$.

Definition 2.2. A qubit (*quantum bit*) is the fundamental unit of quantum information. The (pure) state $|\psi\rangle$ of a qubit is represented mathematically as the linear combination of the computational basis states:

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle, \quad |\alpha|^2 + |\beta|^2 = 1, \quad \alpha, \beta \in \mathbb{C}. \quad (8)$$

Thus, $|\psi\rangle$ is an element of the Hilbert space $\mathcal{H} = \mathbb{C}^2$, of unit norm. We need the condition $|\alpha|^2 + |\beta|^2 = 1$, because of the probabilistic interpretation of α and β as coefficients of $|\psi\rangle$; it must be $\langle\psi|\psi\rangle = 1$. The result of a measurement on the computational basis finds the qubit in the state $|0\rangle$ or in the state $|1\rangle$ with probability $|\alpha|^2$, $|\beta|^2$ respectively, because of the Born rule. This interpretation does not distinguish between the states $|\psi\rangle$ and $e^{i\gamma}|\psi\rangle$, for any phase γ .

Remark 2.3. *We need a Hilbert space structure and not just the inner product space. This is because completeness ensures that Cauchy sequences have limits. If we drop the completeness assumption, a very spectacular consequence is the inability to describe state evolution by the Schrödinger equation. Completeness ensures that the solutions produced by Schrödinger equation are valid quantum states, i.e. they do not lie outside the space. Also, quantum states are often described by infinite sums. Incomplete spaces may fail to contain the limits of convergent infinite sums.*

We use the Dirac ket notation to write the qubit $|\psi\rangle$, as customary in quantum mechanics to represent a quantum mechanical state.

Now, our goal is to pass from $\mathbb{C}^2$ to the complex projective space $\mathbb{P}^1(\mathbb{C})$. We think of $|\psi\rangle$ as a complex line in the Hilbert space $\mathcal{H} = \mathbb{C}^2$, pursuing a geometrical interpretation. Therefore, in Equation (8), we choose a representative of such line. We look at the intersections of the line with the sphere (the condition $|\alpha|^2 + |\beta|^2 = 1$) and then identify all points that differ just by a phase factor $e^{i\gamma}$.

Indeed, let it be $|\psi\rangle$, with $\langle\psi|\psi\rangle = 1$, so $|\psi\rangle$ lives on $\mathbb{S}^2$, and $|\psi'\rangle = e^{i\gamma}|\psi\rangle$. It is

$$\langle\psi'|\psi'\rangle = (e^{i\gamma}\langle\psi|)(e^{i\gamma}|\psi\rangle) = \langle\psi|\psi\rangle = 1.$$

Hence, $|\psi\rangle$ and $|\psi'\rangle$ have the same length. Let $|\psi\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$ as vector of $\mathbb{C}^2$. Then,

$$|\psi'\rangle = e^{i\gamma} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} e^{i\gamma}\alpha \\ e^{i\gamma}\beta \end{pmatrix}.$$

Thus, they lie on the same ray of the Hilbert space $\mathbb{C}^2$, because $|\psi'\rangle$ is a scalar multiple of $|\psi\rangle$. That means, every coefficient of $|\psi\rangle$ (i.e. α , β) is multiplied by the same number. Hence, in projective space $\mathbb{P}^1(\mathbb{C})$, we have that $|\psi\rangle, |\psi'\rangle$ are the same. Essentially, $|\psi\rangle$ and $|\psi'\rangle$ are different mathematical objects with the same length and direction.

Definition 2.4. *A qubit $|\psi\rangle$ is represented parametrically as*

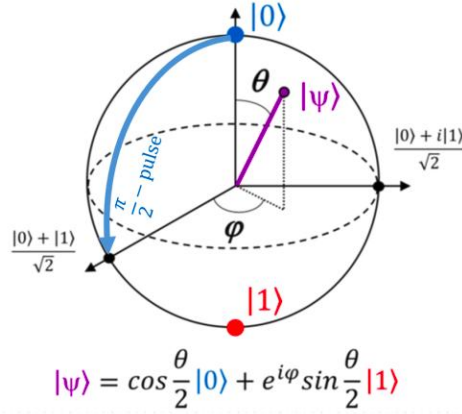
$$|\psi\rangle = e^{i\gamma} \left[\cos\left(\frac{\theta}{2}\right) |0\rangle + e^{i\phi} \sin\left(\frac{\theta}{2}\right) |1\rangle \right], \quad (9)$$

where γ is the global phase and ϕ the relative phase of the state.

Remark 2.5. Here, $|\psi\rangle$ is seen as object of $\mathbb{S}^3$, and the global phase γ acts on $|\psi\rangle$, as $\{e^{i\gamma} : \gamma \in \mathbb{R}\} \cong \text{U}(1)$. We see the action $e^{i\gamma}$ like the state "moves" on one $\mathbb{S}^1$ circle around the 3-dimensional $\mathbb{S}^3$ sphere. Then, we take the quotient space $\mathbb{S}^3/\mathbb{S}^1 \cong \mathbb{S}^2 \cong \mathbb{P}^1(\mathbb{C})$, that is the Riemann sphere.

In quantum mechanics and computing, we use the following definition for $\mathbb{S}^2$ in the sense we discussed before.

Definition 2.6. The Bloch sphere is the geometrical representation of the pure state space of a qubit, as points on the unit sphere $\mathbb{S}^2 \cong \mathbb{P}^1(\mathbb{C})$ as shown below.



Indeed, the basis vectors $|0\rangle$ and $|1\rangle$ are the north and south poles of the Bloch sphere respectively. Moreover, the numbers θ and ϕ define a point on the Bloch sphere, and a qubit is viewed as an element of the sphere.

Also, from the previous calculations we observe that the global phase γ does not change the position of the qubit on the Bloch sphere, since it is $\mathbb{P}^1(\mathbb{C})$. Hence, we typically set the phase $\gamma = 0$, because it has no observable effects on $\mathbb{P}^1(\mathbb{C})$.

The Bloch sphere can be equipped with a natural metric, the *Fubini-Study* metric, coming from its identification with the $\mathbb{P}^1(\mathbb{C})$. We will discuss this further in Chapter 3, when we introduce quantum information geometry.

When we measure the qubit, its quantum superposition $|\psi\rangle$ collapses, forcing it to "choose" a definite classical state -either $|0\rangle$ or $|1\rangle$ - based on probabilistic amplitudes. Therefore, measuring affects the state itself, and hence, the original superposed information. This is a problem in quantum computing, in comparison with classical computing, because we cannot measure a state with arbitrary accurate precision.

As a consequence of this quantum mechanical phenomenon, the quantum algorithms (see Remark 2.22) need to be suitably adapted and are not simple generalizations of the classical ones. Moreover, they are more prone to errors. Also, the correction of errors leads to fault tolerant algorithms. This will concern us later, in Subsection 2.3.3 [17, 41, 40].

We introduced the single or 1-qubit above. Besides that, we can have a 2-qubit system, which states are elements of the Hilbert space $\mathbb{C}^2 \otimes \mathbb{C}^2$.

Definition 2.7. *The computational basis of the 2-qubit space,*

$$|00\rangle, \quad |01\rangle, \quad |10\rangle, \quad |11\rangle \quad (10)$$

is the standard basis of $\mathbb{C}^2 \otimes \mathbb{C}^2$, where we use the notation $|ij\rangle := |i\rangle \otimes |j\rangle$, with $i, j \in \{0, 1\}$.

A 2-qubit state is represented by

$$|\psi\rangle = \alpha_0 |00\rangle + \alpha_1 |01\rangle + \alpha_2 |10\rangle + \alpha_3 |11\rangle, \quad \sum_{i=0}^3 |\alpha_i|^2 = 1, \quad \alpha_i \in \mathbb{C}.$$

This generalizes naturally to an n -qubit system in a similar way; states are expressed as linear combination - superposition of the computational basis states.

Definition 2.8. *The computational basis of the n -qubit space,*

$$|00 \dots 00\rangle, \quad |00 \dots 01\rangle, \quad |00 \dots 10\rangle, \quad \dots \quad |11 \dots 11\rangle$$

is the standard basis of $(\mathbb{C}^2)^{\otimes n}$. A state of an n -qubit system is an element of the Hilbert space $\mathcal{H} = \mathbb{C}^{2^n}$:

$$|\psi\rangle = \alpha_0 |00 \dots 0\rangle + \alpha_1 |00 \dots 1\rangle + \dots + \alpha_{2^n-1} |11 \dots 1\rangle, \quad \sum_{i=0}^{2^n-1} |\alpha_i|^2 = 1, \quad \alpha_i \in \mathbb{C}.$$

The final condition reflects the previous property of measurements in quantum mechanics: after a measurement, we find the state $|\psi\rangle$ in the i -th state of the computational basis with probability $|\alpha_i|^2$. We should also not forget the phase condition γ .

Moreover, we notice that on the computational basis the 2^n states are ordered from $00 \dots 0$ to $11 \dots 1$, making use of the binary representation of the integer i , with $i = 0, 1, \dots, 2^n - 1$ enumerating them.

Definition 2.9. *A vector $\psi \in \mathbb{C}^2 \otimes \dots \otimes \mathbb{C}^2$ describing the state of n -qubit is called entangled if we cannot write it as $|\psi\rangle = u_1 \otimes \dots \otimes u_n$, $u_i \in \mathbb{C}^2$.*

Definition 2.10. *A mixed state is a statistical mixture defined by an ensemble of pure states $\{|\psi_i\rangle\}_{i=1}^m$, that can occur with a probability p_i respectively, with $0 \leq p_i \leq 1$ and $\sum_{i=1}^m p_i = 1$.*

Definition 2.11. *A pure state is a particular example of a mixed state with $p_1 = 1$ and $p_2 = \dots = p_m = 0$.*

2.2 The Density Operator

Up to this point the states we considered were pure states. Now we will work with pure and mixed states. We will introduce the concept of the density operator.

Definition 2.12. Suppose a pure or mixed state $\{|\psi_i\rangle\}_{i=1}^m$, with $0 \leq p_i \leq 1$, $\sum_{i=1}^m p_i = 1$. We define the density operator for the system by the equation

$$\rho = \sum_i p_i |\psi_i\rangle \langle \psi_i|. \quad (11)$$

Remark 2.13. $\langle \psi_i| \in \mathcal{H}^*$ is the dual vector of $|\psi_i\rangle$, that is, $|\psi_i\rangle^\dagger = \langle \psi_i|$.

The density operator is a tool for an alternate formulation, mathematically equivalent to the state vector approach. The difference is that it provides a much more convenient language for thinking about some commonly encountered scenarios in quantum mechanics. It is helpful that someone can easily take the statistical information from density operator or matrix, when the same time this can be complicated for mixed states.

Once an orthonormal basis is fixed, someone can express ρ as a matrix and speak of *density matrix*. We will use ρ_ψ for both density matrix and operator, making an abuse of notation.

Proposition 2.14. The density operator ρ is a self-adjoint, (semi)positive definite operator with unit trace.

Proof. It is

$$\rho^\dagger = \left(\sum_i p_i |\psi_i\rangle \langle \psi_i| \right)^\dagger = \sum_i p_i (|\psi_i\rangle \langle \psi_i|)^\dagger = \sum_i p_i (\langle \psi_i|)^\dagger (|\psi_i\rangle)^\dagger = \sum_i p_i |\psi_i\rangle \langle \psi_i| = \rho.$$

Hence, the density operator is self-adjoint. Also, suppose an arbitrary vector $|\phi\rangle$ in state space. Then,

$$\langle \phi | \rho | \phi \rangle = \langle \phi | \left(\sum_i p_i |\psi_i\rangle \langle \psi_i| \right) | \phi \rangle = \sum_i p_i \langle \phi | \psi_i \rangle \langle \psi_i | \phi \rangle = \sum_i p_i |\langle \phi | \psi_i \rangle|^2 \geq 0.$$

For the unit trace,

$$\text{Tr}(\rho) = \sum_i p_i \text{Tr}(|\psi_i\rangle \langle \psi_i|) = \sum_i p_i = 1,$$

using that $\text{Tr}(|\psi\rangle \langle \psi|) = \langle \psi | \psi \rangle = 1$. □

Theorem 2.15. *The density operator ρ is idempotent (hence it is a projection operator) if and only if the state is pure.*

Proof. Let $|\psi\rangle$ be a pure state. Thus, $\langle\psi|\psi\rangle = 1$. We have

$$\rho^2 = (|\psi\rangle\langle\psi|)^2 = |\psi\rangle\langle\psi|\psi\rangle\langle\psi| = |\psi\rangle\langle\psi| = \rho,$$

hence, ρ is idempotent. We consider a qubit with density operator

$$\rho = \frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|1\rangle\langle 1|.$$

The qubit is mixed since it cannot be represented by a point on the surface of the Bloch sphere, but it lives inside.

It is

$$\begin{aligned}\rho^2 &= \left(\frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|1\rangle\langle 1|\right)^2 \\ &= \frac{1}{4}|0\rangle\langle 0||0\rangle\langle 0| + \frac{1}{4}|0\rangle\langle 0||1\rangle\langle 1| + \frac{1}{4}|1\rangle\langle 1||0\rangle\langle 0| + \frac{1}{4}|1\rangle\langle 1||1\rangle\langle 1| \\ &= \frac{1}{4}|0\rangle\langle 0| + \frac{1}{4}|1\rangle\langle 1| = \frac{1}{2}\rho \\ &\neq \rho\end{aligned}$$

and therefore, ρ is not idempotent. □

Once an orthonormal basis is fixed, someone can express ρ as a matrix and speak of *density matrix*. We will use ρ_ψ for both density matrix and operator, making an abuse of notation.

Example 2.16. *Suppose a 1-qubit system in the pure state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$. We have in the computational basis:*

$$\rho_\psi = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \begin{pmatrix} \bar{\alpha} & \bar{\beta} \end{pmatrix} = \begin{pmatrix} |\alpha|^2 & \alpha\bar{\beta} \\ \bar{\alpha}\beta & |\beta|^2 \end{pmatrix}.$$

For the mixed state defined by the set $\{|0\rangle, |1\rangle\}$ with $p_0 = |\alpha|^2$, $p_1 = |\beta|^2$ we have:

$$\rho = p_0|0\rangle\langle 0| + p_1|1\rangle\langle 1| = \begin{pmatrix} |\alpha|^2 & 0 \\ 0 & |\beta|^2 \end{pmatrix}.$$

The diagonal elements of the density matrix are real numbers that sum to one. They are often called *populations* of the two states $|0\rangle$ and $|1\rangle$. The off-diagonal elements are complex conjugates of each other and can be called *coherences*. They are restricted by the requirement that ρ is a positive semi-definite operator.

2.3 Logic Gates

The circuits of classical computers consist of wires and logic gates. The wires carry information around the circuit. The logic gates manipulate the information, while converting it from one form to another.

Definition 2.17. *The classical single bit logic gates are Boolean functions of the form $f : \mathbb{B} \rightarrow \mathbb{B}$, where $\mathbb{B} = \{0, 1\}$.*

Generalizing this, the classical multi-bit logic gates with n inputs and m outputs are of the form $f : \mathbb{B}^n \rightarrow \mathbb{B}^m$.

Example 2.18. *The following are all the classical single bit logic gates:*

Identity	NOT	Reset 0	Reset 1
$0 \rightarrow 0$	$0 \rightarrow 1$	$0 \rightarrow 0$	$0 \rightarrow 1$
$1 \rightarrow 1$	$1 \rightarrow 0$	$1 \rightarrow 0$	$1 \rightarrow 1$

We will represent the gates as matrices acting on the basis vectors $|0\rangle, |1\rangle$. We recall that the computational basis is identified with the standard basis of $\mathbb{C}^2$:

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

It is easy to make the matrices for cases with one input and one output, but in the next sections things will be much more complicated. So,

$$M_{\text{id}} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad M_{\text{NOT}} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad M_{R_0} = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} \quad M_{R_1} = \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}.$$

We notice that id and NOT are permutations. These two are the only invertible classical 1-bit gates. On the other hand, R_0 and R_1 are irreversible classical single-bit gates, having zero determinant, so their matrices are not unitary, and of course they are not permutations. Gates like the above, that erase information, are allowed in classical computing.

Remark 2.19. *In Section 2.3.1 we will see that reset gates, like R_0 and R_1 from Example 2.18, and other irreversible gates cannot be quantum logic gates. Quantum gates are always represented by unitary matrices. Among the single bit gates, the only non-trivial reversible gate is the NOT gate, that interchanges the 0 and 1 states.*

We mention in the following example some cases of logic gates with two bit inputs and single output.

Example 2.20.

AND	OR	NAND
0 0 → 0	0 0 → 0	0 0 → 1
0 1 → 0	0 1 → 1	0 1 → 1
1 0 → 0	1 0 → 1	1 0 → 1
1 1 → 1	1 1 → 1	1 1 → 0

We note that AND gives 1 only if all input digits are 1, OR gives 1 if at least one digit is 1. NAND (i.e. NOT – AND) gives 0 only if all digits are 1, that is, it flips AND. Their matrix representations are the following:

$$M_{\text{AND}} = \begin{pmatrix} 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad M_{\text{OR}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 \end{pmatrix} \quad M_{\text{NAND}} = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix}.$$

Observation 2.21. Using the NAND gate, we can construct the AND, OR and NOT gates as follows, where (x, y) and z are input bits, respectively.

$$\begin{aligned} \text{AND}(x, y) &= (x \text{ NAND } y) \text{ NAND } (x \text{ NAND } y), \\ \text{OR}(x, y) &= (x \text{ NAND } x) \text{ NAND } (y \text{ NAND } y), \\ \text{NOT}(z) &= z \text{ NAND } z \end{aligned}$$

We can generalize the above gates to construct logic gates with more than two inputs.

Quantum logic gates perform operations on states that are linear combinations of the computational basis elements, as in Definition 2.1 for 1-qubits and in Definition 2.8 for n -qubits. They are represented by unitary matrices U ($U^\dagger U = I$), in order to preserve the probabilities of the states.

Remark 2.22. A quantum algorithm is a step-by-step procedure, where each of the steps can be performed on a quantum computer. It consists of quantum logic gates, i.e. unitary matrices, each of which acts on some finite number of qubits. Quantum algorithms are most commonly described by quantum circuits (see Example 4.1). A quantum circuit is a model for quantum computation, where the steps to solve the problem are quantum gates performed on one or more qubits.

2.3.1 Pauli operators

In this subsection we will discuss Pauli operators. We see them as example of logic gates for single qubits. Pauli operators are relevant elements of ZX-calculus, the graphical language we use to represent and simplify quantum circuits, which we further deal in detail in Chapter 4.

As we said before, the state space of a qubit is $\mathcal{H} := \mathbb{C}^2$ and the vectors $|0\rangle, |1\rangle$ constitute the computational basis.

Definition 2.23. The Z -basis of $\mathcal{H}$ is the computational basis $\{|0\rangle, |1\rangle\}$. The X -basis of $\mathcal{H}$, also called Hadamard basis is

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle), \quad |-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle).$$

On the Bloch sphere Z and X bases can be represented as follows, where the elements of the Z -basis are represented by the green dots and those of the X -basis are represented by the red dots.

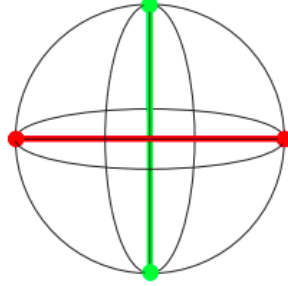


Figure 1: Z and X axes on the Bloch sphere

Z and X bases correspond to the possible outcomes upon measuring the spin of the electron along the Z and X axes respectively. Our interest in these particular spins stems from the fact that they are the simplest example of complementary observables.

Now, we take advantage of the *Pauli spin matrices* X, Y, Z and the *Hadamard operator* H :

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}. \quad (12)$$

Proposition 2.24. Z and X bases consist of the eigenvectors of the Pauli spin matrices Z and X respectively.

Proof. We calculate the eigenvectors solving for $u = (\alpha \ \beta)^t$:

$$Zu = \lambda u \Rightarrow \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \lambda \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \Rightarrow \begin{pmatrix} \alpha \\ -\beta \end{pmatrix} = \begin{pmatrix} \lambda\alpha \\ \lambda\beta \end{pmatrix}.$$

For $\lambda = 1$, it is $u_1 = (1 \ 0)^t = |0\rangle$. For $\lambda = -1$, it is $u_2 = (0 \ 1)^t = |1\rangle$. Thus, the normalized eigenvectors of Z Pauli spin is exactly the Z basis. Similarly,

$$Xu = \lambda u \Rightarrow \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \lambda \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \Rightarrow \begin{pmatrix} \beta \\ \alpha \end{pmatrix} = \begin{pmatrix} \lambda\alpha \\ \lambda\beta \end{pmatrix}.$$

For $\lambda = 1$, it is $u_1 = (1 \ 1)^t$, with unit eigenvector $\frac{1}{\sqrt{2}}(1 \ 1)^t = |+\rangle$. For $\lambda = -1$, $u_2 = (1 \ -1)^t$ and its unit eigenvector is $\frac{1}{\sqrt{2}}(1 \ -1)^t = |-\rangle$ which is the X basis. $\square$

We observe that the Hadamard operator transforms the computational basis $|0\rangle, |1\rangle$ to Hamadard basis, $|+\rangle, |-\rangle$. These operators are graphically given in Figure 2. All of the above X, Y, Z, H are examples of quantum logic gates for a 1-qubit system.

$$\begin{aligned} \text{Pauli-X} & \text{---} \boxed{X} \text{---} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \\ \text{Pauli-Y} & \text{---} \boxed{Y} \text{---} \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \\ \text{Pauli-Z} & \text{---} \boxed{Z} \text{---} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \end{aligned}$$

Figure 2: Quantum logic gates for 1-qubit

2.3.2 CNOT gate and no-cloning theorem

We now introduce the CNOT (*controlled-NOT*) logic gate, as an example of a quantum logic gate for a 2-qubit system.

Definition 2.25. *The quantum logic gate $\text{CNOT} : \mathbb{B}^2 \rightarrow \mathbb{B}^2$ is defined as*

$$\text{CNOT}(|x\rangle \otimes |y\rangle) = |x, x \oplus y\rangle,$$

where $x \oplus y := (x + y) \bmod 2$.

On the CNOT gate, we consider the first value, $|x\rangle$, as *control* and the second value, $|y\rangle$, as *target* value. This logic gate is called *controlled not* because $|x\rangle$ controls the $|y\rangle$ value, flipping it regarding $|x\rangle$:

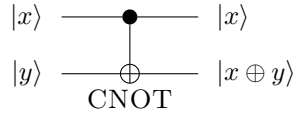
$$\begin{aligned} \text{CNOT}(|0\rangle \otimes |y\rangle) &= |0, 0 \oplus y\rangle = |0, y\rangle, \\ \text{CNOT}(|1\rangle \otimes |y\rangle) &= |1, 1 \oplus y\rangle = |1, \neg y\rangle. \end{aligned}$$

Therefore, CNOT acts as a NOT gate for $|\psi\rangle$ if the control state is *on*. It also leaves the control qubit $|x\rangle$ unchanged, since $|x\rangle \mapsto |x\rangle$.

The following matrix represents the unitary transformation expressing the CNOT in the computational basis (see Definition 2.8):

$$U_{\text{CNOT}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \quad \begin{array}{l} |00\rangle \mapsto |00\rangle \\ |01\rangle \mapsto |01\rangle \\ |10\rangle \mapsto |11\rangle \\ |11\rangle \mapsto |10\rangle \end{array}$$

Below, we give the graphical representation of CNOT gate.



The first line of the diagram, $\bullet$ says that the control digit $|x\rangle$ "passes" through the identity node $\bullet$ and it remains the same. The second line of the diagram, $\oplus$, represents that the target $|y\rangle$ flips with respect to $|x\rangle$, that is, $|x \oplus y\rangle$. The vertical line segment that connects the horizontal lines, determines which states take part in the operation. We note this, because later, we will have circuits with n qubit states, $n \geq 2$ (see Example 4.1).

Despite that the quantum gates described above are specific, the following theorem is a beautiful theoretical result, which says that $\{U(2), \text{CNOT}\}$ is a universal gate set.

Theorem 2.26. *The group $U(2^n)$ is generated by the set consisting of all single-qubit operators $U(2)$ together with the CNOT gate.*

The proof was an extended and detailed work by nine authors in [6]. Any n -qubit gate is described by a $U \in U(2^n)$. They first prove the case for $U(4)$, so every 2-qubit logic gate is a composition of 1-qubit gates and CNOT. By linear algebra, every $U \in U(2^n)$ can be written as composition of two-level unitary matrices $\{\tilde{U}_i\}_i$. A two-level unitary matrix is a unitary matrix which acts non-trivially only on two vector components. Using CNOTs, they bring close the non-trivial elements of $\tilde{U}_i$, until they form a 2×2 non-trivial submatrix, and this is the only non-trivial submatrix of the matrix. The 2×2 non-trivial submatrix can be constructed by $\{U(2), \text{CNOT}\}$. In the continue, they undo the CNOTs, to restore the initial matrix $\tilde{U}_i$. So, all the matrices in $\{\tilde{U}_i\}_i$ are constructed by compositions of 1-qubit gates, CNOTs and trivial gates. Hence, the initial matrix U is constructed by $\{U(2), \text{CNOT}\}$. Therefore, any unitary operator $U \in U(2^n)$ can be expressed as compositions of $U(2)$ operators, and the 2-qubit gate CNOT.

Remark 2.27. *The physical meaning of the theorem is that the single qubit gates, and CNOT is all we need to construct all the quantum logic gates. Hence, CNOT is fundamental to quantum computing. We mention here that the 1-qubit gates are the $U(2)$ operations, that is, all the possible rotations on the Bloch sphere (plus a global phase). So, Theorem 2.26 is a fundamental result, since from $\{U(2), \text{CNOT}\}$ one can construct the needed operators and build any quantum algorithm.*

Remark 2.28. *This is analogous to the fact that with classical NAND (NOT-AND) gates, we can realize all logic functions: AND, OR, NOT and thus, any classical algorithm (see Observation 2.21).*

It is not clear still how -and if- a quantum computer can solve any classical computability question. There are two main reasons that essentially cause obstruction. The first reason is

that the classical NAND gate is non-reversible, while quantum gates are necessarily unitary and hence invertible. The second problem is that we have the so-called *no-cloning theorem* (2.29). This theorem forbids the existence of a quantum gate that allows to copy a generic state of a qubit.

Theorem 2.29 (No-cloning theorem). *There is no single unitary operator U on $\mathcal{H} \otimes \mathcal{H}$ such that for all pure states $|\phi\rangle$ and $|B\rangle$ in $\mathcal{H}$, it is:*

$$U(|\phi\rangle \otimes |B\rangle) = e^{i\alpha(\phi,B)} |\phi\rangle \otimes |\phi\rangle,$$

for some real number α depending on ϕ and B .

Proof. Suppose that such U exists, and assume for simplicity that $\alpha = 0$. We recall that a pure state is a quantum state $|\psi\rangle$ that is described by a single vector in $\mathcal{H}$, and they have the capacity $\langle\psi|\psi\rangle = 1$ (see Definition 2.11). Given the states $|\phi\rangle, |\psi\rangle, |B\rangle \in \mathcal{H}$ we write:

$$U : \mathcal{H} \otimes \mathcal{H} \rightarrow \mathcal{H} \otimes \mathcal{H}, \quad U(|\phi\rangle \otimes |B\rangle) = |\phi\rangle \otimes |\phi\rangle, \quad U(|\psi\rangle \otimes |B\rangle) = |\psi\rangle \otimes |\psi\rangle.$$

It is

$$\begin{aligned} (U(|\phi\rangle \otimes |B\rangle))^\dagger U(|\psi\rangle \otimes |B\rangle) &= (|\phi\rangle \otimes |\phi\rangle)^\dagger (|\psi\rangle \otimes |\psi\rangle) \stackrel{1.18}{=} (|\phi\rangle^\dagger \otimes |\phi\rangle^\dagger) (|\psi\rangle \otimes |\psi\rangle) \\ &\stackrel{1.16}{=} (\langle\phi| \otimes \langle\phi|) (|\psi\rangle \otimes |\psi\rangle) \stackrel{1.18}{=} (\langle\phi|\psi\rangle) (\langle\phi|\psi\rangle) = (\langle\phi|\psi\rangle)^2 \end{aligned}$$

and also,

$$\begin{aligned} (U(|\phi\rangle \otimes |B\rangle))^\dagger (U(|\psi\rangle \otimes |B\rangle)) &\stackrel{1.17}{=} (|\phi\rangle \otimes |B\rangle)^\dagger U^\dagger U (|\psi\rangle \otimes |B\rangle) \stackrel{U^\dagger U = I}{=} (|\phi\rangle \otimes |B\rangle)^\dagger I (|\psi\rangle \otimes |B\rangle) \\ &\stackrel{1.18}{=} (|\phi\rangle^\dagger \otimes |B\rangle^\dagger) (|\psi\rangle \otimes |B\rangle) \stackrel{1.16}{=} (\langle\phi| \otimes \langle B|) (|\psi\rangle \otimes |B\rangle) \\ &\stackrel{1.18}{=} \langle\phi|\psi\rangle \langle B|B\rangle \stackrel{\langle B|B\rangle=1}{=} \langle\phi|\psi\rangle. \end{aligned}$$

So, $(\langle\phi|\psi\rangle)^2 = \langle\phi|\psi\rangle$, that means that either $\langle\phi|\psi\rangle = 0$, thus ϕ, ψ are orthogonal, or that $\langle\phi|\psi\rangle = 1$, that is $\phi = \psi$.

Hence, if U clones the states $|\phi\rangle$ and $|\psi\rangle$, these are either identical or orthogonal. Thus, there is no single unitary U that clones all the states. $\square$

Essentially, the theorem says that it is impossible to design a single device that clones arbitrary unknown quantum states.

Remark 2.30. *We are trying to clone the pure state $|\psi\rangle$. The expected output of cloning is a scalar of $|\phi\rangle \otimes |\phi\rangle$, that lives in $\mathcal{H} \otimes \mathcal{H}$. We are working with quantum gates, therefore we need to use unitary operators $\mathcal{H} \otimes \mathcal{H} \rightarrow \mathcal{H} \otimes \mathcal{H}$. Hence, the input must live on $\mathcal{H} \otimes \mathcal{H}$, that is, it must be of the form $|\phi\rangle \otimes |B\rangle$, for some $|B\rangle$. The state $|B\rangle$ represents the "blank copy register", where the clone will be "printed".*

The no-cloning theorem deals only with a unitary operator and this is enough, because of the Stinespring's dilation theorem; this theorem states that any quantum channel arises from a unitary evolution on a larger system (i.e. system and environment). One can read more about Stinespring's dilation theorem in [45]. Thus, any general operation can be modeled as a unitary operator on a larger system. Therefore, since due to the no-cloning theorem we cannot clone a state using one unitary operator, then we cannot clone it using more unitary operators, because of the Stinespring's dilation theorem.

2.3.3 CCNOT gate

In order to overcome the above difficulty of no-cloning we introduce a suitable 3-qubit gate. It is called the *Toffoli* or CCNOT (*controlled-controlled-NOT*) gate. The name is after Tommaso Toffoli, who introduced the gate in 1980.

Definition 2.31. *The quantum logic gate CCNOT : $\mathbb{B}^3 \rightarrow \mathbb{B}^3$ is defined as*

$$\text{CCNOT}(|x\rangle \otimes |y\rangle \otimes |z\rangle) = |x, y, (x \wedge y) \oplus z\rangle.$$

CCNOT is a 3-qubit universal reversible logic gate. It has two control qubits; the qubits that correspond to $|x\rangle, |y\rangle$, in Definition 2.31. Also, it has one target qubit; the qubit corresponding to $|z\rangle$. The target qubit $|z\rangle$ is flipped, i.e. NOT is applied, only if both control qubits are in $|1\rangle$, that is, if $x \wedge y = 1$. Finally, the three inputs on $|x y z\rangle$ are determined by the state of the system and are not arbitrary variables.

CCNOT

$$U_{\text{CCNOT}} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}$$

As we can see on the above representation diagram of CCNOT, it works like a more complex CNOT. The realization as a diagram is the same as in CNOT; the $\bullet$ means that the qubit passes through the identity gate, while the CCNOT is applied on the qubit that goes through $\oplus$.

Gates like Toffoli allow us to realize a copy of the state of a qubit and implement the NAND operation between two qubits, choosing the appropriate entry of the third qubit (see Examples 2.32 and 2.33). It doesn't mean that we can do cloning, because quantum states are not just qubits. It does not violate the no-cloning theorem because it cannot clone an arbitrary superposition state, $\alpha|0\rangle + \beta|1\rangle$, into independent qubits, but rather entangles them [20].

Example 2.32. Having as input in CCNOT the qubit $|x\rangle \otimes |1\rangle \otimes |0\rangle$ we take:

$$\text{CCNOT}(|x\rangle \otimes |1\rangle \otimes |0\rangle) = |x, 1, (x \wedge 1) \oplus 0\rangle = |x, 1, x\rangle,$$

that is, x clones in the third qubit.

Example 2.33. Having as input in CCNOT the qubit $|x\rangle \otimes |y\rangle \otimes |1\rangle$, we take:

$$\text{CCNOT}(|x\rangle \otimes |y\rangle \otimes |1\rangle) = |x, y, (x \wedge y) \oplus 1\rangle = |x, 1, \text{NAND}(x, y)\rangle,$$

We care about NAND because of its universality (see Observation 2.21).

So far, we have described ideal quantum circuits. In practice, however, physical implementations are affected by noise and decoherence. We briefly discuss about the difficulties that arise, in the following section.

2.4 Noise and Error Correction

When we perform a quantum algorithm on a real physical platform, the information can be degraded. This is because the system also interacts with the environment.

The *wavefunction* of a qubit is its state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$. This encodes the probabilities of its possible states, and we explain it right away. Remember the line segment between the two basis states, and that a qubit state can "be" anywhere between them:



The measurement  of the state will return the value ($|0\rangle$ or $|1\rangle$) that is closer to the state. In the example of the above diagram, we would get $|1\rangle$ after the measurement:



The probability the qubit with the above wavefunction collapses at $|0\rangle$ is $|\alpha|^2$, while the probability the qubit collapses at $|1\rangle$ is $|\beta|^2$. Hence, the qubit state is influenced by the probabilities established prior to measurement.

The collapse of measurement is both fundamental and irreversible; once a measurement is made, the superposed state and the information it held are lost, since we lose the values α, β . So, if a qubit is measured by mistake or prematurely, the superposition is destroyed.

Except measurement, there are other obstacles that cause errors in quantum computing. We call *noise* all unwanted interactions, such as thermal fluctuations, electromagnetic interference,

imperfections in quantum gates. These interactions with the environment cause *decoherence*, which mean that qubit loses its quantum situation.

In addition, the implementation of error correction schemes is more difficult because of the No-cloning theorem. That is, there is no way to copy the unknown quantum state, to protect the information.

Fortunately, fault tolerant algorithms can still be developed to minimize the damage. There is the following important theoretical result, called the *threshold theorem* [29, 59]. The theorem states that when a quantum computer has a physical error rate below a certain threshold, it can reduce the error rate to arbitrarily low levels, applying methods of quantum error correction. Below, $p(n)$ is a polynomial on n .

Theorem 2.34 (Threshold theorem). *A quantum circuit containing $p(n)$ gates may be simulated with probability of error at most ϵ , using*

$$\mathcal{O}(\log^c(p(n)/\epsilon) p(n)), \quad c > 0 \text{ constant}$$

gates on hardware, whose components fail with probability at most p , provided p is below some constant threshold, $p < p_{th}$, and given reasonable assumptions about the noise in the underlying hardware.

Remark 2.35. *We observe that as ϵ decreases, the number of gates required on the hardware are increased. It is important that the number of gates is increased by just $\log^c(p(n)/\epsilon)$. In particular, achieving very small error requires only a modest increase in the circuit size.*

Also, polynomial time, that is the running time $\mathcal{O}(n^c)$ for a constant c , i.e. the number of gates, is typically adopted as the criterion of efficiency in computational complexity. For this reason, we consider quantum circuits of polynomial size $p(n)$ in the theorem.

Remark 2.36. *It is not possible to protect a quantum computation against the effect of completely arbitrary noise [59]. This is why the last assumption of the theorem requires a "well-behaved" noise. The Threshold theorem relies that we have a small number of physically reasonable assumptions about the type of noise occurring in the quantum computer in order to reach its powerful conclusion. Of course, this has to do with the available architect of quantum computers; possibly, a quantum computer with better isolation will be designed in the future. Then, the system will have less unwanted interactions with the environment, thereby the noise will decrease.*

The threshold theorem is an analog of the classical Von Neumann Theorem, which informally states that reliable computations can be achieved from unreliable components if the error probability is below a threshold [69]. Thanks to the threshold theorem, we have the hope to create fault-tolerant algorithms and feasible quantum computing in general.

Chapter 3

Classical and Quantum Information Geometry

Information geometry is a new field at the intersection of statistics, probability, and geometry. It uses techniques from Riemannian and differential geometry in order to study statistical manifolds, that is, manifolds whose points represent families of probability distributions. In this chapter, we focus on the most important notion of the classical Fisher Information Matrix. Then, we introduce its quantum counterpart, the Quantum Fisher Matrix. Finally, we arrive to define the Quantum Geometric Tensor by providing a natural Hermitian metric on the qubit space [1].

3.1 The Fisher Information Matrix

Definition 3.1. [47] A statistical manifold is a triple (M, g, D) , where M is an n -dimensional Riemannian C^∞ -manifold, g is a metric tensor and $D : \mathcal{X}(M) \times \mathcal{X}(M) \times \mathcal{X}(M) \rightarrow C^\infty(M)$ is a symmetric and covariant tensor of order 3.

Definition 3.2. Let $p(\theta)$ be a discrete empirical probability distribution, depending on parameters $\theta \in \mathbb{R}^n$ (or defined on a statistical manifold embedded in $\mathbb{R}^n$). We define the Fisher information matrix or Fisher matrix for short, as F , with

$$F_{ij}(\theta) = \mathbb{E}_p [\partial_{\theta_i} \log p(\theta) \partial_{\theta_j} \log p(\theta)], \quad (13)$$

where $\mathbb{E}_p$ represents the expected value with respect to the probability distribution p .

Another way to express more concisely the Fisher matrix, is to write it as:

$$F(\theta) = \mathbb{E}_p [\nabla \log p(\theta) (\nabla \log p(\theta))^t]$$

where we consider $\nabla \log p(\theta)$ as a column vector and t denotes the transpose vector.

Definition 3.3. The information loss $I(\theta)$ of a probability distribution $p(\theta)$ is $I(\theta) = -\log(p(\theta))$.

Definition 3.4. The Shannon entropy $H(\theta)$ of a probability distribution $p(\theta)$ is

$$H(\theta) = -\sum_{\theta} p(\theta) \log(p(\theta)).$$

Our goal is to connect the sense of Fisher matrix to information loss. Also, we observe that the information loss is linked to Shannon entropy $H(\theta)$ as the negative mean value of self-information:

$$H(\theta) = -\mathbb{E}_p[I(\theta)].$$

Shannon entropy quantifies the unevenness in the probability distribution p . It measures how much the distribution p , and hence the information, is spread.

In particular, the minimum $H(\theta) = 0$ is reached for a variable with a determined outcome, that is $p(\theta)$ is a probability mass distribution: $p_i(\theta) = \delta_{ij}$ for a fixed i and $j = 1, \dots, n$. At the opposite extreme, $H(\theta)$ is maximal, for a uniform distribution: $p_j(\theta) = \frac{1}{n}$ [49].

Note that entropy is a functional of the distribution of θ . It does not depend on the actual values taken by the random variable θ , but only on the probabilities [16].

We now look more closely at the Fisher matrix.

Proposition 3.5. The Fisher matrix is the covariance matrix of the gradient of the information loss.

Proof. The gradient of the information loss is

$$\nabla_{\theta} I(\theta) = \nabla(-\log p(\theta)) = -\nabla(\log p(\theta)) = -\frac{\nabla p(\theta)}{p(\theta)}.$$

Notice that, since $\sum_i p_i$ is fixed, independent of θ :

$$\mathbb{E}_p(\nabla_{\theta} I) = \sum_i p_i \frac{\nabla_{\theta} p_i}{p_i} = \sum_i \nabla_{\theta} p_i = \nabla_{\theta} \left(\sum_i p_i \right) = 0.$$

The covariance matrix of $\nabla_{\theta} I(\theta)$ is by definition:

$$\text{Cov}(\nabla_{\theta} I) = \mathbb{E}_p[(\nabla_{\theta} I - \mathbb{E}_p(\nabla_{\theta} I))^t (\nabla_{\theta} I - \mathbb{E}_p(\nabla_{\theta} I))] = \mathbb{E}_p[(\nabla_{\theta} I)^t (\nabla_{\theta} I)] = F(\theta).$$

□

This overview on Information Geometry closes with an observation for the metric on the parameter space. It is about the relation of Fisher metric and Kullback-Leibler divergence, so we give the required definition:

Definition 3.6. [25] For two (discrete) probability distributions p and q , we define their Kullback-Leibler divergence as:

$$\text{KL}(p\|q) = \sum_i p_i \log \frac{p_i}{q_i} = \mathbb{E}_p \left[\log \frac{p}{q} \right].$$

Notice that $\text{KL}(p\|q)$ gives a measure of how well q approximates p . It is like a statistical "distance" between two probability distributions, answering the question: "If the true distribution is p , how bad is it to pretend it is q ?"

We can write:

$$\text{KL}(p\|q) = \sum_i p_i \log \frac{p_i}{q_i} = \sum_i p_i \log p_i - \sum_i p_i \log q_i = -H(p) + H(p, q),$$

where $H(p)$ is the Shannon entropy and $H(p, q) = -\sum_i p_i \log q_i$ is the *cross-entropy*.

Observation 3.7. We observe that the Fisher metric and the Kullback-Leibler divergence are related to each other. Indeed, we consider

$$\text{KL}(p(\theta + \delta\omega)\|p(\theta))$$

where $\delta\omega$ is an infinitesimal perturbation of a parameter vector ω . The Kullback-Leibler divergence measures how much information is lost when $p(\theta + \delta\omega)$ is approximated by $p(\theta)$. It is:

$$\text{KL}(p(\theta + \delta\omega)\|p(\theta)) = \mathbb{E}_{x \sim p(x|\theta + \delta\omega)} \left[\log \frac{p(x|\theta + \delta\omega)}{p(x|\theta)} \right].$$

Expanding the Taylor series of the Kullback-Leibler divergence in the variable $\theta + \delta\omega$ around θ , we get

$$\begin{aligned} \text{KL}(p(\theta + \delta\omega)\|p(\theta)) &\cong \text{KL}(p(\theta)\|p(\theta)) + \nabla_{\theta} \text{KL}(p(\theta + \delta\omega)\|p(\theta))^t \Big|_{\delta\omega=0} \delta\omega \\ &\quad + \frac{1}{2} (\delta\omega)^t \nabla_{\theta}^2 \text{KL}(p(\theta + \delta\omega)\|p(\theta)) \Big|_{\delta\omega=0} \delta\omega + \mathcal{O}(\|\delta\omega\|^3). \end{aligned}$$

We have that $\text{KL}(p(\theta)\|p(\theta)) = 0$. Also, the Kullback-Leibler divergence is non-negative and vanishes at $\delta\omega = 0$, hence, it minimizes there. Thus, its derivative vanishes at $\delta\omega = 0$:

$$\nabla_{\theta} \text{KL}(p(\theta + \delta\omega)\|p(\theta)) \Big|_{\delta\omega=0} = 0.$$

Furthermore,

$$\begin{aligned}
 \frac{\partial^2 \text{KL}(p(\theta + \delta\omega) \| p(\theta))}{\partial \theta^2} &= \frac{\partial^2}{\partial \theta^2} \sum_x p(x|\theta + \delta\omega) \log \frac{p(x|\theta + \delta\omega)}{p(x|\theta)} \Big|_{\delta\omega=0} \\
 &= \frac{\partial^2}{\partial \theta^2} \sum_x p(x|\theta + \delta\omega) (\log p(x|\theta + \delta\omega) - \log p(x|\theta)) \Big|_{\delta\omega=0} \\
 &= 0 + \sum_x p(x|\theta + \delta\omega) \frac{\partial^2}{\partial \theta^2} (\log p(x|\theta + \delta\omega) - \log p(x|\theta)) \Big|_{\delta\omega=0} \\
 &= - \sum_x p(x|\theta + \delta\omega) \frac{\partial^2 \log p(x|\theta)}{\partial \theta^2} \Big|_{\delta\omega=0} \\
 &= - \sum_x p(x|\theta) \frac{\partial^2 \log p(x|\theta)}{\partial \theta^2} \\
 &= F(\theta),
 \end{aligned}$$

and the remaining terms are of order $\mathcal{O}(\|\delta\omega\|^3)$. Thus, in the end we have:

$$\text{KL}(p(\theta + \delta\omega) \| p(\theta)) \cong \frac{1}{2} (\delta\omega)^t F(\theta) (\delta\omega) + \mathcal{O}(\|\delta\omega\|^3).$$

It is $\langle \delta\omega, \delta\omega \rangle = (\delta\omega)^t F(\theta) (\delta\omega)$, in the Fisher metric. That is,

$$\text{KL}(p(\theta + \delta\omega) \| p(\theta)) \cong \frac{1}{2} \langle \delta\omega, \delta\omega \rangle + \mathcal{O}(\|\delta\omega\|^3)$$

and hence, we can interpret the Fisher matrix as a metric on the statistical manifold.

3.2 The Quantum Information Matrix

Quantum metrology is the study of physical measurements via quantum theory. There, the quantum information matrix or quantum Fisher matrix is the analog of the classical Fisher matrix, that was introduced in the previous section. We will give its definition and the main properties.

First we define the symmetric logarithmic derivative. It is a generalization of the usual notion of logarithmic derivative. In classical statistics, the Fisher information is built from the logarithmic derivative of a probability distribution: $\partial_\theta \log p(x|\theta)$. This is well defined because for $p > 0$, $\log p$ and $\frac{1}{p}$ are well-defined numbers and $\frac{1}{p} \partial_\theta p = \partial_\theta \log p$. So, $\partial_\theta \log p$ is a unique expression, i.e. there is only one possible operator ordering.

The problem in the quantum case is that the scalar that expresses the probability distribution p in the definition of Fisher matrix (see Equation (13)) is replaced by the density operator

ρ (see Equation 2.12). In quantum theory, probabilities are not fundamental objects, since quantum theory predicts probabilities of measurement outcomes, not probabilities of states [66].

Example 3.8. Consider the pure state $|\psi\rangle = |0\rangle$, with corresponding density operator $\rho = |0\rangle\langle 0|$. When measuring in the computational Z -basis, the state can be written as

$$|\psi\rangle = 1|0\rangle + 0|1\rangle,$$

which yields the measurement probabilities $p(0) = 1$ and $p(1) = 0$. In contrast, expressing the same state in the X basis,

$$|\psi\rangle = \frac{1}{\sqrt{2}}|+\rangle + \frac{1}{\sqrt{2}}|-\rangle,$$

leads to the probabilities $p(+) = \frac{1}{2}$ and $p(-) = \frac{1}{2}$. This illustrates that the outcome probabilities depend on the measurement basis, even though the quantum state remains the same.

On the other hand, the state of a system is described by the density operator ρ , which is not a number, but a linear operator acting on Hilbert space $\mathbb{C}^2$. ρ is generally non-commuting with its derivative. Because of this non-commutativity, the $\partial_\theta \log \rho$ is not well-defined, as it happens in the classical case. Hence, we are in need of the new concept of symmetric logarithmic derivative.

We assume that ρ is depending on a parameter θ ; when ρ is defined for pure states (a special, but important case), this parameter comes from a parametrization of the vectors in the Hilbert space [30].

Definition 3.9. The symmetric logarithmic derivative L of a given density operator ρ is implicitly defined as:

$$\partial_\theta \rho = \frac{L\rho + \rho L}{2}. \quad (14)$$

It is immediately noticed that, whenever L and ρ are scalars, we have

$$\begin{aligned} \partial_\theta \rho &= \frac{L\rho + \rho L}{2} = \frac{L\rho + L\rho}{2} = \rho L \Rightarrow \\ L &= \frac{\partial_\theta \rho}{\rho} = \partial_\theta(\log(\rho)). \end{aligned} \quad (15)$$

So, L is the gradient of the logarithm of ρ as it appears in Equation (13). We can derive the explicit expression of L , when we fix an orthonormal basis $\{|\psi_j\rangle\}$ for qubits.

Also, the Equation (15) ensures that the symmetric logarithmic derivative is Hermitian ($L^\dagger = L$), as ρ is also Hermitian.

Observation 3.10. Let $\{|\psi_j\rangle\}$ be an orthonormal basis for Hilbert space $\mathcal{H}$. Using the Hermiticity property of the density matrix ρ (see Proposition 2.14) the spectral theorem guarantees the existence of an orthonormal basis of eigenvectors $\{|\psi_j\rangle\} \subset \mathcal{H}$, with real eigenvalues p_j :

$$\rho |\psi_j\rangle = p_j |\psi_j\rangle, \quad j = 1, \dots, \dim(\mathcal{H}) = N.$$

Moreover, positivity of ρ implies $p_j \geq 0$ and the unit trace $\text{Tr}(\rho) = 1$ (Proposition 2.14) ensures $\sum p_j = 1$. We can also assume that only the first s values are different than zero, where $s = \text{rank}(\rho)$. Now, for any $N \times N$ matrix A , it is

$$\langle \psi_i | A | \psi_j \rangle = \sum_k A_{kj} \langle \psi_i | \psi_k \rangle$$

and since the basis is orthonormal, $\langle \psi_i | \psi_k \rangle = \delta_{ik}$. Hence, we can express the (i, j) entry of A as

$$A_{ij} = \langle \psi_i | A | \psi_j \rangle.$$

Applying this observation to Definition 14 and using the Definition (2.12) for ρ , we get:

$$\begin{aligned} (\partial_\theta \rho)_{ij} &= \langle \psi_i | \partial_\theta \rho | \psi_j \rangle = \frac{1}{2} \langle \psi_i | (L\rho + \rho L) | \psi_j \rangle \\ &= \frac{1}{2} (\langle \psi_i | L\rho | \psi_j \rangle + \langle \psi_i | \rho L | \psi_j \rangle) \\ &= \frac{p_i + p_j}{2} L_{ij}. \end{aligned}$$

We notice that $p_i + p_j = 0$ if both $i, j > s$, so that we can assume $L_{ij} = 0$, for $i, j = s+1, \dots, N$ while the other matrix elements of L are given (see Equation (15)) by

$$L_{ij} = \frac{2(\partial_\theta \rho)_{ij}}{p_i + p_j}. \quad (16)$$

Now we can define the quantum Fisher information matrix (QFIM). In this thesis we deal with single-parameter quantum states, thus, we give the corresponding definition.

Definition 3.11. Let ρ be a density operator as in Definition (2.12). We define the quantum Fisher information matrix as

$$F_Q = \text{Tr}(\rho L^2)$$

where L is the symmetric logarithmic derivative of ρ .

Remark 3.12. For n -parameter family of quantum states, the QFIM is given by the matrix with entries

$$(F_Q)_{ij} = \frac{1}{2} \text{Tr}[\rho(L_{\theta_i} L_{\theta_j} + L_{\theta_j} L_{\theta_i})],$$

where L_{θ_i} are the symmetric logarithmic derivatives of ρ . Hence, the QFIM is the matrix consisting of the corresponding traces, and the dimension of QFIM is $n \times n$. Moreover, in the single-parameter case, the QFIM is a 1×1 matrix.

Proposition 3.13. *Let $\rho(\theta)$ be a smooth one-parameter family of quantum states, where $\rho(\theta)$ is a pure state, for all θ . Then the quantum Fisher information matrix satisfies*

$$F_Q = 4 \operatorname{Tr} [\rho(d\rho)^2] .$$

Proof. It is $\rho = |\psi\rangle\langle\psi|$. Since $\rho(\theta)$ is a pure state, it is

$$\rho^2 = (|\psi\rangle\langle\psi|)^2 = |\psi\rangle\langle\psi|\psi\rangle\langle\psi| = |\psi\rangle \cdot 1 \cdot \langle\psi| = \rho .$$

Differentiating this identity, yields $\partial\rho = \partial(\rho^2) = \partial\rho \cdot \rho + \rho \cdot \partial\rho$. From Definition of symmetric logarithmic derivative, 3.9, it is

$$\frac{L\rho + \rho L}{2} = \partial\rho = \partial\rho \cdot \rho + \rho \cdot \partial\rho \Rightarrow L = 2\partial\rho . \quad (17)$$

So, from the Definition of quantum Fisher information, 3.11, and Equation (17) we obtain

$$F_Q = \operatorname{Tr} [\rho(2d\rho)^2] = 4 \operatorname{Tr} [\rho(d\rho)^2] .$$

□

With an abuse of notation, we write F_Q also for the matrix that represents F_Q in a fixed orthonormal basis.

Remark 3.14. *There is a strict analogy between F_Q and the Fisher information matrix F , Equation (13). This is because, the symmetric logarithmic derivative L is the quantum analogue of $\partial_\theta \log p$. Also, the trace over ρ is the quantum analogue of expectation over $p(\theta)$. And last, the formula has exactly the same mathematical structure: they represent the variance of a score function ($\partial_\theta \log p(x|\theta)$ and ρL) .*

Observation 3.15. *Let the notation be as above and let us fix the orthonormal basis of eigenvectors of ρ . Then, we have:*

$$F_Q = \sum_{k=1}^s \sum_{l=1}^N p_k L_{kl} L_{lk} . \quad (18)$$

Indeed,

$$\begin{aligned} F_Q &= \operatorname{Tr} [\rho L^2] = \sum_k \langle \psi_k | \rho L^2 | \psi_k \rangle = \sum_k \langle \psi_k | \rho \cdot 1 \cdot L^2 | \psi_k \rangle \\ &= \sum_k \langle \psi_k | \rho \cdot \left(\sum_l |\psi_l\rangle\langle\psi_l| \right) \cdot L^2 | \psi_k \rangle = \sum_{k,l} \langle \psi_k | \rho | \psi_l \rangle \langle \psi_l | L^2 | \psi_k \rangle \\ &= \sum_k p_k \langle \psi_k | L^2 | \psi_k \rangle = \sum_{k,l} p_k \langle \psi_k | L | \psi_l \rangle \langle \psi_l | L | \psi_k \rangle , \end{aligned}$$

which is the Equation (18). We notice that the sum over k is practically restricted to $k = 1, \dots, N$, because $p_k = 0$ if $k > s$. We readily see that it is the analog of the expression (13) for F .

Now, we will continue by giving another interesting and useful expression of F_Q .

Proposition 3.16. *We use the previous notation and orthonormal basis $\{|\psi_i\rangle\}$. Then,*

$$F_Q = \sum_{i=1}^s \frac{1}{p_i} (\partial_\theta p_i)^2 + \sum_{i=1}^s \sum_{j=1}^N \frac{4 p_i (p_i - p_j)^2}{(p_i + p_j)^2} |\langle \psi_i | \partial_\theta \psi_j \rangle|^2 \quad (19)$$

where s is the rank of ρ .

Proof. Substituting the Equation (16) for L_{ij} in (18), we get:

$$F_Q = \sum_{i=1}^s \sum_{j=1}^N 4 p_i \frac{(\partial_\theta \rho)_{ij} (\partial_\theta \rho)_{ji}}{(p_i + p_j)^2}.$$

From the Definition 2.12 of ρ :

$$\rho(\theta) = \sum_k p_k(\theta) |\psi_k(\theta)\rangle \langle \psi_k(\theta)|$$

we get

$$\partial_\theta \rho = \sum_k (\partial_\theta p_k) |\psi_k\rangle \langle \psi_k| + \sum_k p_k (|\partial_\theta \psi_k\rangle \langle \psi_k| + |\psi_k\rangle \langle \partial_\theta \psi_k|).$$

It is [50]:

$$\begin{aligned} (\partial_\theta \rho)_{ij} &= \langle \psi_i | \partial_\theta \rho | \psi_j \rangle \\ &= \langle \psi_i | \left(\sum_k (\partial_\theta p_k) |\psi_k\rangle \langle \psi_k| + \sum_k p_k (|\partial_\theta \psi_k\rangle \langle \psi_k| + |\psi_k\rangle \langle \partial_\theta \psi_k|) \right) | \psi_j \rangle \\ &= \sum_k (\partial_\theta p_k) \langle \psi_i | \psi_k \rangle \langle \psi_k | \psi_j \rangle + \sum_k p_k \langle \psi_i | \partial_\theta \psi_k \rangle \langle \psi_k | \psi_j \rangle + \sum_k p_k \langle \psi_i | \psi_k \rangle \langle \partial_\theta \psi_k | \psi_j \rangle \\ &= \delta_{ij} \partial_\theta p_i + p_j \langle \psi_i | \partial_\theta \psi_j \rangle + p_i \langle \partial_\theta \psi_i | \psi_j \rangle. \end{aligned}$$

Since

$$0 = \partial_\theta (\langle \psi_i | \psi_j \rangle) = \langle \partial_\theta \psi_i | \psi_j \rangle + \langle \psi_i | \partial_\theta \psi_j \rangle,$$

finally it is

$$(\partial_\theta \rho)_{ij} = \delta_{ij} \partial_\theta p_i + (p_j - p_i) \langle \psi_i | \partial_\theta \psi_j \rangle.$$

If $i = j$, it is $(\partial_\theta \rho)_i = \partial_\theta p_i$. Else, if $i \neq j$, it is $(\partial_\theta \rho)_{ij} = -(\partial_\theta \rho)_{ji}$. Since ρ is Hermitian, we have that $\partial_\theta \rho$ is Hermitian, because differentiation preserves Hermiticity. So, $(\partial_\theta \rho)_{ij}^* = (\partial_\theta \rho)_{ji}$. Hence,

$$F_Q = \sum_{i=1}^s \frac{1}{p_i} (\partial_\theta p_i)^2 + \sum_{i=1}^s \sum_{\substack{j=1 \\ j \neq i}}^N \frac{|(\partial_\theta \rho)_{ij}|^2}{(p_i + p_j)^2} = \sum_{i=1}^s \frac{1}{p_i} (\partial_\theta p_i)^2 + \sum_{i=1}^s \sum_{j=1}^N \frac{4 p_i (p_i - p_j)^2}{(p_i + p_j)^2} |\langle \psi_i | \partial_\theta \psi_j \rangle|^2.$$

□

The next observation compares, once again, the classical Fisher information matrix to the quantum F_Q .

Observation 3.17. *We notice that the first term of the Quantum Fisher matrix (19) coincides with the Classical Fisher matrix (13). It is given by changes only in the eigenvalues, i.e. of the probabilities p_k , of the density matrix. Instead, the second term of Equation (19) arises from how the eigenvectors $|\psi_k\rangle$ change, that is, their relative phase and coherence, and it is therefore quantum in nature.*

3.3 Quantum Geometric Tensor

Geometric Quantum Mechanics is a novel and prospecting approach motivated by the belief that our world is ultimately geometrical. At the heart of that is a quantity called *Quantum Geometric Tensor* (QGT), which is a complex tensor. It characterizes mathematically the local geometry of quantum states and also, it is gauge-invariant [52].

The real part of QGT, is the *quantum metric*. It is gauge-invariant and it is a proper measure of the local distance between physically inequivalent states in the quantum phase space $\mathbb{P}(\mathcal{H})$ [32]. That is, it determines the quantum distance between two quantum states that are infinitely close. The quantum metric is strongly important to understand many complex quantum phenomena, such as superfluid stiffness, orbital magnetic susceptibility and fractional Chern insulators. The measure of quantum metrics and their geometric effects' unveiling are a very active area of research nowadays. A variety of quantum platforms have been suggested, in these studies. They include cold atoms in optical lattices, superconducting quantum circuits, nitrogen-vacancy centers in diamond, and crystalline solid systems.

The imaginary part of QGT gives the *Berry curvature*. It encodes the gradient information of the geometric phase in parameter space. In the past decades, it was proven that the Berry curvature is required to understand many macroscopic quantum phenomena, such as the Aharonov-Bohm effect and the anomalous Hall effect. The Berry curvature is deeply connected with the band topology of Bloch states. It has been widely adopted in the study of topological properties of quantum matter, such as the first Chern number and integer quantum anomalous Hall effect.

It is extremely interesting that new papers are consecutively being published on this topic. For instance, [52] was just published in March 2026, and concerns the unveil of a new inequality between the Fubini-Study quantum distance and the Berry phase, that is, between the parts of the QGT.

The Quantum Geometric Tensor is a key object in quantum information geometry and it is strictly related to the Quantum Information Matrix introduced in the previous section. We will construct QGT after defining the needed notions we will use for that.

3.3.1 Linear Symplectic Geometry

This subsection contains useful definitions and results of symplectic and complex structures. We will give the very needed notions since it is a semester course by itself and some examples to understand the main concepts (refs. [27, 33, 71]). The following is a preprocessing for the next ones as they construct Kähler manifolds.

We start with the definition of symplectic forms and manifolds.

Definition 3.18. *Let M be a smooth manifold, and ω is a smooth 2-form on M . That is, for any $p \in M$, $\omega_p : T_p M \times T_p M \rightarrow \mathbb{R}$ is a skew-symmetric bilinear map, and ω_p depends smoothly on p . We call ω a symplectic form on M if the following properties are satisfied:*

1. (closedness) ω is a closed 2-form, i.e. $d\omega = 0$,
2. (non-degeneracy) $\forall p \in M$, ω_p is a linear symplectic form on $T_p M$, i.e. $\omega_p \in \Lambda^2(T_p M)$ (antisymmetric) and it is non-degenerate.

Then, the pair (M, ω) is a symplectic manifold.

Example 3.19. *Suppose the manifold $M = \mathbb{R}^{2n}$. Let the coordinates be $(x_1, \dots, x_n, y_1, \dots, y_n)$ and let $\omega = \sum_{i=1}^n dx_i \wedge dy_i$. We will show (M, ω) is a symplectic manifold.*

First, we show that ω is a symplectic form. ω is sum of terms $dx_i \wedge dy_i$ and hence, $\omega \in \Omega^2(\mathbb{R}^{2n})$. Also,

$$d\omega = d\left(\sum_{i=1}^n dx_i \wedge dy_i\right) = \sum_{i=1}^n d(dx_i \wedge dy_i) = \sum_{i=1}^n \left(d(dx_i) \wedge dy_i + dx_i \wedge d(dy_i)\right) = 0,$$

so, ω is a closed 2-form. Now, for $p \in M$, let $u, v \in T_p M$. We denote

$$u = \sum \left(u_{x_i} \frac{\partial}{\partial x_i} + u_{y_i} \frac{\partial}{\partial y_i} \right), \quad v = \sum \left(v_{x_i} \frac{\partial}{\partial x_i} + v_{y_i} \frac{\partial}{\partial y_i} \right).$$

It is

$$\begin{aligned} \omega_p(u, v) &= \sum_{i=1}^n (dx_i \wedge dy_i)(u, v) = \sum_{i=1}^n \left(dx_i(u) dy_i(v) - dx_i(v) dy_i(u) \right) = \sum_{i=1}^n \left(u_{x_i} v_{y_i} - v_{x_i} u_{y_i} \right) \\ &= -\omega_p(v, u) \end{aligned}$$

and $\omega_p(u, v) = 0, \forall v$ thus, $u = 0$. So, ω is non-degenerate and hence it is a symplectic form on M . Therefore, (M, ω) is a symplectic manifold.

We continue with the notions of (almost) complex structures in order to construct the so-called manifolds.

Definition 3.20. A complex structure on a real vector space V is a linear endomorphism $J : V \rightarrow V$ such that $J^2 = -\text{id}$.

Example 3.21. In $M = \mathbb{R}^{2n}$ with $(x_1, \dots, x_n, y_1, \dots, y_n)$ we define the bundle map $J : TM \rightarrow TM$ with

$$J \left(\frac{\partial}{\partial x_i} \right) = \frac{\partial}{\partial y_i}, \quad J \left(\frac{\partial}{\partial y_i} \right) = -\frac{\partial}{\partial x_i}.$$

J is a linear endomorphism on TM . We compute

$$J^2 \left(\frac{\partial}{\partial x_i} \right) = J \left(\frac{\partial}{\partial y_i} \right) = -\frac{\partial}{\partial x_i}, \quad J^2 \left(\frac{\partial}{\partial y_i} \right) = J \left(-\frac{\partial}{\partial x_i} \right) = -\frac{\partial}{\partial y_i}.$$

Let the tangent vector $u \in TM$ with $u = \sum_{i=1}^n \left(u_{x_i} \frac{\partial}{\partial x_i} + u_{y_i} \frac{\partial}{\partial y_i} \right)$. It is

$$J^2(u) = J \left(\sum_{i=1}^n \left(u_{x_i} \frac{\partial}{\partial y_i} - u_{y_i} \frac{\partial}{\partial x_i} \right) \right) = \sum_{i=1}^n \left(-u_{x_i} \frac{\partial}{\partial x_i} - u_{y_i} \frac{\partial}{\partial y_i} \right) = -u.$$

Hence, $J^2 = -\text{id}$, so J is a complex structure. Schematically, applying J once is a rotation of 90° while applying it twice is a rotation of 180° , i.e. multiplication by -1 .

Definition 3.22. An almost complex structure on a real differentiable manifold M is a tensor field J such that $\forall p \in M$, $J_p : T_p M \rightarrow T_p M$ is an endomorphism such that $J_p^2 = -\text{id}_p$. A manifold with such a structure is called an almost complex manifold.

Example 3.23. We take the same example of $M = \mathbb{R}^{2n}$ again, with J as in 3.21. J is an $(1, 1)$ tensor field as it belongs to $\Gamma(TM \otimes T^*M)$ and $\forall p \in \mathbb{R}^{2n}$ it is an endomorphism with $J_p^2 = -\text{id}_p$. Hence, $\mathbb{R}^{2n}$ is an almost complex manifold.

Definition 3.24. Let (M, ω) be a symplectic manifold and J an almost complex structure on M . At each tangent space $T_p M$ we have the linear symplectic structure ω_p and the linear complex structure J_p . It is said that J_p is compatible with ω_p if

$$g_p(u, v) := \omega_p(u, J_p v)$$

is an inner product on $T_p M$. We say that ω is the associated form to the structure.

Example 3.25. Having the same symplectic manifold (M, ω) as in Example 3.19 and almost complex structure J as in Example 3.21, we will show that for $p \in M$, $g_p(u, v) := \omega_p(u, J_p v)$ is an inner product. Actually, for $u, v \in T_p M$ as in 3.19, we have:

$$Jv = J \sum_{i=1}^n \left(v_{x_i} \frac{\partial}{\partial x_i} + v_{y_i} \frac{\partial}{\partial y_i} \right) = \sum_{i=1}^n \left(v_{x_i} \frac{\partial}{\partial y_i} - v_{y_i} \frac{\partial}{\partial x_i} \right).$$

Therefore,

$$g_p(u, v) := \omega_p(u, J_p v) = \sum_{i=1}^n (dx_i \wedge dy_i)(u, Jv) = \sum_{i=1}^n (u_{x_i} v_{y_i} + v_{x_i} u_{y_i})$$

g_p is bilinear, symmetric and positive definite, so it is an inner product on $T_p M$ and hence, J_p is compatible with ω_p .

Definition 3.26. A biholomorphic map is a map $\phi : U \rightarrow V$, defined on an open subset $U \subset \mathbb{C}^n$ which is holomorphic and one-to-one, such that its image is an open set $V \subset \mathbb{C}^n$ and the inverse $\phi^{-1} : V \rightarrow U$ is also holomorphic.

Definition 3.27. Let M be a real differentiable manifold of dimension $2n$. Let $\{(U_\alpha, \phi_\alpha)\}$ be an atlas on M , with $\phi : U \rightarrow \phi(U) \subset \mathbb{C}^n$ homeomorphisms, such that every transition map

$$\phi_2 \circ \phi_1^{-1} : \phi_1(U_1 \cap U_2) \rightarrow \phi_2(U_1 \cap U_2)$$

is holomorphic, for $U_1 \cap U_2 \neq \emptyset$. The almost complex structure J given on each $p \in M$ by

$$J_p = (d\phi_\alpha|_p)^{-1} \circ i \circ d\phi_\alpha|_p,$$

is a complex structure and is called integrable. (M, J) is locally biholomorphic to $(\mathbb{C}^n, \sqrt{-1})$ and with abuse of the notation, we write that $(M, J) \cong (\mathbb{C}^n, \sqrt{-1})$ locally.

Example 3.28. $(\mathbb{R}^{2n}, J) \cong (\mathbb{C}^n, \sqrt{-1})$ where J is as in example 3.21, hence J is integrable and $\mathbb{R}^{2n}$ is a complex structure.

Thus, a complex structure is an integrable almost complex structure. It is hard to use the definition above to detect whether an almost complex structure is complex or not. However, we do have a very useful criteria (3.30).

Definition 3.29. The Nijenhuis tensor N_J is

$$N_J(u, v) = [Ju, Jv] - J[Jv, u] - J[u, Jv] - [u, v].$$

Theorem 3.30. (Newlander-Nirenberg). Let J be an almost complex structure on M . Then

$$J \text{ is integrable} \iff N_J = 0.$$

Example 3.31. We will compute the Nijenhuis tensor for J of example 3.21 in the basis vectors. It is

$$\begin{aligned} N_J(\partial_{x_i}, \partial_{x_j}) &= [J\partial_{x_i}, J\partial_{x_j}] - J[J\partial_{x_i}, \partial_{x_j}] - J[\partial_{x_i}, J\partial_{x_j}] - [\partial_{x_i}, \partial_{x_j}] \\ &= [\partial_{y_i}, \partial_{y_j}] - J[\partial_{y_i}, \partial_{x_j}] - J[\partial_{x_i}, \partial_{y_j}] = 0. \end{aligned}$$

So, $N_J = 0$ everywhere on M , and hence, J is integrable.

Definition 3.32. A complex manifold of complex dimension n is a manifold that is locally homeomorphic to open subsets in $\mathbb{C}^n$, with biholomorphic transition maps.

Example 3.33. The complex projective space $\mathbb{P}^n(\mathbb{C})$ is the quotient space of $\mathbb{C}^{n+1} \setminus \{0\}$ by the equivalent relation:

$$(z_0, \dots, z_n) \sim (\lambda z_0, \dots, \lambda z_n), \quad \lambda \in \mathbb{C}^*.$$

We denote the equivalence class by the homogeneous coordinate $[z_0 : \dots : z_n]$. We define $U_0, \dots, U_n$ by $U_i = \{[z_0 : \dots : z_n] | z_i \neq 0\}$. U_i is the set of complex lines in $\mathbb{C}^{n+1}$ which pass through the origin and do not belong to the hyperplane $z_i = 0$. Define $\phi_j : U_j \rightarrow \mathbb{C}^n$ by

$$\phi_j[z_0 : \dots : z_n] = \left(\frac{z_0}{z_j}, \dots, \frac{z_{j-1}}{z_j}, \frac{z_{j+1}}{z_j}, \dots, \frac{z_n}{z_j} \right).$$

The transition function

$$\phi_{01}(u_1, \dots, u_n) = \phi_1([1 : u_1 : \dots : u_n]) = \left(\frac{1}{u_1}, \frac{u_2}{u_1}, \dots, \frac{u_n}{u_1} \right)$$

is holomorphic on $\phi_0(U_0 \cap U_1)$, making $\mathbb{P}^n(\mathbb{C})$ a complex manifold of dimension n . Here, the integrable structure J is the canonical complex structure on $\mathbb{P}^n(\mathbb{C})$, that is in local coordinates $(z_1, \dots, z_n) \in \mathbb{C}^{n+1} \setminus \{0\}$, with $z_k = x_k + iy_k$, it is

$$J \left(\frac{\partial}{\partial x_k} \right) = \frac{\partial}{\partial y_k}, \quad J \left(\frac{\partial}{\partial y_k} \right) = -\frac{\partial}{\partial x_k}.$$

Observation 3.34. An alternative way to construct $\mathbb{P}^n(\mathbb{C})$ begins with normalizing the vectors of the states and obtaining a $(2n+1)$ -dimensional real sphere $\mathbb{S}(\mathcal{H}) := \{|\psi\rangle \in \mathcal{H} \mid \langle\psi|\psi\rangle = 1\} \cong \mathbb{S}^{2n+1}$. Two points in $\mathbb{S}(\mathcal{H})$ are physically equivalent $|\phi\rangle \cong |\psi\rangle$ iff they differ only by a phase factor $|\psi\rangle = e^{i\theta} |\phi\rangle$. Hence, $\mathbb{P}(\mathcal{H})$ is the quotient space $\mathbb{S}(\mathcal{H})/\text{U}(1) \cong \mathbb{P}^n(\mathbb{C})$.

Remark 3.35. The Hopf fibration $f : \mathbb{S}^3 \rightarrow \mathbb{S}^2$ is the composition of the map

$$\mathbb{S}^3 \rightarrow \mathbb{P}^1(\mathbb{C}) : (z, w) \mapsto [z : w],$$

where $\mathbb{S}^3$ is understood as the unit sphere in $\mathbb{C}^2$, with the diffeomorphism from $\mathbb{P}^1(\mathbb{C})$ to $\mathbb{S}^2$:

$$f(z, w) = (2 \operatorname{Re}(\bar{z}w), 2 \operatorname{Im}(\bar{z}w), |z|^2 - |w|^2).$$

Hence, the Hopf fibration $p : \mathbb{S}^3 \rightarrow \mathbb{S}^2$ corresponds to the $n = 1$ case of $\mathbb{S}(\mathcal{H})/\text{U}(1) \cong \mathbb{P}^1(\mathbb{C})$, since $\text{U}(1) \cong \mathbb{S}^1$ and $\mathbb{P}^1(\mathbb{C}) \cong \mathbb{S}^2$. [31, 62]

Definition 3.36. Let (M, J) be an almost complex manifold. A Hermitian metric on (M, J) is a Riemannian metric h , such that

$$h(X, Y) = h(JX, JY), \quad \forall X, Y \in TM.$$

The fundamental form of a Hermitian metric is defined by $\Omega(X, Y) := h(JX, Y)$ and it is uniquely determined by the metric and the complex structure.

Definition 3.37. Let $J : TM \rightarrow TM$ be an almost complex structure with $J^2 = -1$. A 2-form Ω is called J -invariant if

$$\Omega(JX, JY) = \Omega(X, Y),$$

for all tangent vectors $X, Y \in TM$.

Proposition 3.38. Let (M, J) an almost complex structure and h a Hermitian metric on (M, J) . The fundamental form $\Omega(X, Y) := h(JX, Y)$ is J -invariant.

Proof. Indeed, let $X, Y \in TM$. It is

$$\begin{aligned} \Omega(JX, JY) &= h(J(JX), JY) \stackrel{J^2=-1}{=} h(-X, JY) \stackrel{h \text{ Hermitian}}{=} h(J(-X), J^2Y) \\ &\stackrel{J^2=-1}{=} h(-JX, -Y) \stackrel{h \text{ metric}}{=} h(JX, Y) = \Omega(X, Y). \end{aligned}$$

□

Definition 3.39. A complex differential form ω , is said to be real if $\bar{\omega} = \omega$.

Definition 3.40. Let (M, J) be a complex manifold of complex dimension n . A complex differential k -form ω is said to be of type (p, q) , with $p + q = k$, if

$$\omega \in \Omega^{p,q}(M) := \Gamma(\Lambda^p(T^{1,0}M)^* \otimes \Lambda^q(T^{0,1}M)^*).$$

Equivalently, in local holomorphic coordinates,

$$\omega = \sum \omega_{i_1 \dots i_p \bar{j}_1 \dots \bar{j}_q} dz_{i_1} \wedge \dots \wedge dz_{i_p} \wedge d\bar{z}_{j_1} \wedge \dots \wedge d\bar{z}_{j_q},$$

where $\omega_{i_1 \dots i_p \bar{j}_1 \dots \bar{j}_q}$ are complex functions.

3.3.2 Kähler manifolds

Kähler structures were introduced by Erich Kähler in his article [36] with the following motivation. Given any Hermitian metric h on a complex manifold, we can express the fundamental 2-form Ω in local holomorphic coordinates as follows:

$$\Omega = i \sum_{j,k} h_{j\bar{k}} dz_j \wedge d\bar{z}_k, \quad \text{where } h_{j\bar{k}} := h\left(\frac{\partial}{\partial z_j}, \frac{\partial}{\partial \bar{z}_k}\right). \quad (20)$$

Kähler then noticed that the very restrictive condition $d\Omega = 0$ is equivalent to the local existence of some function u such that

$$h_{j\bar{k}} = \frac{\partial^2 u}{\partial z_j \partial \bar{z}_k}.$$

In other words, the whole metric tensor is defined by a unique function. Comparing to the existence of the large number of independent functions in generic Hermitian cases (n^2 functions for a $2n$ -dimensional manifold), this only one function in Kähler structures is very special. The Kähler condition turns a complicated tensor into something generated by a single scalar potential. This remarkable property of the metric allows one to obtain simple explicit expressions for the Christoffel symbols and the Ricci and curvature tensors, and "a long list of miracles occur then".

Also, on a complex manifold with Hermitian metric, the existence of a specialized holomorphic coordinate system around each point, named normal holomorphic coordinates, is equivalent to the metric being Kähler. Kähler manifolds have found many applications in various domains like Differential Geometry, Complex Analysis, Algebraic Geometry or Theoretical Physics [56].

Now, we will continue with some well known facts regarding Kähler manifolds.

Definition 3.41. *A Hermitian metric h on a complex manifold (M, J) is called a Kähler metric if the associated 2-form Ω is closed, i.e. $d\Omega = 0$. The associated form Ω is the fundamental form of M and it is given by Equation (20). A local Kähler potential of the metric h is a local real function u that satisfies $\Omega = i\partial\bar{\partial}u$, where*

$$i\partial\bar{\partial}u := i \sum_{j,k} \frac{\partial^2 u}{\partial z_j \partial \bar{z}_k} dz_j d\bar{z}_k.$$

We are ready to give the definition of Kähler manifolds. There are three equivalent definitions, each of a different aspect.

Definition 3.42.

1. **Symplectic viewpoint.** *A Kähler manifold is a symplectic manifold (M, ω) with a complex structure J which is compatible with the symplectic form ω . Equivalently, the bilinear form $g(u, v) = \omega(u, Jv)$ is a Riemannian metric on M [33]. This is the aftermath of Definition 3.41.*
2. **Complex viewpoint.** *A Kähler manifold is a complex manifold M with an Hermitian metric h and a closed 2-form ω such that $\omega(u, v) := \operatorname{Re} h(iu, v) = \operatorname{Im} h(u, v)$, for tangent vectors u, v on every point of M [20].*
3. **Riemannian viewpoint.** *A Kähler manifold is a Riemannian manifold M of even dimension $2n$ whose holonomy group is contained in the unitary group $U(n)$ [44].*

Proposition 3.43. *On a Kähler manifold M we have the following structures, compatible with each other:*

1. *A complex structure J , i.e. $J^2 = -\text{id}$ on the tangent bundle.*
2. *A symplectic (real) structure ω , i.e. a non degenerate closed 2-form.*
3. *A Riemannian metric g , compatible with the symplectic form and the complex structure, $g(X, Y) = \omega(X, JY)$.*
4. *A Hermitian metric h such that $h = g - i\omega$, i.e. the real and imaginary parts of h are g and $-\omega$.*

On the following subsection we will supply the complex projective space $\mathbb{P}^n(\mathbb{C})$ with a Kähler metric, the *Fubini–Study* metric. We will discuss this in detail in Subsection 3.3.4, since it will play the role of a very important tensor in Quantum Geometry, the so-called *Quantum Geometric Tensor*.

3.3.3 The Fubini–Study metric on $\mathbb{P}(\mathcal{H})$

As it was remarked in Section 2.1, the space of qubits is identified with the projective space $\mathbb{P}(\mathcal{H}) = \mathbb{P}^{N-1}(\mathbb{C})$, that is the space of complex rays or lines in $\mathcal{H} = \mathbb{C}^2 \otimes \cdots \otimes \mathbb{C}^2 = (\mathbb{C}^2)^{\otimes n} = \mathbb{C}^{2^n}$, where $N = 2^n$. From the Example 3.33, we have that the $\mathbb{P}^{N-1}(\mathbb{C})$ is a complex manifold. In order to prove that $\mathbb{P}^{N-1}(\mathbb{C})$ is also a Kähler manifold, we need a Hermitian metric and a closed 2-form as in Definition 3.42. We first construct the needed form [20, 56].

We remind that we are working with the standard Hermitian product (see Example 1.13).

Proposition 3.44. *We can define a 2-form Ω on $\mathbb{P}^n(\mathbb{C})$ that satisfies*

$$\pi^*(\Omega(\mathbf{z})) = i\partial\bar{\partial}(\log |\mathbf{z}|^2), \quad (21)$$

where $\pi : \mathbb{C}^{n+1} \setminus \{0\} \rightarrow \mathbb{P}^n(\mathbb{C})$ is the canonical projection.

Proof. Consider the holomorphic atlas $\{(U_j, \phi_j)\}$ on $\mathbb{P}^n(\mathbb{C})$, as in Example 3.33. where

$$U_j = \{\mathbf{z} = [z_0 : \cdots : z_n] \mid z_j \neq 0\} \quad \text{and} \quad \phi_j[z_0 : \cdots : z_n] = \left(\frac{z_0}{z_j}, \dots, \frac{z_{j-1}}{z_j}, \frac{z_{j+1}}{z_j}, \dots, \frac{z_n}{z_j} \right) =: \mathbf{w}.$$

Let $\pi : \mathbb{C}^{n+1} \setminus \{0\} \rightarrow \mathbb{P}^n(\mathbb{C})$ be the canonical projection $\pi(z_0, \dots, z_n) = [z_0 : \cdots : z_n]$. This map is surjective. It is moreover a principal $\mathbb{C}^*$ -fibration, with local trivializations $\psi_j : \pi^{-1}U_j \rightarrow U_j \times \mathbb{C}^*$ given by

$$\psi_j(\mathbf{z}) = ([\mathbf{z}], z_j),$$

and satisfying $\psi_j \circ \psi_k^{-1}([\mathbf{z}], \alpha) = ([\mathbf{z}], \frac{z_j}{z_k} \alpha)$, $\forall \alpha \in \mathbb{C}^*$. Consider the functions

$$\begin{aligned} u : \mathbb{C}^n &\rightarrow \mathbb{R}, & u(\mathbf{w}) &= \log(1 + |\mathbf{w}|^2), \\ v : \mathbb{C}^{n+1} \setminus \{0\} &\rightarrow \mathbb{R}, & v(\mathbf{z}) &= \log(|\mathbf{z}|^2). \end{aligned}$$

For every $j \in \{0, \dots, n\}$, we define $f_j = \phi_j \circ \pi$. Hence, for $\mathbf{z} = (z_0, \dots, z_n)$, it is

$$f_j(\mathbf{z}) = f_j(z_0, \dots, z_n) = \phi_j \circ \pi(z_0, \dots, z_n) = \phi_j([z_0 : \dots : z_n]) = \left(\frac{z_0}{z_j}, \dots, \frac{z_{j-1}}{z_j}, \frac{z_{j+1}}{z_j}, \dots, \frac{z_n}{z_j} \right),$$

and the following diagram is commutative:

$$\begin{array}{ccc} \mathbb{C}^{n+1} \setminus \{0\} & \xrightarrow{\pi} & U_j \subset \mathbb{P}^n(\mathbb{C}) \\ \downarrow f_j & \swarrow \phi_j & \\ \mathbb{C}^n & & \end{array}$$

The map f_j is holomorphic as a quotient with holomorphic numerator and nonvanishing denominator. Also, we have

$$\begin{aligned} u \circ f_j(\mathbf{z}) &= u\left(\frac{z_0}{z_j}, \dots, \frac{z_{j-1}}{z_j}, \frac{z_{j+1}}{z_j}, \dots, \frac{z_n}{z_j}\right) = \log\left(1 + \left(\left(\frac{z_0}{z_j}\right)^2 + \dots + \left(\frac{z_n}{z_j}\right)^2\right)\right) \\ &= \log\left(1 + \frac{|\mathbf{z}|^2}{z_j^2} - \frac{z_j^2}{z_j^2}\right) = \log|\mathbf{z}|^2 - \log|z_j|^2 = v(\mathbf{z}) - \log|z_j|^2. \end{aligned}$$

It is $\partial\bar{\partial}\log|z_j|^2 = \partial\bar{\partial}(\log z_j + \log \bar{z}_j) = \partial(\bar{\partial}\log z_j + \bar{\partial}\log \bar{z}_j) = \partial\left(0 + \frac{1}{\bar{z}_j}\right) = \frac{\partial\left(\frac{1}{\bar{z}_j}\right)}{\partial z_j} = 0$, which shows that

$$(f_j)^*(\partial\bar{\partial}u) = \partial\bar{\partial}(u \circ f_j) = \partial\bar{\partial}v - \partial\bar{\partial}\log|z_j|^2 = \partial\bar{\partial}v,$$

which means that $\pi^* \circ \phi_j^*(\partial\bar{\partial}u) = f_j^*(\partial\bar{\partial}u) = \partial\bar{\partial}v$ on $\pi^{-1}(U_j)$. We thus can define a 2-form Ω on $\mathbb{P}^n(\mathbb{C})$ by $\Omega|_{U_j} := i(\phi_j)^*(\partial\bar{\partial}u)$, which satisfies

$$\pi^*(\Omega) = i \pi^*((\phi_j)^*(\partial\bar{\partial}u)) = i\partial\bar{\partial}v.$$

□

Lemma 3.45. *The form Ω is closed, real, J -invariant (where the complex structure $J = i$) and antisymmetric $(1, 1)$ -form.*

Proof. Ω is a $(1, 1)$ -form, since $\partial\bar{\partial}u = \sum_{j,k} \frac{\partial^2 u}{\partial z_j \partial \bar{z}_k} dz_j \wedge d\bar{z}_k$. Also, Ω is closed, because it is $d\Omega = i d(\partial\bar{\partial}u) = 0$. Moreover,

$$\bar{\Omega} = \overline{i \partial\bar{\partial}u} = -i \overline{\partial\bar{\partial}u} = -i(-\partial\bar{\partial}u) = i\partial\bar{\partial}u = \Omega,$$

thus, Ω is also real form. For the complex structure i we have

$$\begin{aligned} \Omega(iX, iY) &= i \sum_{j,k} \frac{\partial^2 u}{\partial z_j \partial \bar{z}_k} (dz_j \wedge d\bar{z}_k)(iX, iY) = i \sum_{j,k} \frac{\partial^2 u}{\partial z_j \partial \bar{z}_k} \left(dz_j(iX)d\bar{z}_k(iY) - dz_j(iY)d\bar{z}_k(iX) \right) \\ &= i \sum_{j,k} \frac{\partial^2 u}{\partial z_j \partial \bar{z}_k} \left(i dz_j(X)(-i d\bar{z}_k(Y)) - i dz_j(Y)(-i d\bar{z}_k(X)) \right) \\ &= i \sum_{j,k} \frac{\partial^2 u}{\partial z_j \partial \bar{z}_k} \left(-i^2 dz_j(X)d\bar{z}_k(Y) + i^2 dz_j(Y)d\bar{z}_k(X) \right) \\ &= i \sum_{j,k} \frac{\partial^2 u}{\partial z_j \partial \bar{z}_k} \left(dz_j(X)d\bar{z}_k(Y) - dz_j(Y)d\bar{z}_k(X) \right) \\ &= i \sum_{j,k} \frac{\partial^2 u}{\partial z_j \partial \bar{z}_k} (dz_j \wedge d\bar{z}_k)(X, Y) = \Omega(X, Y). \end{aligned}$$

Hence, Ω is invariant under the action of i . Finally, Ω is antisymmetric, since

$$\begin{aligned} \Omega(Y, X) &= i \sum_{j,k} \frac{\partial^2 u}{\partial z_j \partial \bar{z}_k} (dz_j \wedge d\bar{z}_k)(Y, X) = i \sum_{j,k} \frac{\partial^2 u}{\partial z_j \partial \bar{z}_k} \left(dz_j(Y)d\bar{z}_k(X) - dz_j(X)d\bar{z}_k(Y) \right) \\ &= -i \sum_{j,k} \frac{\partial^2 u}{\partial z_j \partial \bar{z}_k} \left(dz_j(X)d\bar{z}_k(Y) - dz_j(Y)d\bar{z}_k(X) \right) = -i \sum_{j,k} \frac{\partial^2 u}{\partial z_j \partial \bar{z}_k} (dz_j \wedge d\bar{z}_k)(X, Y) \\ &= -\Omega(X, Y). \end{aligned}$$

□

From Ω , we define the tensor $h : T\mathbb{P}^n(\mathbb{C}) \times T\mathbb{P}^n(\mathbb{C}) \rightarrow \mathbb{R}$, $(X, Y) \mapsto h(X, Y) := \Omega(X, iY)$. Let the vector fields $X = \sum_{j=1}^n \left(X_j \frac{\partial}{\partial z_j} + \bar{X}_j \frac{\partial}{\partial \bar{z}_j} \right)$ and $Y = \sum_{j=1}^n \left(Y_j \frac{\partial}{\partial z_j} + \bar{Y}_j \frac{\partial}{\partial \bar{z}_j} \right)$. Then,

$$\begin{aligned} h(X, Y) &= \Omega(X, iY) = i \sum_{j,k} \frac{\partial^2 u}{\partial z_j \partial \bar{z}_k} (dz_j \wedge d\bar{z}_k)(X, iY) \\ &= i \sum_{j,k} \frac{\partial^2 u}{\partial z_j \partial \bar{z}_k} \left(dz_j(X)d\bar{z}_k(iY) - dz_j(iY)d\bar{z}_k(X) \right) \\ &= i \sum_{j,k} \frac{\partial^2 u}{\partial z_j \partial \bar{z}_k} \left(-i X_j \bar{Y}_k - i Y_j \bar{X}_k \right) = \sum_{j,k} \frac{\partial^2 u}{\partial z_j \partial \bar{z}_k} \left(X_j \bar{Y}_k + Y_j \bar{X}_k \right). \end{aligned}$$

Lemma 3.46. *The tensor h is symmetric and i -invariant.*

Proof. It is

$$h(Y, X) = \Omega(Y, iX) = \Omega(iY, i^2X) = \Omega(iY, -X) = -\Omega(-X, iY) = \Omega(X, iY) = h(X, Y),$$

so, h is symmetric. Also, $h(iX, iY) = \Omega(iX, i^2Y) = \Omega(X, iY) = h(X, Y)$, that is, h is i -invariant. $\square$

The next lemma proves that h defines a Kähler metric on $\mathbb{P}^n(\mathbb{C})$.

Proposition 3.47. *The tensor h is positive definite on $\mathbb{P}^n(\mathbb{C})$.*

Proof. We will work on $\mathbb{C}^n$ instead of $\mathbb{P}^n(\mathbb{C})$, as it is easier to do so. We first fix some local holomorphic chart $\phi_j : U_j \rightarrow \mathbb{C}^n$, by the standard affine coordinate chart of projective space:

$$U_j = \{\mathbf{z} = [z_0 : \cdots : z_n] \mid z_j \neq 0\} \quad \text{and} \quad \phi_j[z_0 : \cdots : z_n] = \left(\frac{z_0}{z_j}, \dots, \frac{z_{j-1}}{z_j}, \frac{z_{j+1}}{z_j}, \dots, \frac{z_n}{z_j} \right).$$

We define the tensor $\hat{h} : T\mathbb{C}^n \times T\mathbb{C}^n \rightarrow \mathbb{R}$, $\hat{h}(X, Y) := i\partial\bar{\partial}u(X, iY)$, $\forall X, Y \in T\mathbb{C}^n$. Clearly, $\hat{h}$ is a symmetric tensor, since

$$\begin{aligned} \hat{h}(Y, X) &= i\partial\bar{\partial}u(Y, iX) = -i\partial\bar{\partial}u(iX, Y) = -i\partial\bar{\partial}u(i^2X, iY) = -i\partial\bar{\partial}u(-X, iY) \\ &= i\partial\bar{\partial}u(X, iY) = \hat{h}(X, Y). \end{aligned}$$

It is naturally $h = (\phi_j)^*(\hat{h})$. We will prove that $\hat{h}$ is positive definite, so we will have the expected result. Now, $U(n)$ is a group of holomorphic transformations of $\mathbb{C}^n$ that preserves u , since for $A \in U(n)$,

$$u(A\mathbf{z}) = \log(1 + |A\mathbf{z}|^2) = \log(1 + AA^\dagger|\mathbf{z}|^2) = \log(1 + |\mathbf{z}|^2) = u(\mathbf{z}),$$

and also

$$A^*(\partial\bar{\partial}u) = \partial\bar{\partial}(u \circ A) = \partial\bar{\partial}u.$$

Hence, $U(n)$ preserves $\hat{h}$:

$$\begin{aligned} (A^*\hat{h})_{\mathbf{z}}(X, Y) &= \hat{h}_{A\mathbf{z}}(A_*X, A_*Y) = i\partial\bar{\partial}u(A_*X, iA_*Y) = i\partial\bar{\partial}u(A_*X, A_*(iY)) \\ &= i\partial\bar{\partial}u(X, iY) = \hat{h}_{\mathbf{z}}(X, Y). \end{aligned}$$

Moreover, $U(n)$ is acting transitively on the unit sphere of $\mathbb{C}^n$. So it is enough to prove that $\hat{h}$ is positive definite at a point $p = (p_1, 0, \dots, 0) \in \mathbb{C}^n$, for some positive real number p_1 . We have

$$\begin{aligned}
 \partial\bar{\partial}u(\mathbf{z}) &= \partial\bar{\partial}\log(1 + |\mathbf{z}|^2) = \partial\left(\frac{\sum_{i=1}^n z_i d\bar{z}_i}{1 + |\mathbf{z}|^2}\right) \\
 &= \frac{(1 + |\mathbf{z}|^2) \sum_{i=1}^n dz_i \wedge d\bar{z}_i - \left(\sum_{j=1}^n \bar{z}_j dz_j\right) \wedge \left(\sum_{i=1}^n z_i d\bar{z}_i\right)}{(1 + |\mathbf{z}|^2)^2} \\
 &= \sum_{i,j=1}^n \frac{(1 + |\mathbf{z}|^2)\delta_{ij} - \bar{z}_j z_i}{(1 + |\mathbf{z}|^2)^2} dz_j \wedge d\bar{z}_i.
 \end{aligned}$$

Hence, at p ,

$$\hat{h}_p = i\partial\bar{\partial}u|_p = \frac{i}{(1 + p_1^2)^2} \left(dz_1 \wedge d\bar{z}_1 + (1 + p_1^2) \sum_{i=2}^n dz_i \wedge d\bar{z}_i \right).$$

Let the vector fields $X = \sum_{i=1}^n \left(X_i \frac{\partial}{\partial z_i} + \bar{X}_i \frac{\partial}{\partial \bar{z}_i} \right)$ and $Y = \sum_{i=1}^n \left(Y_i \frac{\partial}{\partial z_i} + \bar{Y}_i \frac{\partial}{\partial \bar{z}_i} \right)$. It is

$$\begin{aligned}
 i(dz_i \wedge d\bar{z}_i)(X, iY) &= i[dz_i(X)d\bar{z}_i(iY) - dz_i(iY)d\bar{z}_i(X)] \\
 &= i(-iX_i \bar{Y}_i - iY_i \bar{X}_i) = X_i \bar{Y}_i + Y_i \bar{X}_i = 2\operatorname{Re}(X_i \bar{Y}_i).
 \end{aligned}$$

Therefore, $\hat{h}$ is positive definite, since

$$\begin{aligned}
 \hat{h}_p(X, Y) &= \frac{1}{(1 + p_1^2)^2} \left(2\operatorname{Re}(X_1 \bar{Y}_1) + (1 + p_1^2) \sum_{i=2}^n 2\operatorname{Re}(X_i \bar{Y}_i) \right) \\
 &= \frac{2}{(1 + p_1^2)^2} \operatorname{Re} \left(X_1 \bar{Y}_1 + (1 + p_1^2) \sum_{i=2}^n X_i \bar{Y}_i \right) > 0.
 \end{aligned}$$

□

Lemma 3.48. *The tensor h is a Kähler metric on $\mathbb{P}^n(\mathbb{C})$.*

Proof. The tensor h is symmetric (Lemma 3.46) and positive definite (Lemma 3.47), so h is Riemannian metric. Also, h is i -invariant (3.46), hence, h is Hermitian metric. Since i is integrable and the associated form Ω is closed, we conclude that h is a Kähler metric. □

The Kähler metric h on $\mathbb{P}^n(\mathbb{C})$ constructed in the above way is called *Fubini-Study*. It is the natural Kähler metric of the projective space, and also, it is very important for Quantum Geometry. On the next subsection we will see that it is strongly connected to the so-called *Quantum Geometric Tensor*.

Definition 3.49. The Fubini-Study metric is defined as the following 2-tensor, for a fixed holomorphic frame on the tangent space of $\mathbb{P}^n(\mathbb{C})$:

$$ds^2 := h_{i\bar{j}} dz_i d\bar{z}_j = \frac{(1 + z_i \bar{z}_i) dz_j d\bar{z}_j - \bar{z}_j z_i dz_j d\bar{z}_i}{(1 + z_i \bar{z}_i)^2}, \quad (22)$$

where, as customary, we sum over repeated indices. The matrix in this frame is explicitly given by:

$$(h_{i\bar{j}}) = \frac{1}{(1 + |\mathbf{z}|^2)^2} \begin{pmatrix} 1 + |\mathbf{z}|^2 - |z_1|^2 & -\bar{z}_1 z_2 & \cdots & -\bar{z}_1 z_n \\ -\bar{z}_2 z_1 & 1 + |\mathbf{z}|^2 - |z_2|^2 & \cdots & -\bar{z}_2 z_n \\ \vdots & \vdots & \ddots & \vdots \\ -\bar{z}_n z_1 & -\bar{z}_n z_2 & \cdots & 1 + |\mathbf{z}|^2 - |z_n|^2 \end{pmatrix}, \quad (23)$$

where $|\mathbf{z}|^2 = \sum_{i=1}^n |z_i|^2$.

Lemma 3.50. The Kähler potential of the Fubini-Study metric is $K = \log(1 + |\mathbf{z}|^2)$.

3.3.4 Quantum Geometric Tensor

Now, we are ready for the key definition of Quantum Geometric Tensor [11, 32].

Definition 3.51. The Quantum Geometric Tensor (QGT) on $\mathbb{P}(\mathcal{H})$ is defined as the Kähler metric:

$$Q_{\mu\nu} = \frac{\langle \partial_\mu \psi | \partial_\nu \psi \rangle}{\langle \psi | \psi \rangle} - \frac{\langle \partial_\mu \psi | \psi \rangle \langle \psi | \partial_\nu \psi \rangle}{\langle \psi | \psi \rangle^2}. \quad (24)$$

Remark 3.52. We call rays in $\mathcal{H}$ the complex lines that pass from the origin of $\mathcal{H}$, i.e. the elements of $\mathbb{P}(\mathcal{H})$.

Observation 3.53. For QGT we have that $Q_{\mu\nu}^* = Q_{\nu\mu}$, that is, QGT is Hermitian. Now, [11]

$$Q_{\mu\nu} = \text{Re}(Q_{\mu\nu}) + i \text{Im}(Q_{\mu\nu}),$$

where $\text{Re}(Q_{\mu\nu}) = \frac{1}{2}(Q_{\mu\nu} + Q_{\nu\mu})$ and $\text{Im}(Q_{\mu\nu}) = \frac{1}{2}(Q_{\mu\nu} - Q_{\nu\mu})$. It is $Q_{\mu\nu} = g_{\mu\nu} + \frac{i}{2}F_{\mu\nu}$.

The real and imaginary parts of QGT on $\mathbb{P}(\mathcal{H})$, i.e. in rays of $\mathcal{H}$, are the Fubini-Study metric (see Proposition 3.54) and the Berry curvature, respectively. We briefly give the elements constructing it, as we will not deal with it on this thesis. The Berry connection is defined as

$$A_j = i \langle \psi | \partial_j \psi \rangle$$

and the Berry curvature is the associated curvature 2-form $F = dA$. Therefore,

$$F_{j\bar{l}} = i \frac{\bar{w}_j w_l - (1 + \bar{w}_k w^k) \delta_{jl}}{(1 + \bar{w}_k w^k)^2} dw^j \wedge d\bar{w}^l = i \frac{\bar{w}_j w_l - (1 + |\mathbf{w}|^2) \delta_{jl}}{(1 + |\mathbf{w}|^2)^2} dw^j \wedge d\bar{w}^l = i h_{j\bar{l}} dw^j \wedge d\bar{w}^l,$$

where h is the Fubini-Study metric.

The following result gives the connection between QGT and Fubini-Study metric [31].

Proposition 3.54. *The QGT is equivalent to the Fubini-Study Hermitian metric.*

Proof. Since the restriction $|\psi\rangle \in \mathbb{S}(\mathcal{H})$ is relaxed in Definition 3.51, we have $\langle\psi|\psi\rangle \neq 0$ in general. It is

$$d\psi = \sum_i \partial_i \psi dw_i,$$

and from Equation (24) we have

$$ds^2 = \sum_{i,j} A_{i\bar{j}} d\bar{w}_i dw_j,$$

where

$$A_{i\bar{j}} = \frac{\langle\partial_i \psi|\partial_j \psi\rangle \langle\psi|\psi\rangle - \langle\partial_i \psi|\psi\rangle \langle\psi|\partial_j \psi\rangle}{\langle\psi|\psi\rangle^2}. \quad (25)$$

We choose an orthonormal basis of $\mathbb{C}^{n+1}$, under which the coordinates of $|\psi\rangle \in \mathbb{C}^{2^m}$, $n = 2^m - 1$ can be represented by a generic complex vector $(z_0, \dots, z_n)$, such that

$$\langle\psi|\psi\rangle = \left\langle \sum_{k=0}^n z_k e_k \left| \sum_{l=0}^n z_l e_l \right. \right\rangle = \sum_{k=0}^n \sum_{l=0}^n \bar{z}_k z_l \langle e_k | e_l \rangle = \sum_{k=0}^n \sum_{l=0}^n \bar{z}_k z_l \delta_{kl} = \sum_{k=0}^n \bar{z}_k z_k = \sum_{k=0}^n |z_k|^2.$$

Since $\mathbb{P}(\mathcal{H}) = \mathbb{P}^{2^m-1}(\mathbb{C}) = (\mathbb{C}^{2^m} \setminus \{0\})/\mathbb{C}^*$, every point in $\mathbb{P}(\mathcal{H})$ represents a class in $\mathbb{C}^{2^m} \setminus \{0\}$ with respect to any equivalence (gauge) transformation $c \in \mathbb{C}^*$. Explicitly, $|\psi\rangle \sim |\phi'\rangle$ if $|\psi\rangle = c|\phi'\rangle$ or $z_i = cz'_i$.

We take $z_0 \neq 0$ without loss of generality, and so we work on $U_0 = \{[z_0 : \dots : z_n] | z_0 \neq 0\}$. The coordinates of $[\psi] \in \mathbb{P}(\mathcal{H})$ are thereby obtained via the projection $f_0 = \phi_0 \circ \pi$, as it is defined in Lemma 3.44, with $w_i = \frac{z_i}{z_0}$, with $i = 1, \dots, n$, since z_0 is omitted from the expression of f_0 .

Hence, for fixed z_0 it is $z_i = z_0 w_i$ and by the chain rule we have $\frac{\partial}{\partial w_i} = z_0 \frac{\partial}{\partial z_i}$ and $|\psi\rangle = (z_0, z_1, \dots, z_n) = z_0(1, w_1, \dots, w_n)$. Thus,

$$|\partial_i \psi\rangle = z_0(0, \dots, 1, \dots, 0) = z_0 e_i,$$

that is, only the i -th component is nonzero. Therefore, we obtain

$$\begin{aligned} \langle\partial_i \psi|\partial_j \psi\rangle &= \left\langle \sum_{k=0}^n z_0 \delta_{ik} e_k \left| \sum_{l=0}^n z_0 \delta_{jl} e_l \right. \right\rangle = \sum_{k=0}^n \sum_{l=0}^n \overline{z_0 \delta_{ik}} z_0 \delta_{jl} \langle e_k | e_l \rangle \\ &= \sum_{k=0}^n \sum_{l=0}^n \bar{z}_0 z_0 \delta_{ik} \delta_{jl} \langle e_k | e_l \rangle = |z_0|^2 \sum_{k=0}^n \sum_{l=0}^n \delta_{ik} \delta_{jl} \delta_{kl} = |z_0|^2 \delta_{ij}, \end{aligned}$$

also,

$$\langle \partial_i \psi | \psi \rangle = \left\langle z_0 e_i \left| \sum_{k=0}^n z_k e_k \right. \right\rangle = \sum_{k=0}^n \bar{z}_0 z_k \langle e_i | e_k \rangle = \sum_{k=0}^n \bar{z}_0 \delta_{ik} z_k = \bar{z}_0 z_i,$$

and its complex conjugate

$$\langle \psi | \partial_i \psi \rangle = \overline{\langle \partial_i \psi | \psi \rangle} = \overline{\bar{z}_0 z_i} = z_0 \bar{z}_i.$$

Substituting them into Equation (25), we get

$$\begin{aligned} A_{i\bar{j}} &= \frac{(|z_0|^2 \delta_{ij} \sum_{k=0}^n |z_k|^2) - |z_0|^2 z_i \bar{z}_j}{(\sum_{k=0}^n |z_k|^2)^2} = \frac{(1 + \sum_{k=1}^n |w_k|^2) \delta_{ij} - w_i \bar{w}_j}{(1 + \sum_{k=1}^n |w_k|^2)^2} \\ &= \frac{(1 + |\mathbf{w}|^2) \delta_{ij} - w_i \bar{w}_j}{(1 + |\mathbf{w}|^2)^2}, \end{aligned}$$

thus, $A_{i\bar{j}}$ is exactly the $(i, \bar{j})$ -component of the Fubini-Study metric, Equation (23). $\square$

We now show that the QGT is strictly related to F_Q , the QFIM [21].

Proposition 3.55. *The real part of the Quantum Geometric Tensor is equivalent to the metric defined by the quantum Fisher information matrix F_Q on rays in $\mathcal{H}$, up to constant scale.*

Proof. We will work with normalized pure states. Then, the result also holds for rays on $\mathcal{H}$. Let $|\psi\rangle \in \mathcal{H}$ be a normalized pure state, and let $\rho = |\psi\rangle \langle \psi|$ be the associated density operator. Since $|\psi\rangle$ is normalized, it is $\langle \psi | \psi \rangle = 1$. Hence, $d\langle \psi | \psi \rangle = 0$, and deriving that, we take

$$\langle d\psi | \psi \rangle + \langle \psi | d\psi \rangle = 0 \quad (26)$$

and also it is

$$d\rho = |d\psi\rangle \langle \psi| + |\psi\rangle \langle d\psi|. \quad (27)$$

Now, it is from Equation (27),

$$\begin{aligned} (d\rho)^2 &= (|d\psi\rangle \langle \psi| + |\psi\rangle \langle d\psi|)^2 \\ &= |d\psi\rangle \langle \psi| d\psi \rangle \langle \psi| + |d\psi\rangle \langle \psi | \psi \rangle \langle d\psi| + |\psi\rangle \langle d\psi | d\psi \rangle \langle \psi| + |\psi\rangle \langle d\psi | \psi \rangle \langle d\psi| \\ &= \langle \psi | d\psi \rangle |d\psi\rangle \langle \psi| + |d\psi\rangle \langle d\psi| + \langle d\psi | d\psi \rangle |\psi\rangle \langle \psi| + \langle d\psi | \psi \rangle |\psi\rangle \langle d\psi|. \end{aligned}$$

Thus,

$$\begin{aligned} \rho(d\rho)^2 &= |\psi\rangle \langle \psi| (\langle \psi | d\psi \rangle |d\psi\rangle \langle \psi| + |d\psi\rangle \langle d\psi| + \langle d\psi | d\psi \rangle |\psi\rangle \langle \psi| + \langle d\psi | \psi \rangle |\psi\rangle \langle d\psi|) \\ &= \langle \psi | d\psi \rangle |\psi\rangle \langle \psi | d\psi \rangle \langle \psi| + |\psi\rangle \langle \psi | d\psi \rangle \langle d\psi| + \langle d\psi | d\psi \rangle |\psi\rangle \langle \psi | \psi \rangle \langle \psi| + \langle d\psi | \psi \rangle |\psi\rangle \langle \psi | \psi \rangle \langle d\psi| \\ &= \langle \psi | d\psi \rangle^2 |\psi\rangle \langle \psi| + |\psi\rangle \langle \psi | d\psi \rangle \langle d\psi| + \langle d\psi | d\psi \rangle |\psi\rangle \langle \psi| + \langle d\psi | \psi \rangle |\psi\rangle \langle d\psi| \\ &= (\langle \psi | d\psi \rangle^2 + \langle d\psi | d\psi \rangle) |\psi\rangle \langle \psi| + (\langle \psi | d\psi \rangle + \langle d\psi | \psi \rangle) |\psi\rangle \langle d\psi| \\ &\stackrel{Eq.(26)}{=} (\langle d\psi | d\psi \rangle + \langle \psi | d\psi \rangle^2) \rho. \end{aligned} \quad (28)$$

From Equation (26) and the fact that $\langle \psi | d\psi \rangle = \overline{\langle d\psi | \psi \rangle}$, we have that $\overline{\langle \psi | d\psi \rangle} = -\langle \psi | d\psi \rangle$. Hence, $\langle \psi | d\psi \rangle$ is purely imaginary, that is, we can write $\langle \psi | d\psi \rangle^2 = -|\langle \psi | d\psi \rangle|^2$. Therefore, Equation (28) becomes

$$\rho(d\rho)^2 = (\langle d\psi | d\psi \rangle - |\langle \psi | d\psi \rangle|^2) \rho.$$

Using Proposition 3.13, for normalized pure states it is:

$$F_Q = 4 \operatorname{Tr}(\rho(d\rho)^2) = 4 (\langle d\psi | d\psi \rangle - |\langle \psi | d\psi \rangle|^2) \operatorname{Tr}(\rho) = 4 (\langle d\psi | d\psi \rangle - \langle \psi | d\psi \rangle \langle d\psi | \psi \rangle).$$

The right part corresponds to the associated quadratic form $\mathcal{Q}_{i\bar{j}} dz^i d\bar{z}^j$, which is real due to Hermiticity. Therefore, $F_Q = 4 \operatorname{Re}(Q)$. $\square$

In other words, the imaginary part of QGT makes no contribution to the quantum distance.

Remark 3.56. *For mixed states, the real part is the Bures metric [31]:*

$$g_{\mu\nu}^B = \frac{1}{2} \sum_{i,j} \frac{\langle i | \partial_\mu \rho | j \rangle \langle j | \partial_\nu \rho | i \rangle}{\lambda_i + \lambda_j}$$

and the current relation is $F_Q = 4g_{Bures}$.

Concluding this chapter, we finish dealing with the deeper geometric structure of quantum theory. The Kähler geometry of projective Hilbert spaces unifies the geometric framework for the metric, symplectic and complex structures underlying quantum theory. We used it to determine important notions in quantum computing, such as the distance between quantum states, their curvature, geometric phase and probabilistic structure of the states. In the last chapter of this thesis, we turn to the ZX -calculus, a diagrammatic language to encode quantum processes through interacting algebraic structures. The geometric formulation of Chapter 3 emphasizes the differential and symplectic aspects of quantum theory, while the ZX -calculus captures quantum behavior through algebraic relations between the elements.

Chapter 4

ZX-calculus and Hopf algebras

In Section 2.3 we introduced the quantum logic gates. They are used for *quantum circuits*, which consist of a sequence of quantum logic gates and measurements.

Example 4.1. Suppose the following quantum circuit.

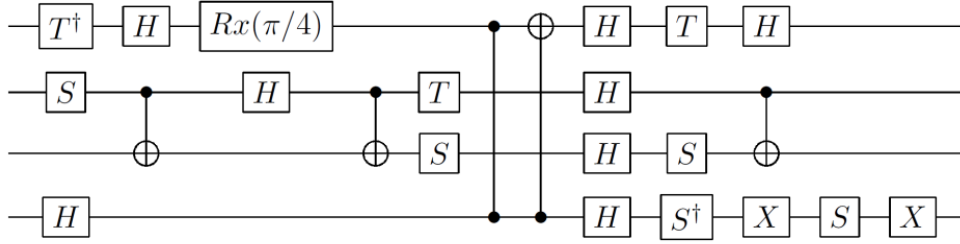


Figure 3: A quantum circuit

The lines denote the wires of the circuit, and $\square$, $\bullet$, $\oplus$, denote the quantum logic gates that take place, that is, the operations, as we saw them in Section 2.3.

The circuit consists of four lines, hence, we are in a 4-qubit system. We consider that the qubits are ordered as $|q_1q_2q_3q_4\rangle$, where $|q_1\rangle$ is the qubit of the upper line, $|q_2\rangle$ is the qubit of the second line, etc. We see the evolution of the algorithm, reading the picture from left to right. When the circuit begins, at time $t = 0$, we are at the left side of the picture. Hence, the initial 4-qubit is entered from the wires on the left; $|q_1\rangle$ enters from the upper left wire, below enters the $|q_2\rangle$, then the $|q_3\rangle$ and finally, the $|q_4\rangle$ enters from the bottom left wire.

The default value for a quantum register depends on the hardware. In most of them, including IBM quantum hardware, the initial value is typically set to the ground state, $|0\rangle$. It is a standard assumption across most hardware platforms and simulation frameworks, that all qubits in an n -qubit register start in the state $|0\rangle^{\otimes n}$.

Hence, in the example we consider the initial value to be $|\psi_0\rangle = |0000\rangle \in \mathbb{C}^{16}$.

We now see which are the gates of the picture. We denote as X, Y, Z the Pauli matrices

gates and H the Hadamard gate as in Equation (12). The $R_x(\pi/4)$ represents a rotation of $\pi/4$ around the x -axis of the Bloch sphere. Also, the phase gates S and T represent rotations around the z -axis by $\pi/2$ and $\pi/4$, respectively, up to a global phase: $S \sim R_z(\pi/2)$, $T \sim R_z(\pi/4)$. It is

$$S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} \quad \text{and} \quad T = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}.$$

The $\dagger$ as usual indicates the adjoint. Hence,

$$S^\dagger = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}^\dagger = \begin{pmatrix} 1 & 0 \\ 0 & -i \end{pmatrix}, \quad T^\dagger = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}^\dagger = \begin{pmatrix} 1 & 0 \\ 0 & e^{-i\pi/4} \end{pmatrix}.$$

The S and T gates are related to each other by the relationship $S = T^2$. We mention here that the reason we use matrices in $\mathbb{C}^2$ to represent gates is because the qubit lives in $\mathbb{C}^2$. The Bloch sphere is a very effective tool to show the evolution of a qubit, but it loses fundamental information of its actual state, such as the global phase. In particular, the Bloch sphere represents physical pure states modulo global phase.

The gate represented by $\bullet \oplus$ is the CNOT gate. Moreover, $\bullet \bullet$ represents the Controlled-Z (CZ) gate. The CZ gate is given by the following matrix:

$$U_{\text{CZ}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

Intuitively, the CZ gate applies a phase flip only to the computational basis state $|11\rangle$. Equivalently, it multiplies the state by -1 when both qubits are in the state $|1\rangle$, and leaves the other computational basis states unchanged: $\text{id} \otimes |0\rangle \langle 0| + Z |1\rangle \langle 1|$.

While the circuit works, we have the following changes for the 4-qubit:

$$\begin{aligned} |0000\rangle &\rightarrow (T^\dagger \otimes S \otimes \text{id} \otimes H) |0000\rangle \\ &\rightarrow \text{CNOT}_{2 \rightarrow 3}(H \otimes \text{id} \otimes \text{id} \otimes \text{id})(T^\dagger \otimes S \otimes \text{id} \otimes H) |0000\rangle \\ &\rightarrow (R_x(\pi/4) \otimes H \otimes \text{id} \otimes \text{id})\text{CNOT}_{2 \rightarrow 3}(H \otimes \text{id} \otimes \text{id} \otimes \text{id})(T^\dagger \otimes S \otimes \text{id} \otimes H) |0000\rangle \\ &\rightarrow \text{CNOT}_{2 \rightarrow 3}(R_x(\pi/4) \otimes H \otimes \text{id} \otimes \text{id})\text{CNOT}_{2 \rightarrow 3}(H \otimes \text{id} \otimes \text{id} \otimes \text{id})(T^\dagger \otimes S \otimes \text{id} \otimes H) |0000\rangle \\ &\rightarrow \dots \rightarrow \\ &\rightarrow (\text{id} \otimes \text{id} \otimes \text{id} \otimes X)(\text{id} \otimes \text{id} \otimes \text{id} \otimes S)\text{CNOT}_{2 \rightarrow 3}(H \otimes \text{id} \otimes \text{id} \otimes X)(T \otimes \text{id} \otimes S \otimes S^\dagger) \\ &\quad (H \otimes H \otimes H \otimes H)\text{CNOT}_{4 \rightarrow 1}\text{CZ}_{1 \rightarrow 4}(\text{id} \otimes T \otimes S \otimes \text{id})\text{CNOT}_{2 \rightarrow 3} \\ &\quad (R_x(\pi/4) \otimes H \otimes \text{id} \otimes \text{id})\text{CNOT}_{2 \rightarrow 3}(H \otimes \text{id} \otimes \text{id} \otimes \text{id})(T^\dagger \otimes S \otimes \text{id} \otimes H) |0000\rangle, \end{aligned}$$

which is very complicated to compute. To be mathematically accurate, one should construct the particular 16×16 matrices that correspond to the operations on the 4-qubit states.

The previous example shows why it is very useful to depict the quantum circuits as in Figure 3 instead of using mathematical operations. Still, the issue of an efficient way to do the operations remains.

Thus, in this chapter we show the fundamental principles of ZX -calculus, a graphical language originally developed by Bob Coecke and Ross Duncan in 2008, in [13]. They introduced the fundamental concepts of spiders, strong complementarity and most of the standard rewrite rules. We present how we can effectively use ZX -calculus to understand and optimize simple quantum circuits. Our main references are [14, 54, 68]. Moreover, we explore the connection between the language of ZX -calculus and quantum groups, as it is expressed in [54]. For the representation of the ZX -diagrams on LaTeX we are thankful that we were guided by the detailed work of Léo Colisson [60].

In Sections 4.1, 4.2, we present a graphical calculus, specific to the Z and X -spin observables, which is a special case of the general theory of observables (theory that defines the physical properties -position, momentum, spin, energy- as Hermitian operators that act on a Hilbert space of states) [14].

As well as demonstrating the main features of the entire theory, this simplified calculus is sufficiently powerful in order to carry out many calculations, that are useful in the quantum computation context, as the examples in ([14], Sec. 3).

This framework refers exclusively to the mathematics that underlie quantum computation and not to any details of how the operations are implemented. This is ideal for unifying various approaches to quantum computation. For example, we can demonstrate that two different quantum computational models are equivalent.

4.1 Quantum circuits and ZX -diagrams

ZX -calculus is a graphical language to describe linear maps $\mathcal{H}_n \rightarrow \mathcal{H}_m$ on qubits. It consists of components joined by wires, similar to electronic circuit diagrams or flow charts.

The input and output qubits on ZX -diagrams can be denoted by wires either from left to right or from top to bottom. These two diagram choices are equivalent and in this thesis we will use the left $\rightarrow$ right as input $\rightarrow$ output. So, the simplest non-trivial diagram in the language is simply a wire running from left to right (see Definition 4.2).

Definition 4.2. *The identity $\text{id}_{\mathcal{H}} : \mathcal{H} \rightarrow \mathcal{H}$, where $\mathcal{H} = \mathbb{C}^2$, is written on ZX -calculus as a single string:*

$$\text{id}_{\mathcal{H}} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \textit{in} \left[\begin{array}{c} \text{---} \end{array} \right] \textit{out}$$

We think of diagrams as being enclosed in a box as in Definition 4.2 with a certain number of points through which wires enter and leave; that is, each diagram has a fixed interface. Exactly one wire must be present at each point of the interface, and we must distinguish which wire is

connected to which point. Indeed, this is the only difference between the two diagrams below, in Definitions 4.3, 4.4:

Definition 4.3. The identity $\text{id}_{\mathcal{H}_2} : \mathcal{H}_2 \rightarrow \mathcal{H}_2$, where $\mathcal{H}_2 = \mathcal{H} \otimes \mathcal{H}$ corresponds to two strings flowing from left to right:

$$\text{id}_{\mathcal{H}_2} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \begin{array}{c} \text{in}_1 \\ \text{in}_2 \end{array} \boxed{\begin{array}{c} \text{---} \\ \text{---} \end{array}} \begin{array}{c} \text{out}_1 \\ \text{out}_2 \end{array}$$

Definition 4.4. The SWAP $\sigma : \mathcal{H}_2 \rightarrow \mathcal{H}_2$ is given by $\sigma(|\phi\rangle \otimes |\psi\rangle) = |\psi\rangle \otimes |\phi\rangle$. Equivalently,

$$\sigma_{\mathcal{H}_2} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \begin{array}{c} \text{in}_1 \\ \text{in}_2 \end{array} \boxed{\begin{array}{c} \text{---} \backslash \\ \text{---} / \end{array}} \begin{array}{c} \text{out}_1 \\ \text{out}_2 \end{array}$$

It is not important whether crossing wires pass over or under, i.e. we are in a symmetric setting, not a braided one [34]. Wires may bend, linking two outputs to form a $\subset$ or two inputs to form a $\supset$ like in the next example.

Example 4.5. The (unnormalized) Bell state is a diagram with 0 inputs and 2 outputs and it is represented by

$$|00\rangle + |11\rangle = \boxed{\begin{array}{c} \text{---} \text{---} \\ \text{---} \text{---} \end{array}} \begin{array}{c} \text{out}_1 \\ \text{out}_2 \end{array}$$

Its corresponding projection, or so-called Bell effect is the following, with 2 inputs and 0 outputs:

$$\langle 00| + \langle 11| = \begin{array}{c} \text{in}_1 \\ \text{in}_2 \end{array} \boxed{\begin{array}{c} \text{---} \text{---} \\ \text{---} \text{---} \end{array}}$$

From here on, the inputs and outputs will not be named, and they are distinguished simply by their ordering from up to down. We write $\mathfrak{D} : n \rightarrow m$ to indicate that the diagram $\mathfrak{D}$ has n inputs and m outputs.

Aside from wires, the ZX-calculus contains the following four kinds of components:

Lemma 4.8. *In $\mathcal{H}_1 \rightarrow \mathcal{H}_1$ (that is one input to one output), the green Z-spider with phase α represents the linear operator:*

$$Z_\alpha := \text{---} \textcircled{\alpha} \text{---} = |0\rangle \langle 0| + e^{i\alpha} |1\rangle \langle 1|.$$

This map is a rotation about the Z-axis of the Bloch sphere by an angle α . For $\alpha = \pi$, it is the Z Pauli matrix. Respectively,

$$X_\alpha := \text{---} \textcircled{\alpha} \text{---} = |+\rangle \langle +| + e^{i\alpha} |-\rangle \langle -|,$$

which is a rotation about the X-axis of the Bloch sphere by angle α . For $\alpha = \pi$, it is the X Pauli matrix.

We continue with some special cases of elementary 1-qubit operations:

Lemma 4.9. *The following generators do not have inputs. We see these cases as state vectors/ state preparations and not as operators:*

$$\textcircled{\alpha} \text{---} = |0\rangle + e^{i\alpha} |1\rangle,$$

and respectively

$$\textcircled{\alpha} \text{---} = |+\rangle + e^{i\alpha} |-\rangle.$$

Example 4.10. *For $\alpha = 0$, the circuits of Lemma 4.9 become:*

$$\textcircled{} \text{---} = |0\rangle + |1\rangle = \sqrt{2} |+\rangle,$$

$$\textcircled{} \text{---} = |+\rangle + |-\rangle = \sqrt{2} |0\rangle.$$

Lemma 4.11. *The following generators do not have outputs. They denote the projection effects:*

$$\text{---} \textcircled{\alpha} = \langle 0| + e^{i\alpha} \langle 1|,$$

$$\text{---} \textcircled{\alpha} = \langle +| + e^{i\alpha} \langle -|.$$

The above spiders act on the input qubit and project it onto the superposition state, producing a scalar.

Example 4.12. *Lemma 4.11 gives us for $\alpha = 0$:*

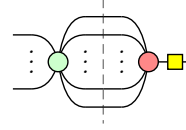
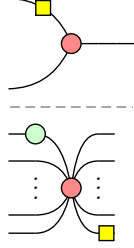
$$\text{---} \textcircled{} = \langle 0| + \langle 1| = \frac{1}{\sqrt{2}} \langle +|,$$

$$\text{---} \textcircled{} = \langle +| + \langle -| = \frac{1}{\sqrt{2}} \langle 0|.$$

So, for example, we can establish the correspondence between elementary operations on 1-qubits and ZX-diagrams as in Examples 4.10 and 4.12. We detail the rules involved in the identifications in the next section.

Definition 4.13. *Diagrams are built from these generators (straight, crossing, and bending wires, and Z, X, H, and $\diamond$ vertices) in two manners:*

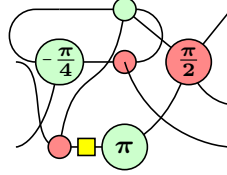
- *Placing them the one below the other:*
- *Connecting outputs to inputs:*



Notation: Given $\mathcal{D} : m \rightarrow n$ and $\mathcal{D}' : m' \rightarrow n'$, their tensor product is denoted $\mathcal{D} \otimes \mathcal{D}' : m + m' \rightarrow n + n'$.

Notation: Given $\mathcal{D}_1 : m \rightarrow n$ and $\mathcal{D}_2 : n \rightarrow k$ their composition is denoted $\mathcal{D}_2 \circ \mathcal{D}_1 : m \rightarrow k$.

The terms of graphical ZX language are networks of vertices of each type, straight, crossing, and bent wires:



In such a network, there can be no "loose" wires: every wire must terminate at a vertex, or else be an input or output.

The following are important examples of spiders' special cases.

Example 4.14. *These spiders are with 2 inputs and 1 output (binary operation), with no input and 1 output (initiation of a value) which we will call a point, with 1 input and 2 outputs (copying) and with 1 input and no output (erasing):*



Shortly, we will see that these unlabelled spiders play a special role in the calculus, as do those labelled e.g. by π .

Given a quantum circuit, we can immediately draw the corresponding ZX-diagram, using all the above. Vice-versa, given a ZX-diagram, one can write the quantum circuit associated with it. This means that the rules of the ZX-calculus are adequate to study quantum circuits

and that we can use them to study fundamental problems, such as quantum circuit optimization and quantum error correction codes.

4.2 The ZX-rules

There are some mathematically obvious rules regarding quantum circuits, that can be translated into operations on the spiders that model them. This originates the ZX-calculus, that we describe below. Such calculus allows to simplify a ZX-diagram and then translate such simpler version to a quantum circuit equivalent to the starting one. This allows for a more effective and efficient realization of simple quantum circuits.

In this section, we will deal with some of such operations, forming the backbone of ZX-calculus rules.

4.2.1 The T-rule or “only topology matters”

The *T-rule* postulates that the intuitive phrase “only the topology matters” suffices: the wires of the diagram may be arbitrarily straightened, stretched, bent, twisted, tied in knots, etc., without altering the meaning of the diagram, provided the connections are maintained.

Using the **T**-rule, the linear operation represented by the ZX-diagram doesn’t change. After identifying (e.g. by enumerating) the inputs and the outputs, any topological deformation of the internal structure of the network yields a network that is equal to the given one. The following are two important cases of such topological rewrites. The rest of them arise similarly.

Axiom 4.15 (**T** – rules - homotopic rewrites).

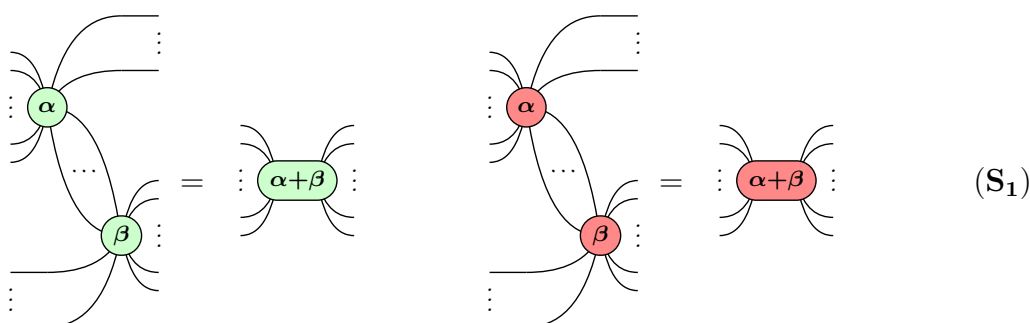
$$\begin{array}{c}
 \begin{array}{ccc}
 \begin{array}{c} \text{spider with green dot and wavy input} \end{array} & = & \begin{array}{c} \text{spider with green dot and straight input} \end{array} \\
 \begin{array}{ccc}
 \begin{array}{c} \text{spider with red dot and wavy input} \end{array} & = & \begin{array}{c} \text{spider with red dot and straight input} \end{array}
 \end{array}
 \end{array}
 \quad (\mathbf{T}_1)$$

$$\begin{array}{ccc}
 \begin{array}{c} \text{bent wire} \end{array} & = & \begin{array}{c} \text{straight wire} \end{array}
 \end{array}
 \quad (\mathbf{T}_2)$$

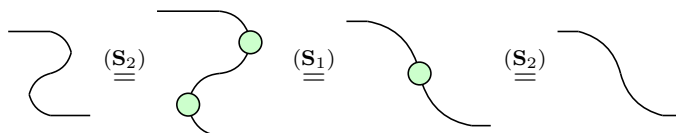
In fact, **T**₁ and **T**₂ rules can also be seen as consequences of the **S**-rules of Subsection 4.2.2, when introducing a green dot on diagrams as in **S**₂; see Example 4.21 below.

Remark 4.16. *Since wires can be stretched without consequence, if one adds a straight length of wire to the input or output of diagram, this has strictly no effect. Hence, bundles of straight wires act as identity elements in the algebra of diagrams.*

Remark 4.17. *While the slogan says “only the topology matters”, it does not imply that the topology is always preserved. The other rules (see 4.2.2, 4.2.3, and their following) may change the topology of the diagram in various ways, for example to remove loops, or to disconnect previously connected vertices.*



$$\text{---} \bigcirc \text{---} = \text{---} = \text{---} \bigcirc \text{---} \quad (\text{S}_2)$$



($\text{id}_{\mathcal{H}} \otimes \Delta_{\bullet}$) $\circ$ $\Delta_{\bullet} =$ $=$ $\stackrel{(\mathbf{s}_1)}{=}$

($\Delta_{\bullet} \otimes \text{id}_{\mathcal{H}}$) $\circ$ $\Delta_{\bullet} =$ $=$ $\stackrel{(\mathbf{s}_1)}{=}$

$$\begin{aligned} (\text{id}_{\mathcal{H}} \otimes \varepsilon_{\bullet}) \circ \Delta_{\bullet} &= \text{diagram} = \text{diagram} \stackrel{(\mathbf{S}_1)}{=} \text{diagram} \stackrel{(\mathbf{S}_2)}{=} \text{id}_{\mathcal{H}}, \\ (\varepsilon_{\bullet} \otimes \text{id}_{\mathcal{H}}) \circ \Delta_{\bullet} &= \text{diagram} = \text{diagram} \stackrel{(\mathbf{S}_1)}{=} \text{diagram} \stackrel{(\mathbf{S}_2)}{=} \text{id}_{\mathcal{H}}. \end{aligned}$$

4.2.3 The B-rules

As a consequence of the previous **S**-rules we have constructed an associative unital algebra structure $(m_{\bullet}, \eta_{\bullet})$ and a coassociative counital coalgebra structure $(\Delta_{\bullet}, \varepsilon_{\bullet})$ on $\mathcal{H}$ (see Propositions 4.22 and 4.23). The **B**-rules will relate such structures with each other to realize an unnormalized Hopf algebra (see Section 1.6). We will first state the rules, that are manifest, once the significance of diagrams is translated into quantum circuits.

Axiom 4.25 (**B**₁-rule - copy rule).

$$\diamond \quad \text{red dot} \text{---} \text{green dot} \text{---} \text{split} = \text{red dot} \text{---} \text{red dot} \quad \diamond \quad \text{split} \text{---} \text{red dot} \text{---} \text{green dot} = \text{green dot} \text{---} \text{green dot} \quad (\mathbf{B}_1)$$

The **B**₁-rule can be read loosely as "green copies red points" and "red copies green points", in both cases "up to a diamond", i.e. up to a scalar $\diamond$. In this thesis it is $\diamond = \sqrt{2}$.

Axiom 4.26 (**B**₂-rule - bialgebra rule).

$$\diamond \quad \text{green dot} \text{---} \text{red dot} \text{---} \text{green dot} \text{---} \text{red dot} = \text{split} \text{---} \text{red dot} \text{---} \text{green dot} \text{---} \text{split} \quad (\mathbf{B}_2)$$

The **B**₂-rule (or *bialgebra law*) is a powerful commutation principle, and generates a whole family of equations, allowing alternating cycles of red and green dots to be replaced with simpler graphs [18].

Remark 4.27. On circuits like the previous, we assume that wires are neither intersected, nor glued, and they do not meet on any vertex - in other case it will be clearly represented like that. It is just one wire above the other.

Example 4.28. An important equation derivable from the **B**-rules is (**B'**) (or Hopf law):

$$\text{loop} = \text{green dot} \text{---} \text{red dot} \quad (\mathbf{B}')$$

where the loop represents two diamonds (see Subsection 4.2.6). The above equation is obtained as follows:

$$\begin{aligned} \text{loop} &\stackrel{(\mathbf{T})}{=} \text{green dot} \text{---} \text{red dot} \text{---} \text{green dot} \text{---} \text{red dot} \stackrel{(\mathbf{S})}{=} \text{green dot} \text{---} \text{green dot} \text{---} \text{red dot} \text{---} \text{red dot} \stackrel{(\mathbf{D}_2)}{=} \diamond \text{---} \text{green dot} \text{---} \text{red dot} \text{---} \diamond \\ &\stackrel{(\mathbf{B}_2)}{=} \diamond \text{---} \text{split} \text{---} \text{red dot} \text{---} \text{green dot} \text{---} \text{split} \stackrel{(\mathbf{B}_1)}{=} \text{green dot} \text{---} \text{red dot} \text{---} \text{red dot} \stackrel{(\mathbf{S})}{=} \text{green dot} \text{---} \text{red dot} \end{aligned}$$

Remark 4.29. Rules $\mathbf{B}'$ and $\mathbf{B}_2$ are known informally as the Hopf law and the bialgebra law. Together, the $\mathbf{B}$ -rules state that the interaction of different coloured spiders produce a structure we call a scaled bialgebra, which differs from a bialgebra only by a normalising factor.

Proposition 4.30. *Let the notation be as above. Then, $(\mathcal{H}, m_{\bullet}, \eta_{\bullet}, \Delta_{\bullet}, \varepsilon_{\bullet})$ is an unnormalized Hopf algebra with trivial antipode.*

Proof. From Propositions 4.22, 4.23 we have that $(\mathcal{H}, m_{\bullet}, \eta_{\bullet})$ is an associative unital algebra and $(\mathcal{H}, \Delta_{\bullet}, \varepsilon_{\bullet})$ is a coassociative counital coalgebra. According to $\mathbf{B}_1$ rule we have that $\Delta_{\bullet}$ is unital with respect to $\eta_{\bullet}$, up to the normalization constant $\diamond$. Also, the $\mathbf{B}_1$ states that $m_{\bullet}$ is counital with respect to $\varepsilon_{\bullet}$, up to $\diamond$. From $\mathbf{B}_2$ we have that $\Delta_{\bullet}$ respects the product $m_{\bullet}$, up to $\diamond$. Thus, the structure is a bialgebra. The antipode axiom (4) with respect to the trivial antipode is automatically satisfied, due to $\mathbf{B}'$ rule. This yields $\mathcal{H}$ is an unnormalized Hopf algebra. $\square$

Thus, Proposition 4.30 gives an unexpected connection between Quantum Computing and Hopf algebra Theory, i.e. Quantum Group Theory (see Section 1.6). We continue to describe the algebraic structure in more detail. One can show, in the exact same way, that also $\Delta_{\bullet}$ respects the product $m_{\bullet}$ (up to the same normalization constant $\diamond$), which implies that also $(\mathcal{H}, m_{\bullet}, \eta_{\bullet}, \Delta_{\bullet}, \varepsilon_{\bullet})$ is an unnormalized Hopf algebra.

A natural question is whether $(\mathcal{H}, m_{\bullet}, \eta_{\bullet}, \Delta_{\bullet}, \varepsilon_{\bullet})$ or $(\mathcal{H}, m_{\bullet}, \eta_{\bullet}, \Delta_{\bullet}, \varepsilon_{\bullet})$ are unnormalized Hopf algebras. This is not the case in general, but in fact, it happens only in trivial situations. Instead, the green (or red) algebra and coalgebra together form an F -algebra, with compatibility condition discussed in Definition 1.57. Together with Proposition 4.30, this is the data of an F -Hopf algebra (see Subsection 1.6.2).

Proposition 4.31. [14] *The pair $(\mathcal{H}, m_{\bullet}, \eta_{\bullet}, \Delta_{\bullet}, \varepsilon_{\bullet})$, $(\mathcal{H}, m_{\bullet}, \eta_{\bullet}, \Delta_{\bullet}, \varepsilon_{\bullet})$ of F -algebras forms an F -Hopf algebra.*

Proof. We have that $(m_{\bullet}, \eta_{\bullet})$, $(m_{\bullet}, \eta_{\bullet})$ are associative unital algebra structures on $\mathcal{H}$. Also, $(\Delta_{\bullet}, \varepsilon_{\bullet})$, $(\Delta_{\bullet}, \varepsilon_{\bullet})$ are coassociative counital coalgebra structures on $\mathcal{H}$. Because of $\mathbf{S}_1$, the Equation (5) of Definition 1.57 holds. So, $(\mathcal{H}, m_{\bullet}, \eta_{\bullet}, \Delta_{\bullet}, \varepsilon_{\bullet})$, $(\mathcal{H}, m_{\bullet}, \eta_{\bullet}, \Delta_{\bullet}, \varepsilon_{\bullet})$ are F -algebras.

Moreover, from rules $\mathbf{B}_1$, $\mathbf{B}_2$, $\mathbf{D}_1$ and Equations (2), (3) we have that $(\mathcal{H}, m_{\bullet}, \eta_{\bullet}, \Delta_{\bullet}, \varepsilon_{\bullet})$, $(\mathcal{H}, m_{\bullet}, \eta_{\bullet}, \Delta_{\bullet}, \varepsilon_{\bullet})$ are bialgebras (up to the normalization constant $\diamond$). So, this pair is an F -bialgebra.

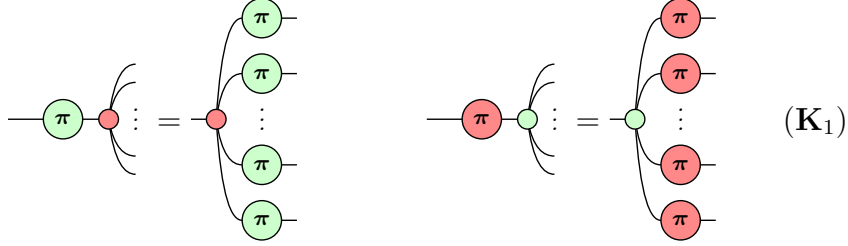
Regarding to Proposition 4.30 and it's dual case, $(\mathcal{H}, m_{\bullet}, \eta_{\bullet}, \Delta_{\bullet}, \varepsilon_{\bullet})$, $(\mathcal{H}, m_{\bullet}, \eta_{\bullet}, \Delta_{\bullet}, \varepsilon_{\bullet})$ are both Hopf algebras. Therefore, the pair is an F -algebra. $\square$

Hence, the structure of ZX -calculus satisfies the interaction of Z - and X -spiders as components of bialgebras and Hopf algebras. This means that we can successfully work on the Hopf algebras of the spaces of qubits, $(\mathbb{C}^2)^{\otimes n}$, using ZX -calculus.

4.2.4 The K-rules

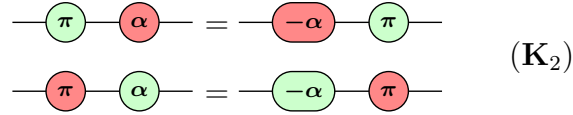
The following are known as the **K-rules**, where **K** stands for *classical* (klassisch in German) and they are concerned with special properties of spiders with phase $\alpha = \pi$. The vertices with phase π are *classical*, in the sense that they can be copied according to the following rules.

Axiom 4.32 (**K**₁-rule - π -copy).



Rule **K**₁ states that dots labeled by π commute with spiders of the other color.

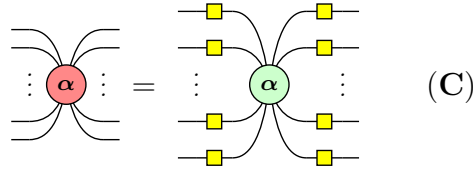
Axiom 4.33 (**K**₂-rule - π -commutation).



Rule **K**₂ states that the vertices of phase π invert the phase of the other color.

4.2.5 The C-rule

Axiom 4.34 (**C**-rule - Hadamard).



The **C-rule** concerns *color*. It permits the H vertex (the Hadamard gate) to commute with colored dots and function as an explicit color changing operation which transforms "green structures" into "red structures" and vice versa. Also, the composition of two Hadamard gates yields the identity operation, that is, **C**-rule asserts that H is self-inverse.

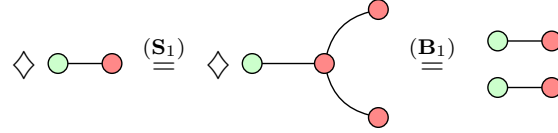
4.2.6 The D-rules

Finally, the **D-rules** determine the normalization constant as the outcome of the following closed circuits.

Axiom 4.35 ($\mathbf{D}_1$ -rule - diamond).

$$\text{green circle} - \text{red circle} = \diamond \quad (\mathbf{D}_1)$$

Remark 4.36. The $\mathbf{D}_1$ rule almost follows from the other rules:



which would yield the desired result if $\text{green circle} - \text{red circle}$ could be cancelled.

Axiom 4.37 ($\mathbf{D}_2$ -rule - loop).

$$\begin{array}{c} \diamond \\ \diamond \end{array} = \bigcirc \quad (\mathbf{D}_2)$$

Remark 4.38. The $\mathbf{D}_2$ rule states that two diamonds are equal to a loop of wire, itself the result of composing a $\subset$ and a $\supset$. The loop represents the dimension of the underlying Hilbert space, which is 2 in case of $\mathcal{H} = \mathbb{C}^2$. The spacial juxtaposition is a form of multiplication, and this justifies the name $\sqrt{D}$ for the diamond [14]. That is, for $\mathcal{H}$ case,

$$\dim(\mathcal{H}) = \bigcirc = \diamond \cdot \diamond = \diamond^2 \Rightarrow \diamond = \sqrt{2}.$$

Remark 4.39. In quantum computing, we can work on spaces with superposition of more than 2 states. We call these digits qudits (quantum dits), where the term "dit" refers to D -ary digit.

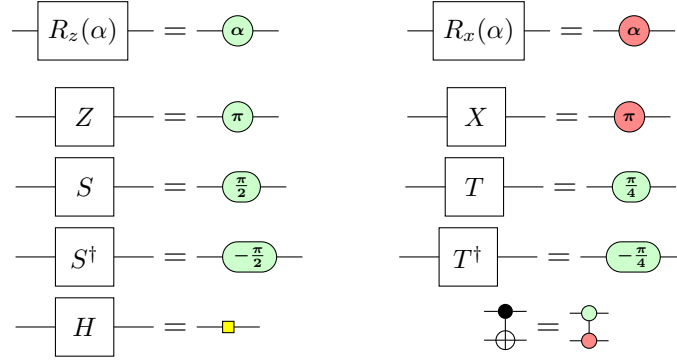
Two cases are the qutrits, for $D = 3$, and the ququarts, for $D = 4$. There, we work on $(\mathbb{C}^3)^{\otimes n}$ and $(\mathbb{C}^4)^{\otimes n}$, $n \in \mathbb{N}_+$, respectively. The computational bases there, are $\{|0\rangle, |1\rangle, |2\rangle\}$ for qutrits, $\{|0\rangle, |1\rangle, |2\rangle, |3\rangle\}$ for ququarts. In these cases, the black diamonds $\blacklozenge$ in ZX-calculus are $\blacklozenge = \sqrt{3}$ and $\blacklozenge = \sqrt{4}$, respectively.

One can generalize it for $D > 4$. The computational basis will be $\{|0\rangle, |1\rangle, \dots, |D\rangle\}$, and the respective Hilbert spaces are $(\mathbb{C}^D)^{\otimes n}$, $n \in \mathbb{N}_+$.

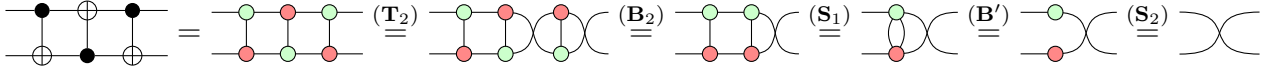
4.3 Applications using ZX-calculus

In this subsection we show some examples of quantum circuits simplification using ZX-calculus rules.

Remark 4.40. First, we give the representation in ZX-calculus of the basic quantum gates.

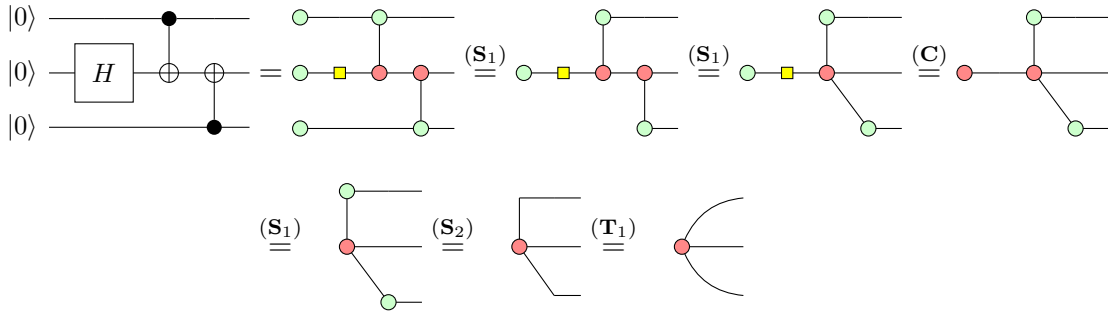


Example 4.41. We simplify the circuit that consists of three CNOT gates, that is given by the following diagram:

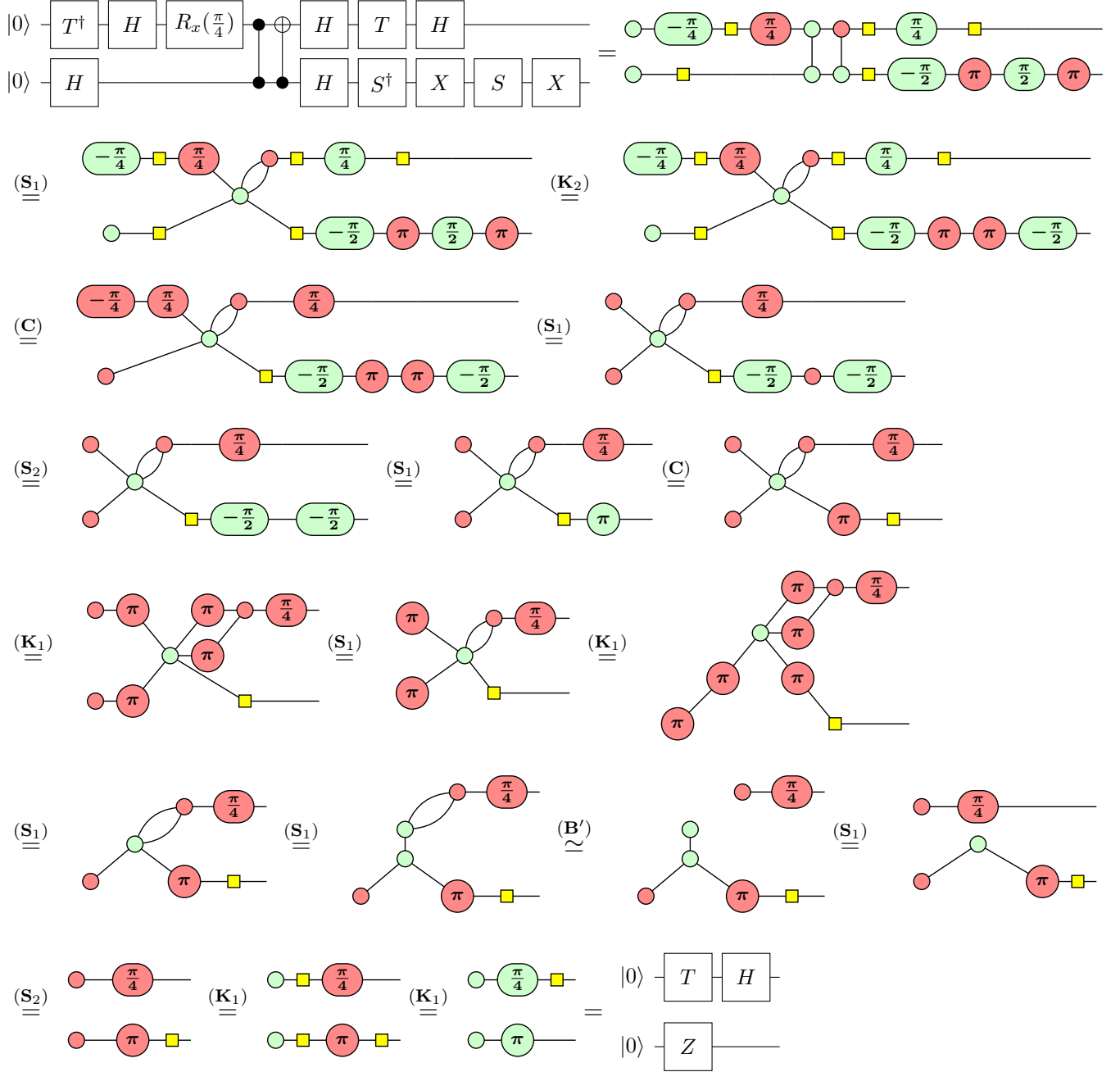


which is the SWAP gate.

Example 4.42. In the following circuit we suppose that we work on a hardware with default initial value $|000\rangle$.



The last example is a part of the circuit of the Example 4.1. We simplify the entangled $|q_1q_4\rangle$ qubit, that is, the upper and lower qubits of the example.

Example 4.43.


Bibliography

- [1] S.-I. Amari. Natural gradient works efficiently in learning. *Neural computation*, 10(2):251–276, 1998.
- [2] Y. Arike. A construction of symmetric linear functions on the restricted quantum group $u_q(sl_2)$. 2010.
- [3] S. Axler. *Linear algebra done right*. Springer Nature, 2024.
- [4] R. Aziz and S. Majid. Co-double bosonisation and dual bases of $c_q[sl_2]$ and $c_q[sl_3]$. *Journal of Algebra*, 518:75–118, 2019.
- [5] W. Ballmann. *Lectures on Kähler manifolds*, volume 2. European mathematical society, 2006.
- [6] A. Barenco, C. H. Bennett, R. Cleve, D. P. DiVincenzo, N. Margolus, P. Shor, T. Sleator, J. A. Smolin, and H. Weinfurter. Elementary gates for quantum computation. *Physical review A*, 52(5):3457, 1995.
- [7] E. J. Beggs and S. Majid. *Quantum Riemannian Geometry*, volume 355. Springer, 2020.
- [8] B. Berche, D. Malterre, and E. Medina. Gauge transformations and conserved quantities in classical and quantum mechanics. *American Journal of Physics*, 84(8):616–625, 2016.
- [9] J. Bichon. The group of bi-galois objects over the coordinate algebra of the frobenius-lusztig kernel of sl_2 . *Glasgow Mathematical Journal*, 58(3):727–738, 2016.
- [10] T. Brzezinski and R. Wisbauer. *Corings and comodules*. Number 309. Cambridge University Press, 2003.
- [11] R. Cheng. Quantum geometric tensor (fubini-study metric) in simple quantum system: A pedagogical introduction. *arXiv preprint arXiv:1012.1337*, 2010.
- [12] B. Coecke. Basic zx-calculus for students and professionals. *arXiv preprint arXiv:2303.03163*, 2023.

- [13] B. Coecke and R. Duncan. Interacting quantum observables. In *International colloquium on automata, languages, and programming*, pages 298–310. Springer, 2008.
- [14] B. Coecke and R. Duncan. Interacting quantum observables: categorical algebra and diagrammatics. *New Journal of Physics*, 13(4):043016, 2011.
- [15] P. Comon, G. Golub, L.-H. Lim, and B. Mourrain. Symmetric tensors and symmetric tensor rank. *SIAM Journal on Matrix Analysis and Applications*, 30(3):1254–1279, 2008.
- [16] T. M. Cover. *Elements of information theory*. John Wiley & Sons, 1999.
- [17] E. Dennis, A. Kitaev, A. Landahl, and J. Preskill. Topological quantum memory. *Journal of Mathematical Physics*, 43(9):4452–4505, 2002.
- [18] R. Duncan and S. Perdrix. Graph states and the necessity of euler decomposition. In *Conference on Computability in Europe*, pages 167–177. Springer, 2009.
- [19] R. Duncan and S. Perdrix. Pivoting makes the zx-calculus complete for real stabilizers. *arXiv preprint arXiv:1307.7048*, 2013.
- [20] E. Ercolessi, R. Fioresi, and T. Weber. The geometry of quantum computing. *International Journal of Geometric Methods in Modern Physics*, 21(10), 2024.
- [21] P. Facchi, R. Kulkarni, V. Man’Ko, G. Marmo, E. Sudarshan, and F. Ventriglia. Classical and quantum fisher information in the geometrical formulation of quantum mechanics. *Physics Letters A*, 374(48):4801–4803, 2010.
- [22] B. L. Feigin, A. M. Gainutdinov, A. M. Semikhatov, and I. Y. Tipunin. Modular group representations and fusion in logarithmic conformal field theories and in the quantum group center. *Communications in mathematical physics*, 265(1):47–93, 2006.
- [23] R. P. Feynman. Simulating physics with computers. In *Feynman and computation*, pages 133–153. cRc Press, 2018.
- [24] F. Finster, S. Kindermann, and J.-H. Treude. Causal fermion systems: An introduction to fundamental structures, methods and applications. *arXiv preprint arXiv:2411.06450*, 2024.
- [25] R. Fioresi and F. Zanchetta. Deep learning and geometric deep learning: An introduction for mathematicians and physicists. *International Journal of Geometric Methods in Modern Physics*, 20(12):2330006, 2023.
- [26] T. Frankel. *The geometry of physics: an introduction*. Cambridge university press, 2004.

BIBLIOGRAPHY

- [27] K. Fritzsche, H. Grauert, and H. Grauert. *From holomorphic functions to complex manifolds*, volume 213. Springer, 2002.
- [28] L. Godinho and J. Natário. *An introduction to Riemannian geometry*. Springer, 2014.
- [29] D. Gottesman. An introduction to quantum error correction and fault-tolerant quantum computation. *arXiv preprint arXiv:0904.2557*, 2009.
- [30] R. B. Griffiths. *Consistent quantum theory*. Cambridge University Press, 2003.
- [31] X.-Y. Hou, Z. Zhou, X. Wang, H. Guo, and C.-C. Chien. Local geometry and quantum geometric tensor of mixed states. *Physical Review B*, 110(3):035144, 2024.
- [32] Y.-M. R. Hu, E. A. Ostrovskaia, and E. Estrecho. Quantum geometric tensor and wavepacket dynamics in two-dimensional non-hermitian systems. *Physical Review Research*, 7(1):L012067, 2025.
- [33] D. Huybrechts. *Complex geometry: an introduction*. Springer, 2005.
- [34] A. Joyal and R. Street. Braided tensor categories. *Advances in Mathematics*, 102(1):20–78, 1993.
- [35] D. Joyce. Compact manifolds with exceptional holonomy. In *Proceedings of the International Congress of Mathematicians 1998*, pages 361–370. European Mathematical Society-EMS-Publishing House GmbH, 1998.
- [36] E. Kähler. Über eine bemerkenswerte hermitesche metrik. In *Abhandlungen aus dem Mathematischen Seminar der Universität Hamburg*, volume 9, pages 173–186. Springer, 1933.
- [37] C. Kassel. Postlude. a universal knot invariant. In *Quantum Groups*, pages 484–505. Springer, 1995.
- [38] A. A. Kirillov. Lie groups and lie algebras. In *Elements of the Theory of Representations*, pages 83–107. Springer, 1976.
- [39] A. Kissinger and J. van de Wetering. Picturing quantum software: An introduction to the zx-calculus and quantum compilation. *Preprint*, 2024.
- [40] A. Kitaev. Periodic table for topological insulators and superconductors. In *AIP conference proceedings*, volume 1134, pages 22–30. American Institute of Physics, 2009.
- [41] A. Y. Kitaev. Fault-tolerant quantum computation by anyons. *Annals of physics*, 303(1):2–30, 2003.

-
- [42] A. Klimyk and K. Schmüdgen. The quantum algebra $u_q(\mathfrak{sl}_2)$ and its representations. In *Quantum Groups and Their Representations*, pages 53–96. Springer, 1997.
 - [43] A. W. Knap and A. W. Knap. *Lie groups beyond an introduction*, volume 140. Springer, 1996.
 - [44] S. Kobayashi and K. Nomizu. *Foundations of differential geometry, volume 2*, volume 2. John Wiley & Sons, 1996.
 - [45] D. Kretschmann, D. Schlingemann, and R. F. Werner. The information-disturbance trade-off and the continuity of stinespring’s representation. *IEEE transactions on information theory*, 54(4):1708–1717, 2008.
 - [46] S. Lang. *Algebra*. Springer Science & Business Media, 2012.
 - [47] S. L. Lauritzen. Statistical manifolds. *Differential geometry in statistical inference*, 10(2):163–216, 1987.
 - [48] J. M. Lee. Smooth manifolds. In *Introduction to smooth manifolds*, pages 1–29. Springer, 2003.
 - [49] A. Lesne. Shannon entropy: a rigorous notion at the crossroads between probability, information theory, dynamical systems and statistical physics. *Mathematical Structures in Computer Science*, 24(3):e240311, 2014.
 - [50] J. Liu, X.-X. Jing, W. Zhong, and X.-G. Wang. Quantum fisher information for density matrices with arbitrary ranks. *Communications in Theoretical Physics*, 61(1):45, 2014.
 - [51] J. Liu, H. Yuan, X.-M. Lu, and X. Wang. Quantum fisher information matrix and multi-parameter estimation. *Journal of Physics A: Mathematical and Theoretical*, 53(2):023001, 2020.
 - [52] T. Liu, Q. Zhang, and C. Qiu. Quantum geometric inequality and its classical wave verification. *Physical Review Letters*, 136(11):116602, 2026.
 - [53] S. Majid. *Foundations of quantum group theory*. Cambridge university press, 2000.
 - [54] S. Majid. Quantum and braided zx calculus. *Journal of Physics A: Mathematical and Theoretical*, 55(25):254007, 2022.
 - [55] S. Montgomery. *Hopf algebras and their actions on rings*. Number 82. American Mathematical Soc., 1993.
 - [56] A. Moroianu. Lectures on kähler geometry. *arXiv preprint math.DG/0402223*, 2004.

BIBLIOGRAPHY

- [57] M. Nakahara. *Geometry, topology and physics*. CRC press, 2018.
- [58] D. Nakamura, T. Shibata, and K. Shimizu. Exact module categories over $\text{Rep}(u_q(\mathfrak{sl}_2))$. *arXiv preprint arXiv:2503.21265*, 2025.
- [59] M. A. Nielsen and I. L. Chuang. *Quantum Computation and Quantum Information*. Cambridge University Press, 10th anniversary edition edition, 2010.
- [60] L. C. Palais. *zx-calculus: zx-calculus with tikz*, 2024.
- [61] A. Ram. Quantum groups: A survey of definitions, motivations, and results. *Geometric analysis and Lie theory in mathematics and physics, Austral Math. Soc. Lect. Ser.*, 1(1):20–104, 1996.
- [62] J. W. Robbin and D. A. Salamon. Introduction to differential topology. *Book project*, 2011.
- [63] J. W. Robbin and D. A. Salamon. *Introduction to differential geometry*. Springer Nature, 2022.
- [64] M. R. Sepanski. *Compact lie groups*. Springer, 2007.
- [65] M. E. Sweedler. Hopf algebras wa benjamin. *Inc., New York*, 1969.
- [66] C. G. Timpson. Probabilities in realist views of quantum mechanics. *Probabilities in physics*, page 201, 2011.
- [67] L. W. Tu. *An Introduction to Manifolds*. Universitext. Springer, 2011.
- [68] J. van de Wetering. Zx-calculus for the working quantum computer scientist. *arXiv preprint arXiv:2012.13966*, 2020.
- [69] J. von Neumann. Probabilistic logics and the synthesis of reliable organisms from unreliable components. In C. E. Shannon and J. McCarthy, editors, *Automata Studies*, pages 43–98. Princeton University Press, 1956.
- [70] L. Wasserman. *All of statistics: a concise course in statistical inference*, volume 26. Springer, 2004.
- [71] R. O. Wells and O. García-Prada. *Differential analysis on complex manifolds*, volume 21980. Springer New York, 1980.
- [72] A. Wu and C. N. Yang. Evolution of the concept of the vector potential in the description of fundamental interactions. *International Journal of Modern Physics A*, 21(16):3235–3277, 2006.